Anatomy of a Safe Asset: Bund Market Structure and Reform

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Abstract

This paper constructs a non-anonymous transaction-level dataset for German Bunds, the euro area's benchmark safe asset, from MiFID I and MiFID II transaction reports since 2008. We show that a concentrated dealer core intermediates this safe asset through persistent dealer–client relationships. Trading is segmented across investor sectors, with order flow fragmented across bilateral OTC and electronic venues. Identified stress flows show foreign official accounts supplying Bunds whether prices fall in a dash for cash or rise in a flight to quality. Hedge funds absorb Bunds in both episodes, while primary dealers absorb Covid selling but distribute Bunds after fiscal supply news. Hedge funds and official sector accounts carry the most permanent price-relevant flow yet pay middling effective half-spreads, while retail households pay by far the highest. Execution costs do not increase monotonically with the price content of flow. We evaluate two reforms targeting this structure: mandatory central clearing would cut dealers' net settlement obligations in Bunds by 35%, less than comparable estimates for U.S. Treasuries and U.K. Gilts. In the benchmark structural calibration, platform centralization lowers welfare because matching gains from broader dealer access do not offset lost relationship surplus under imperfect dealer competition.

Keywords: safe assets, sovereign bonds, German Bunds, market microstructure, OTC markets, dealer intermediation, electronic trading platforms, central clearing, MiFID II transaction reports.

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1 Introduction

Repeated stress episodes have put the design, regulation, and reform of sovereign bond markets back at the center of policy debate.[1] Yet despite the central role of sovereign bonds as safe assets (Gorton, 2017; Caballero, Farhi, and Gourinchas, 2017), their market microstructure remains opaque. To diagnose stress and prescribe reform reliably, policymakers need an anatomy of the market. We build that anatomy from what is, to our knowledge, the most granular and comprehensive transaction-level record of any major sovereign bond market. Covering nearly two decades of German Bund transactions, the record shows who meets whom, through which venues, and at what prices. We use this microdata to document market structure and evaluate two major reforms as counterfactuals.

We first map the market structure. A tight primary dealer core intermediates most activity and connects a broad client periphery; direct client-to-client trading is rare. Clients reach that core through narrow, persistent relationships: most bilateral clients use a single dealer, and electronic platforms broaden access without delivering all-to-all trading. The periphery is segmented across venue types: investor sectors reach the market in systematically different ways. Trading is fragmented across venues and concentrated in a small set of bonds, especially on-the-run and futures-deliverable issues, while most outstanding bonds trade only sporadically on a given venue. The market is therefore hierarchical, relationship-based, and segmented.

Identifying both counterparties shows directly which sectors supply and absorb Bunds in stress. In the 2020 dash for cash and the 2023 flight to quality, despite opposite price moves, foreign official accounts are the dominant net sellers and hedge funds buy, while primary dealers absorb material flow only in the cash shock. After the 2025 fiscal supply news, primary dealers distribute Bunds rather than warehousing them. In normal times, hedge funds and official sector accounts carry the most permanent price-relevant flow, but they do not pay the highest effective half-spreads. Retail households pay by far the most, and, within a given investor, larger trades pay more. The effective half-spread is therefore an execution cost that does not map monotonically to adverse selection.

Both reform counterfactuals target the dealer core, but at different stages. Central clearing keeps observed trades fixed and changes how dealer obligations net after execution. Netting each dealer's own trades at a clearing house would cut dealers' net settlement obligations by 35%, less than comparable estimates for U.S. Treasuries and U.K. Gilts (Fleming and Keane, 2021; Baranova, Holbrook, MacDonald, Rawstorne, Vause, and Waddington, 2023). Platform centralization changes the matching problem before execution: investors may reach more dealers, dealers may compete more aggressively, and bilateral relationship value may be lost. Applying the dealer–investor framework of Allen and Wittwer (2023) to Bunds, we find that forced platform centralization lowers welfare by 7.73% relative to the current access regime. Matching

[1] Duffie (2025) provides a recent overview. Current reform proposals in the United States include mandatory central clearing (SEC) and capital regulation affecting dealer intermediation capacity (Federal Reserve). Reforms in the EU include faster settlement from $T+2$ to $T+1$ (ESMA), a consolidated tape (ESMA), and venue consolidation (Reuters). The Savings and Investments Union adds market integration to the EU agenda (European Commission, COM(2025) 124, 19 March 2025).

gains from broader dealer access do not offset lost relationship surplus under imperfect dealer competition. The sign matches the Canadian estimates (a loss of 1.75%), and in the benchmark calibration the losses are larger. We import only the demand elasticity, which the Bund data cannot identify (Appendix A.6). The loss relative to a fully bilateral market persists across its entire confidence interval.

2 Literature

The paper relates to four aspects of decentralized fixed income markets: dealer cores, relationships, venue segmentation, and market design. The facts show where existing models fit the Bund market and where they do not; no single structural model matches all of them. The first aspect is the dealer core. Bond markets remain more decentralized and relationship-based than equity markets (Bessembinder, Spatt, and Venkataraman, 2020; Biais and Green, 2019), and sovereign bond price formation depends on liquidity and intermediation (Fleming and Remolona, 1999; Pasquariello and Vega, 2007; Cheung, Rindi, and De Jong, 2005; Mizrach and Neely, 2008). A concentrated intermediary layer can improve inventory offsetting and execution capacity, but it can also concentrate market power and balance sheet obligations, as in core–periphery and dealer network theories (Wang, 2016; Li and Schürhoff, 2019; Colliard, Foucault, and Hoffmann, 2021) and related work on dealer specialization, primary market participation, and balance sheet provision (Wittwer and Uthemann, 2025; Allen, Hortaçsu, Richert, and Wittwer, 2026; Huh and Infante, 2021; Duffie, 2023). Our facts bear on both sides: the tight hierarchy and the rarity of direct client-to-client trading measure how concentrated intermediation has become, while the stress episodes show the core absorbing cash shock selling in 2020, taking little net flow in the 2023 safety shock, and distributing Bunds after the 2025 fiscal news; relative to trade-level safe asset stress work on U.K. Gilts (Pintér, 2023; Pintér, Siriwardane, and Walker, 2026), the identified Bund flows span episodes of opposite price sign and a domestic supply contrast (Duffie, 2025). They do not identify whether the Bund core is too thin or too broad. The safe asset literature asks why investors demand safety and which assets supply it (Gorton, 2017; Caballero et al., 2017; He, Krishnamurthy, and Milbradt, 2019). We take the safe asset status of Bunds as given and document the market through which that demand is met: who buys and sells in safety shocks, and what they pay to trade.

The second aspect is relationship trading. Search and relationship models show why clients use a small set of dealers, why platforms need not create all-to-all competition, and why a dealer connection can carry non-price value (Allen and Wittwer, 2023; Wang, 2023, 2022; Allen and Wittwer, 2025; Di Maggio, Kermani, and Song, 2017; Kondor and Pintér, 2022). Our relationship facts show that Bund clients broaden access on platforms but remain far below random matching, which is the market structure input for the centralization exercise.

The third aspect is venue segmentation and information. Trading protocols allocate order flow across venues, and the identity of that flow matters for price discovery (Hasbrouck, 1995; Menkveld, Yueshen, and Zhu, 2017; Dugast, Üslü, and Weill, 2022; Lee and Wang, 2025; Lee, Wang, and Zawadowski, 2025; Hagströmer and Menkveld, 2019; Hagströmer, 2022; Hagströmer

and Menkveld, 2023). Outages provide causal evidence in the same market: when the Bund futures market halts, cash market liquidity evaporates, dealers withdraw, and transaction prices deviate from fundamentals (Kerssenfischer and Helmus, 2025)—time-series evidence that market functioning depends on specific venues. Identified dealer-to-client data also let us separate whose flow has permanent price content from who pays the highest effective half-spread (Czech, Huang, Lou, and Wang, 2021; Kondor and Pintér, 2022; Pasquariello and Vega, 2007; Pintér, Wang, and Zou, 2024, 2022; Lee, Nathan, and Wang, 2024). That separation is central here: the most price-relevant flow is not the flow charged the highest execution cost.

The fourth aspect is market design. Central clearing, platform centralization, and all-to-all trading are distinct reforms. Clearing changes post-trade netting (Fleming and Keane, 2021; Baranova et al., 2023; Duffie and Wang, 2017; Duffie, Singh, and Wang, 2026). Platform centralization changes matching, competition, and relationship value; in that structural model, markups and spreads are transfers, so welfare turns on gains from trade, not on price concessions alone (Allen and Wittwer, 2023; Kutai, Nathan, and Wittwer, 2025; Chaboud, Correia Golay, Cox, Fleming, Huh, Keane, Lee, Schwarz, Vega, and Windover, 2025; Lee and Wang, 2025; Duffie, 2023). We therefore treat clearing as a mechanical settlement counterfactual and platform centralization as a structural welfare counterfactual.[2]

3 Data and Measurement

We assemble a new, non-anonymous transaction-level dataset for the German sovereign bond market. As Figure 1 summarizes, we distill roughly 19 million raw regulatory reports from two reporting regimes into approximately 16.4 million Bund trade records spanning 2008–2025. We then match each transaction—on ISIN, LEI, MIC, and timestamp—to bond, investor, venue, and price benchmarks from more than a dozen external reference datasets. Section D provides details.

The underlying transaction reports are confidential regulatory microdata and cannot be posted publicly. We will maintain the final research dataset and its documentation as a reusable resource for Bundesbank researchers whose projects are authorized to use the underlying data.

3.1 Transaction-Level

Our transaction-level dataset for German sovereign bonds ("Bunds") combines two regulatory reporting regimes: MiFID I (2008–2017) and MiFID II/MiFIR transaction reports (2018–2025). Together they cover approximately 16.4 million trade records; the unit of the cleaned file is the trade record as reported, and each record has two legs, its buyer side and its seller side. One execution can appear as several records. An agency or matched principal report, in which the reporting firm is neither buyer nor seller, becomes two records, one against each side, and on an anonymous platform each member files its own record against the venue. We link the

[2]The same transaction data support a companion paper tracing an on-the-run discount in Bunds to the Finanzagentur's tapping policy (Kerssenfischer, 2026). We keep that pricing question separate and focus here on market structure and trading costs.

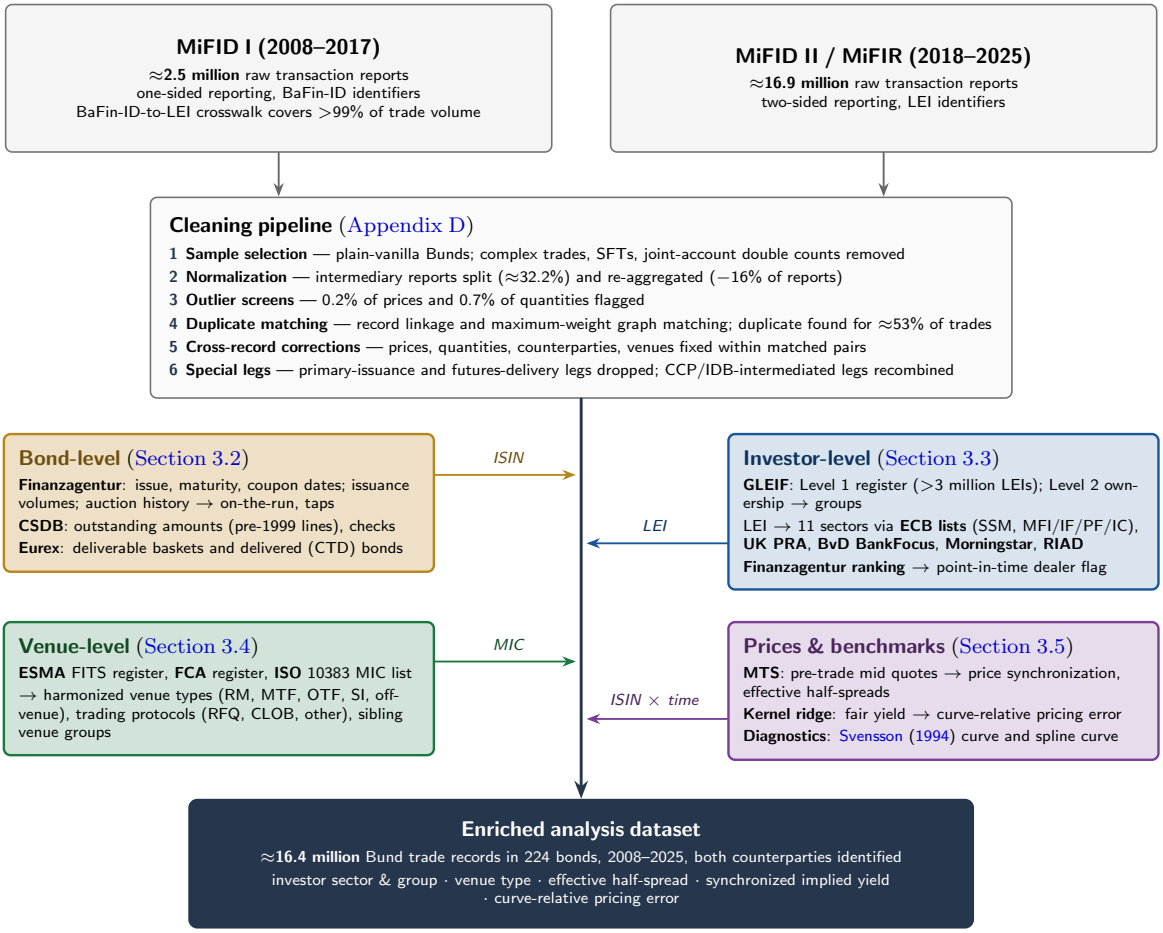


Figure 1: Construction of the analysis dataset. Regulatory transaction reports from the two reporting regimes (top) pass through the cleaning pipeline (middle). The cleaned records are enriched with bond-, investor-, venue-, and price-level reference data (side panels), each merged on the indicated key (bottom). Percentages give the share of reports or trades affected by each cleaning step; references point to the relevant sections and Appendix material.

records of each execution (Section D.8). Statistics of market size—volume, trade counts and sizes, venue shares, and bond and venue concentration—count each execution once; summed over records, average daily MiFID II volume would be €24 bn instead of €20 bn. Statistics that need a buyer or a seller—sectors, dealers, counterparties, net flows, and the price regressions—use records, because the records of one execution name different buyers and sellers, such as an agent and its client. In those analyses, a trade is a record. The reports are regulatory records, not research data, so we normalize reporting formats, aggregate mechanically related reports, screen implausible prices and quantities, and identify duplicates using institutional features of each regime. The full pipeline and diagnostics are in Appendices D.1–D.8.

MiFID II/MiFIR is the core market structure sample. Since 3 January 2018, investment firms have reported German sovereign bond transactions to BaFin irrespective of trading venue, with LEI-based buyer, seller, and executing-firm identifiers. Each report names the buyer and the seller, so we observe both counterparties of bilateral trades, and two-sided reporting lets us match their duplicate reports. On anonymous platforms each member reports only its own side; we link the two reports where both are in the data (Section D.8). MiFID I provides a longer historical window but has more limited client-side coverage because reporting was predominantly one-sided and used BaFin institutional codes rather than LEIs. We resolve those codes to LEIs where possible, but the market structure analysis uses the MiFID II subsample unless otherwise noted.

We cross-check the regulatory reports against commercial transaction records from MTS and other trading venues. The comparisons cover prices, quantities, timestamps, and execution counts within overlapping samples. Appendix D.9 describes the sources and method.

3.2 Bond-Level

We obtain bond-level information from the Finanzagentur: issue date, maturity date, first coupon date, dated date (start of accrual), and gross outstanding volume (cumulative auction and syndication volume, including the Finanzagentur's own-account quota).[3]

Each day, we identify whether each outstanding bond is deliverable into the front contract of the 2-year, 5-year, 10-year, or 30-year Eurex Bund futures, applying the contract eligibility rules detailed in Section D. The cheapest-to-deliver (CTD) bond of each contract is the bond most delivered into it, according to Eurex statistics on notified bonds (Bloomberg data before 2009). The on-the-run bond in each maturity segment is the most recent new issue in the Finanzagentur's auction and syndication history, which also identifies tap reissuances.

[3]Finanzagentur bond metadata are more comprehensive and reliable than the Centralised Securities Database (CSDB). Hence, we use the CSDB to validate monthly outstanding volumes of each bond and, together with the Finanzagentur's year-end statistics, to supply outstanding volumes for bonds issued before the auction history begins in 1999.

3.3 Investor-Level

Our investor reference data cover over three million Legal Entity Identifiers (LEIs) from the Global Legal Entity Identifier Foundation (GLEIF).[4] We map LEIs into eleven sectors: primary dealers (participants in the primary auctions of Bunds); banks (commercial, investment, savings, and cooperative banks); interdealer brokers (IDB; voice and electronic brokers intermediating between dealers); investment funds (UCITS, money market funds, ETFs, and asset managers); hedge funds, including principal trading firms (PTFs)[5]; insurance companies and pension funds (ICPF); central counterparties (CCPs); non-bank financial intermediaries (NBFI; e.g. execution intermediaries, trading venue operators, and payment service providers); official sector entities (e.g. central banks and sovereign wealth funds); non-financial corporations (NFC); and households (individual retail investors).[6] Primary issuance defines the dealer core institutionally. The Finanzagentur sells Bunds in multiple-price auctions in which only members of the Bund Issues Auction Group (Bietergruppe) may bid. Membership requires winning at least 0.05% of the duration-weighted amounts allotted in a calendar year; members that fall short are excluded, and the Finanzagentur publishes a semiannual ranking of members.[7] We distinguish dealer-to-dealer, dealer-to-client, and client-to-client trades using a point-in-time dealer flag constructed from Finanzagentur's semiannual auction group rankings and the entry and exit information reported in their footnotes.[8] This dealer definition is intentionally institutional rather than activity-based: some large banks and market makers outside the auction group are classified as clients, not dealers. The secondary market network and concentration facts therefore validate the dealer core in the data rather than defining it by trading volume ex post. Because the auction member is often one entity of a larger banking group, we define dealers for investor groups under "Dealers and Clients".

Appendix D.10 details the mapping, which combines multiple regulatory, commercial, and manually curated sources. To ensure consistent country codes, we use the GLEIF register to map LEIs to countries. The resulting investor reference dataset has near-complete coverage for the Bund market.

Counterparty Coverage by Regime MiFID I usually identifies reporting firms and regulated counterparties, not end clients. MiFID II names buyers and sellers directly. On the client side of dealer-to-client records, the share of volume with a sector rises from 17% in 2013–2017 to 62% in 2018–2022 (Table A14). Our sector-level analyses therefore use MiFID II.

[4] https://www.gleif.org/en/lei-data/access-and-use-lei-data/level-1-data-who-is-who

[5] Both groups trade in arbitrage, price discovery, and liquidity provision (Aragon and Strahan, 2012; Menkveld, 2013). Hedge funds manage delegated capital; PTFs trade their own balance sheets. The pooled category describes a business type, not a common strategy or funding mechanism.

[6] Natural persons have no retained individual identifier and cannot be linked across trades. They enter sector-level statistics but not group-level counterparty measures. They appear on 32% of MiFID II records (13.1% outside client allocations), but account for 0.16% of notional; the median ticket is €1,441.

[7] See the auction rules.

[8] For example, see Finanzagentur.

Investor Grouping We consolidate LEIs to the group level, using GLEIF ownership relationships augmented with ECB registers, to avoid treating subsidiaries, branches, and parent entities of the same institution as independent investors. We consolidate only operating companies and keep fund vehicles separate, so each fund remains its own group. Consolidating funds would collapse the many separately identified vehicles advised by a single manager into a few mega-groups that mechanically dominate any concentration, dispersion, or network statistic, erasing the cross-fund variation the fund-level reports are designed to capture. The rule keeps investment funds and hedge funds at the entity level (99% are singleton groups), while banks, insurers, interdealer brokers, and primary dealers are consolidated wherever ownership links exist. Asset managers keep the investment fund sector whoever owns them, because they trade as agents for the funds and mandates they manage.

The transaction data are then merged to both entity-level and group-level sector and country identifiers; unless otherwise noted, the analyses use group-level identities. Appendix D.10 reports the data sources, hierarchy rules, and validation checks.

Dealers and Clients Each investor thus carries three labels: the sector of its own LEI, its investor group, and the sector of that group. Apart from asset managers, a group takes the sector of its head entity. A group with an auction member at any point since 2018 is in the primary dealer sector (Appendix D.10), except for its asset management and insurance subsidiaries, which keep their own sectors. The remaining entities of a primary dealer group form its dealer desk: the auction member and its sister trading entities, for example the London entity of a group whose auction member is its EU bank.

We use two dealer definitions. The primary dealer sector is time-invariant and describes what an institution is. The dealer flag is point-in-time and describes whether it acts as an auction group dealer on the day: an entity is a dealer on a trade date if it belongs to the dealer desk of a group with an auction member on that date. Asset management and insurance arms are never flagged; they are clients in their own sectors, also when they trade with their parent. A trade record is dealer-to-dealer, dealer-to-client, or client-to-client by the number of its two legs that carry the flag. A leg without an identified account owner is never flagged, so a dealer's one-sided record of an anonymous platform trade is a dealer-to-client record without an identified client. The client-level analyses—information, execution costs, and the platform and home dealer tests—use the flag and keep records with an identified client outside the primary dealer sector; clients in that sector are dealer groups trading outside their membership windows. The venue type test also reports all dealer-to-client records (Appendix A.2). Statistics on the dealer core as a set of institutions—its concentration, its network, the clearing exercise, and the welfare model—use the primary dealer sector.

Records between two entities of one investor group are intragroup bookings, almost all of them between the entities of one dealer group, such as its EU bank and its U.K. broker-dealer; they carry 26% of MiFID II record volume. The network and dealer concentration statistics

exclude them; the market size and venue statistics include them. Between different groups, dealer-to-dealer records carry 4–7% of record volume a year.[9]

3.4 Venue-Level

We classify trading venues by combining three external reference datasets—the ESMA Financial Instruments Transparency System (FITS) register, the U.K. Financial Conduct Authority (FCA) register, and the ISO 10383 list of Market Identifier Codes (MICs)—each providing venue identifiers (MICs), legal entity identifiers (LEIs), names, and venue types. We harmonize venue types to the ESMA taxonomy of multilateral trading facilities (MTFs), organized trading facilities (OTFs), systematic internalizers (SIs), and regulated markets (RMs), applying the sources hierarchically (ESMA, then FCA, then ISO), which covers all MICs present in MiFIR; Appendix D.11 details the sources, venue group construction, and residual classification of the few venues without an explicit type. Each MIC also receives a protocol family: request-for-quote (RFQ), central limit order book (CLOB), other, or unknown. The family describes the trading model of a venue or segment, not how an individual trade executed; Appendix D.11 states the rules and the coverage of each label.

Venue Grouping Due primarily to Brexit, several major venue operators maintain parallel trading venues in the European Union and the United Kingdom, resulting in distinct MIC codes for what is effectively the same trading infrastructure (for example, Bloomberg, Tradeweb, MarketAxess). These "sibling venues" largely reflect regulatory and reporting requirements, not economically meaningful differences in market structure, and are typically indistinguishable from the perspective of market participants. Hence, we merge MICs of the same venue type that belong to the same operator into a common venue group.

3.5 Price Synchronization and Fair Value

We synchronize transaction prices to the Bundesbank's 11:45 a.m. fixing using changes in MTS mid-quotes, following Duffie, Fleming, Keane, Nelson, Shachar, and Van Tassel (2023). Quote-quality checks and separately flagged fallbacks address unreliable or missing quotes; effective half-spreads always require a pre-trade quote for the same bond. We convert synchronized prices into yields using the settlement cycle and the ICMA coupon convention. Individual-price screening is centered on the bond's own daily yield, with a separate guard against gross errors relative to the Bundesbank curve. This preserves bond-specific yield deviations for analysis. Appendix B details synchronization, sample selection, and curve estimation.

We distinguish three pricing measures. The effective half-spread is the deviation of the transaction price from the pre-trade MTS mid, signed from the client's side and reported in price

[9] This record share understates interdealer trading. On an anonymous broker or order book venue, each member files a one-sided record against the venue, and from 2021 a U.K. member's record is outside the data. Executions with dealers of two groups identified carry 9% of MiFID II execution volume in 2018–2020 and 7% from 2021. Adding executions in which one dealer group faces an interdealer broker or order book venue and no other party is identified raises these shares to at most 29% and 23%.

basis points. The synchronized implied yield is the yield-to-maturity implied by the synchronized price, reported in yield basis points. The curve-relative pricing error is the synchronized implied yield minus the kernel ridge fair yield; positive values indicate that the bond trades cheap relative to the kernel ridge curve.

Our baseline fair value benchmark is the kernel ridge Bund yield curve, estimated day by day from each bond's volume-weighted synchronized price following Filipović, Pelger, and Ye (2025); Appendix B gives implementation details. We also estimate the Svensson (1994) curve published by the Bundesbank and a spline curve for diagnostics, including the yield curve fitting error comparison in Figure A12. The execution cost analyses use effective half-spreads, the price dispersion analysis uses synchronized implied yields directly, and the informed flow and venue price discovery tests use changes in curve-relative pricing errors. The platform centralization model also uses the kernel ridge curve as the fair value reference for markups. The clearing exercises are separate and measured in notional euros and settlement obligations.

4 Market Structure

4.1 Dealer Core and Relationships

The central market structure fact is hierarchy. Figure 2 plots the Bund trading network in the MiFID II subsample.[10] A small set of primary dealers sits at the center, intermediating most activity and connecting a broad client periphery. Direct trading between two identified clients is rare: it carries at most 4.0% of MiFID II execution volume in 2018–2020 and 1.6% from 2021. The dealer core is therefore not a label imposed by the primary market roster; it is visible in secondary market transactions.

A concentrated dealer core The network is not only dealer-centered but highly concentrated within the dealer group itself. Across the 38 primary dealers active in our sample, intermediation is far from evenly distributed: the five most active dealers account for 44% of all dealer-intermediated volume, and the top ten for 71%. The auction design sets who belongs to this group, not how concentrated intermediation within it is: the rules impose a minimum allotment share, and the same core emerges from secondary market behavior alone (Table 2). Figure 3 plots the disclosure-safe concentration curve from rank five onward and, as a benchmark, the analogous curve for the client sector. Concentration is far lower among clients: the top twenty clients account for only 30% of client volume, against 94% for the top twenty dealers. This contrast confirms that the concentration is specific to the dealer core, not a feature of the market as a whole. The effective number of intermediating dealers—the inverse of the activity Herfindahl, $10{,}000/611 \approx 16$—is small, even though primary dealers as a group stand on at least one side of 84% of all volume. This quantifies the visual hierarchy in Figure 2: the Bund market

[10] Throughout the paper, statistics based on protected investor entities are reported only when the underlying cell contains at least five protected entities and the two largest protected entities account for no more than 85% of the reported aggregate. Cells failing these publication restrictions are suppressed or aggregated; exact individual investor statistics are never reported.

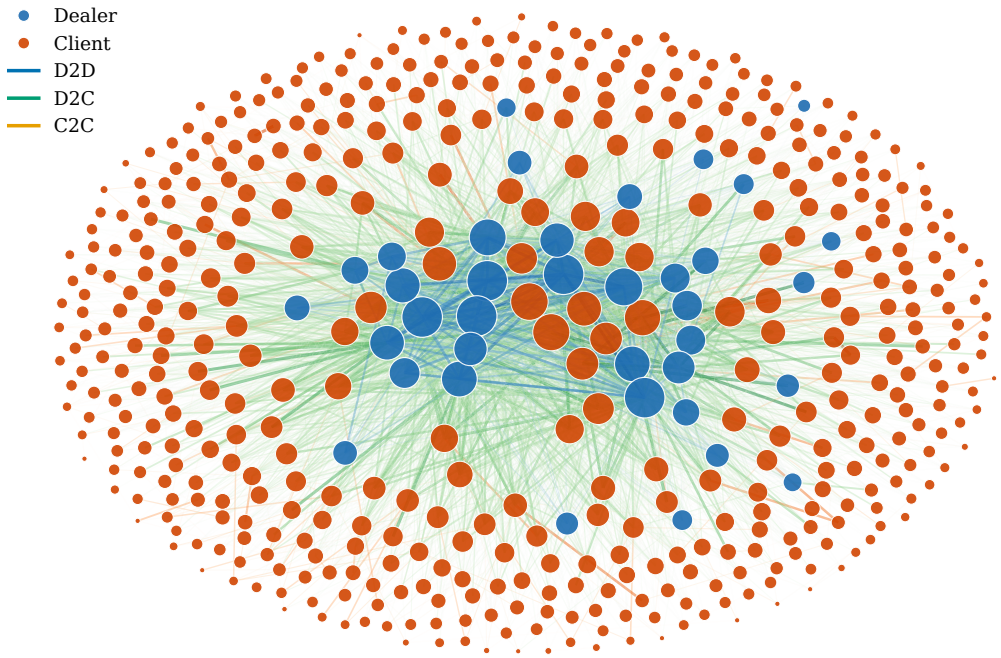


Figure 2: The Bund market network. Undirected graph of investor groups in the MiFID II subsample, restricted to relationships with at least 50 trades per edge. The plotted component is the largest connected component after selecting the top 500 groups by graph degree, with ties broken by weighted degree and group LEI. Node position reflects eigenvector centrality. Node size scales with centrality, with the largest nodes top-coded at a common maximum. Edges scale with the log number of trades between each investor pair. Intragroup trades are omitted. Dealers are investor groups in the primary dealer sector; all other investors are clients.

rests on a thin layer of core intermediaries. The theory interpretation is a competition-versus-inventory tradeoff: a concentrated core can improve inventory offsetting and execution capacity, but it can also concentrate market power and settlement obligations (Li and Schürhoff, 2019; Colliard et al., 2021). The same concentration also sizes the central clearing exercise in Section 6.1: multilateral netting compresses settlement obligations precisely because so few dealers stand on so many trades. Section 4.2 sharpens this picture: the thin layer coincides with the official bidder group itself, and its boundary, internal hierarchy, distribution role, clientele, and stress behavior can be characterized directly.

Sticky dealer–client relationships The periphery connects to the core through narrow, persistent relationships rather than broad search. Figure 4 compares bilateral OTC and MTF trading for active clients.[11] OTC clients usually trade with few dealers and concentrate a large share of volume with a home dealer. Platforms broaden access, but they do not turn the market into all-to-all trading. The persistence diagnostics in Table 1 show that this is not just a small-sample count artifact: home dealer identities are stable across active years, and within-year dealer breadth remains below a trade-count-matched random-matching null. These persistent relationships support the relevance of the relationship value mechanism in the venue

[11] We require at least ten trades per client and venue type: a one-trade client necessarily has one dealer. Without this filter, 86% of OTC clients use one dealer. The filter retains the vast majority of traded volume.

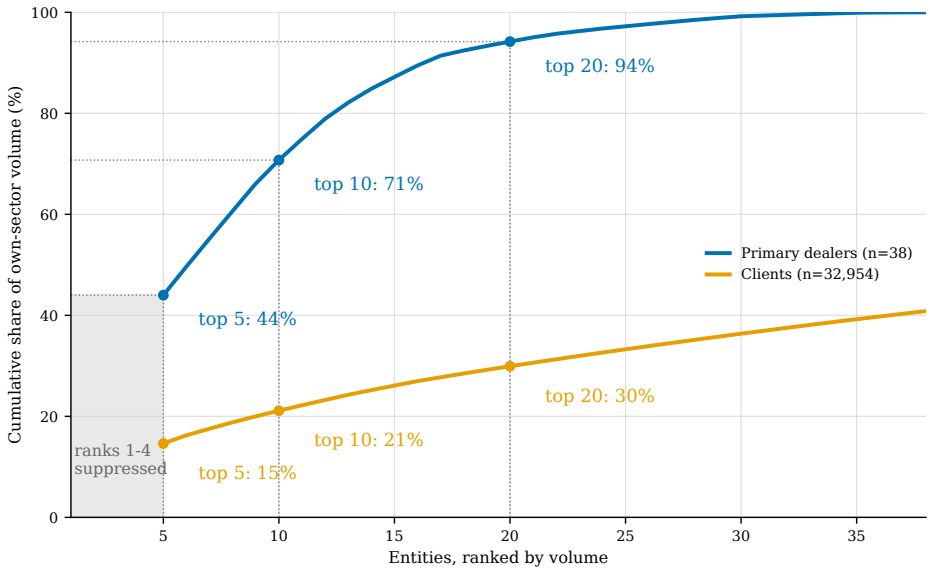


Figure 3: Concentration of dealer intermediation, with the client sector as a benchmark. For each sector, the cumulative share of own-sector volume is plotted against entities ranked from most to least active (blue: primary dealers; orange: clients). A dealer's intermediation volume sums the notional of all trades in which it participates on either side; a client's volume sums its non-dealer legs. Intragroup trades are excluded. The plotted curves start at the top-five aggregate; the shaded left region marks unplotted ranks. The horizontal axis spans the 38 active dealers. Blue markers highlight the top-5, top-10, and top-20 dealer shares; the orange marker the top-20 client share.

centralization analysis of Section 6.2, but do not by themselves identify the relationships' welfare value.

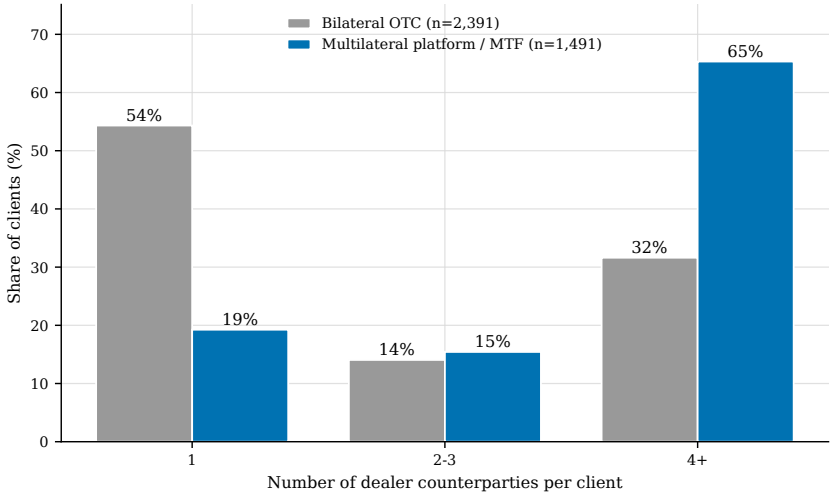


Figure 4: Concentration of dealer–client relationships. Distribution of the number of distinct dealer counterparties per client in dealer-to-client trading, comparing bilateral OTC with MTF execution; bins are 1, 2–3, and 4+ dealers. Computed at the investor-group level on the MiFID II subsample, using all MTF venues and all dealers, and restricted to clients with at least ten trades in the venue.

Bigger clients use more dealers—but far fewer than chance How many dealers does a client use, and does that number grow with trading footprint? Figure 5a answers yes, but

	Bilateral OTC	Platform (MTF)
Distinct dealers per client-year, observed	2.5	6.2
expected, trade-count-matched null	3.2	7.8
ratio (observed / null)	0.78	0.80
Same home dealer in consecutive years (%)	68	42
Active clients ($\geq$ 10 trades)	2391	1491

Table 1: Relationship persistence over time. Active dealer-to-client clients (investor groups with at least ten trades of that venue type), MiFID II. The trade-count-matched null is the expected number of distinct dealers a client would touch in a year if its lifetime trades were randomly assigned across its active years, holding each year's trade count and the client's lifetime dealer mix fixed; a ratio below one indicates within-year concentration beyond the mechanical shorter-window effect. "Same home dealer" is the share of consecutive active-year pairs in which the client's largest-volume dealer is unchanged—an identity measure free of any count dependence.

slowly. Most institutional clients use one dealer; the busiest clients use more. This is the same direction as in the Gilt market evidence of Kondor and Pintér (2022): larger and more active clients are better connected.

The relevant benchmark is not zero. A client with more trades mechanically has more chances to meet more dealers. We therefore compare observed breadth with a random-matching null in which each of the client's trades draws a dealer in proportion to dealer activity. Real clients fall far below that null across the activity distribution. They are not spreading trades across the dealer core; they are adding relationships only gradually. The growth that remains is economically meaningful: among clients with the same trade count, larger tickets are associated with more dealer relationships, which a count-only random model cannot generate.

Dealer breadth is descriptive, not causal. Bund clients are sticky but not fixed: larger footprints buy some additional access while remaining far from random matching. That is the relationship stickiness the platform centralization model needs. Nor is breadth a sophistication index: Figure A15 in Appendix C shows that the cross-sectional gradient of flow informativeness in dealer breadth is mostly sector composition—the share of hedge funds and official sector accounts rises with breadth—and that, once those sectors are excluded, only the most connected clients retain a smaller, significant impact. The Appendix also implements the within-client design of Kondor and Pintér (2022)—time variation in a client's connections as an information proxy—and finds no effect in Bunds in that design either, precisely enough estimated to rule out the Gilt market magnitude. Dealer breadth marks who a client is, not how informed its trading currently is.

Appendix C gives complementary market structure evidence. Bund trading is predominantly cross-border—domestic German counterparties account for only 10% of trade-leg volume—and the Appendix maps where the counterparties that trade Bunds are domiciled. It also reports dealer trading capacity and time-series diagnostics on dealer concentration, Brexit-related venue coverage, retail participation, and algorithmic execution.

(a) Dealer breadth by trading footprint.

(b) Dealer breadth by average ticket size.

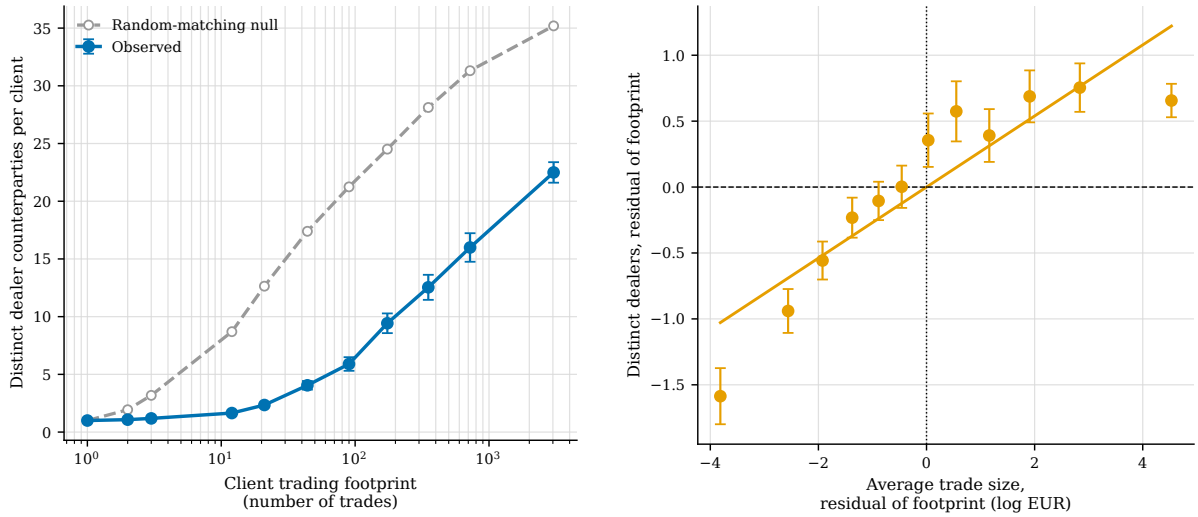


Figure 5: Distinct dealer counterparties per client against trading footprint. The left panel reports the mean number of distinct dealer groups per institutional client by footprint bin (number of dealer-to-client trades, log scale), observed (blue) versus an i.i.d. random-matching null (gray dashed), $\sum_d [1 - (1 - p_d)^N]$, the expected distinct count if each of a client's N trades drew a dealer independently from the dealer-popularity distribution $\{p_d\}$. The right panel reports mean distinct dealers against average trade size, both residualized on footprint-decile fixed effects so that the trade count, and hence the null, is held fixed; the line is the fitted slope. Error bars are 95% confidence intervals. Institutional client groups (a valid legal entity identifier; retail clients and clients in the primary dealer sector excluded), dealer-to-client trades, MiFID II subsample (2018–2025, 13,262 clients).

4.2 The Dealer System

Client identity on every trade, the Finanzagentur's per-auction allotments, and official Bietergruppe rankings let us characterize the dealer system directly: who intermediates, how the hierarchy is organized, and how new supply travels from auctions to investors.

Primary dealers are the intermediaries Defining intermediaries agnostically as groups that buy and sell the same bond within the week, across many bonds and with many distinct counterparties, recovers the Bietergruppe almost exactly (Table 2). Essentially every active member group qualifies; the roughly thirty members handle 78.5% of intermediated client volume after 2020 (71.6–79.6% across screening parameters, stable in every half-year; Figure A1). Primary dealer status is thus a function, not merely a label. By trade count the members' share is 45.6%, so the club's dominance is a large-ticket phenomenon. The residual is not competing intermediation: 84.7% of it is investment funds and hedge funds trading in both directions—the balance sheet absorption that Branger, Muck, and Pütz (2024) document—and the bank-sector two-sided traders outside the club carry 2.0% of intermediated volume. Unlike U.S. Treasuries, where principal trading firms supply a large share of interdealer liquidity (Joint Staff Report, 2015) and proposals for all-to-all trading aim to add intermediation capacity beyond the primary dealers (Duffie, 2020), the Bund market has little intermediation capacity outside the club.

Functional intermediaries	Groups	Share of intermediated volume
Bietergruppe members	32	78.5%
Investment funds & hedge funds	95	18.2%
Banks outside the Bietergruppe	104	2.0%
Other	35	1.3%

Table 2: Functional intermediaries and the Bietergruppe after 2020. Groups qualifying as functional intermediaries under the baseline behavioral screen—same-ISIN round-trips with third parties in at least eight distinct ISIN-weeks and at least ten distinct counterparty groups per half-year—split by Bietergruppe membership and, for non-members, by sector; membership is attached only after the intermediary set is fixed. Group counts are averages per half-year; shares are of intermediated client volume pooled across half-years. On average 33 member groups trade in the screened slice per half-year, of which 32 qualify. Across the six screening configurations the member share ranges from 71.6% to 79.6%. Investment funds and hedge funds are pooled; "Other" pools the remaining client sectors. The screened slice, trades between identified groups with exactly one intermediary side, carries 25.8% of all trading volume outside intragroup bookings, including interdealer trades and anonymous venue, IDB, and CCP legs.

One hierarchy, uncontested Within the club, large dealers are central dealers. Dealers with more secondary market client volume, more trades, more active days, broader client reach, larger auction allotments, and stronger Bietergruppe rank all sit closer to the center of the dealer–client network. A principal component analysis of ranked centrality, secondary market activity, and primary market status measures summarizes this common component: the first component explains 69.7% of their variation.

The more informative question is whether primary market status still matters after holding secondary market size fixed. It does: among dealers with similar secondary market volume, the dealer with the stronger Bietergruppe rank is still more central in the dealer–client network (partial rank correlation 0.28; 0.33 within volume quintiles; Figure 6). Thus the official primary market ranking and the secondary market network ranking are linked, not identical: primary rank contains information about secondary market position that is not explained by secondary market volume alone. The hierarchy is persistent and uncontested in our data: semi-annual changes in rank and centrality are uncorrelated, so dealers do not appear to move up the primary market ranking in the same periods in which they gain secondary market centrality. The over-the-counter intermediation literature documents a stable core-periphery dealer network in which central dealers post lower and less dispersed spreads (Li and Schürhoff, 2019; Hollifield, Neklyudov, and Spatt, 2017; Üslü, 2019) and in which heterogeneous risk-bearing capacity among the core dealers shapes prices (Eisfeldt, Herskovic, Rajan, and Siriwardane, 2023), and the Treasury auction literature documents persistent heterogeneity between primary dealers and other bidders (Hortaçsu, Kastl, and Zhang, 2018) alongside durable dealer-client relationships (Hendershott, Li, Livdan, and Schürhoff, 2020). Those literatures study the primary and secondary markets separately; we show they are one object, and that the resulting hierarchy is uncontested at the frequency our data resolve.

The distribution chain Auction wins show up immediately in client flow. Prior work sees auction cycles in prices or aggregate bidder group flows (Lou, Yan, and Zhang, 2013; Beetsma,

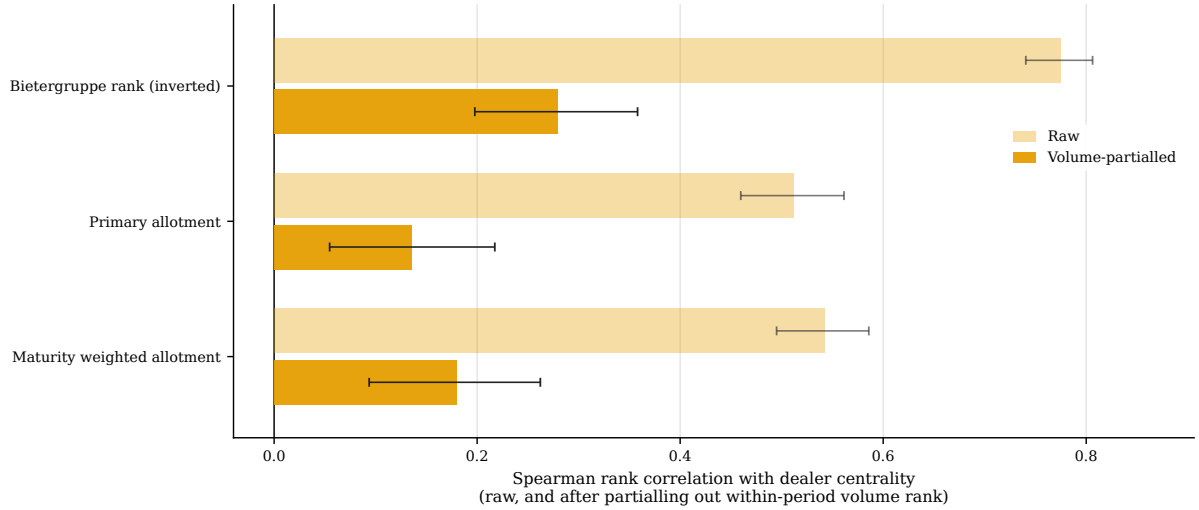


Figure 6: Dealer centrality and primary market status, raw and conditional on scale. Bars are pooled Spearman correlations between half-yearly eigenvector centrality in the dealer–client network and each primary market characteristic, with 95% confidence intervals from resampling half-years. Pale bars are raw correlations; solid bars condition on within-period secondary market volume rank. Bietergruppe rank is inverted, so positive values correspond to stronger official rank.

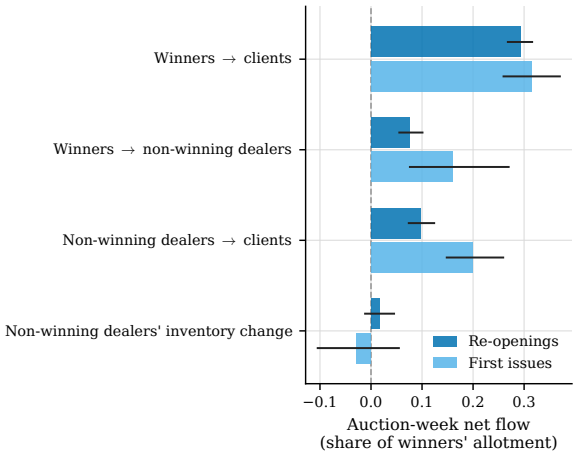
Giuliodori, Hanson, and de Jong, 2018; Green, Hollifield, and Schürhoff, 2007; Branger et al., 2024). Client identity lets us trace the quantity chain dealer by dealer. Winning dealers are the groups with a positive allotment. In the auction week, they sell, net, 29.3% of their allotment to clients at re-openings and 31.5% at first issues (Figure 7a). They pass a further 7.6% and 15.9% to non-winning dealers in the interdealer market. Non-winners sell on to clients within the week. Their net client sales equal 19.9% of the winners' allotment at first issues, against 9.7% at re-openings. Client demand for a new line thus reaches dealers that won none of it, while at a re-opening it sits mostly with the winners. The buyers are mainly investment funds and hedge funds, whose share of winners' auction week sales is 10.6 percentage points above winners' usual client mix (Figure 7b). Distribution is fast. For auctions not followed by another auction of the same bond within twelve weeks, winners' net sales to clients equal 30% of the allotment in the auction week and 45% within twelve weeks. New supply therefore moves through dealers as flow, not as inventory they warehouse.

Two further properties of the dealer system—the geographic division of labor across the hierarchy and the stability of the core's share of client volume in acute stress windows—are documented in Section A.1.

4.3 Fragmentation, Concentration, and Segmentation

Fragmentation Venue structure reinforces the hierarchy; it does not replace it. Figure 8a shows that OTC, MTF, OTF, and SI trading all carry different mixtures of dealer access, venue protocol family (Table A1), execution cost environment, and information environment. OTFs are close to interdealer broker markets. SIs and OTC trading remain centered on dealers: even leaving out bookings between entities of one dealer group, a dealer is party to 97% of SI and 81%

(a) Auction week net flows through winning and non-winning dealers.



(b) Shift in the client sector mix of winners' sales.

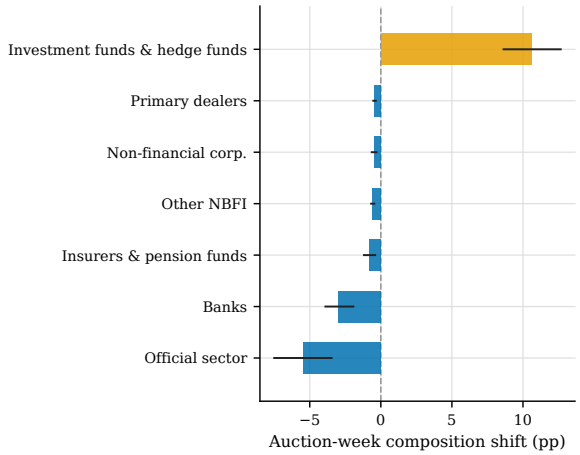


Figure 7: The distribution chain from auction to investors. Panel (a): auction week net flows of the auctioned bond as a share of the winning dealers' allotment, separately for re-openings and first issues: winning dealers' net sales to clients and to non-winning dealers, non-winning dealers' net sales to clients, and non-winning dealers' inventory change, which also includes their net purchases from the Bundesbank or Finanzagentur and from records without an identified counterparty. Winning dealers are the groups with a positive allotment; non-winning dealers are the other auction group members. Panel (b): client sector composition of winners' auction week sales of the auctioned bond relative to their sales of all bonds in the same week. Intervals are 95% auction bootstrap confidence intervals.

of OTC execution volume. MTFs broaden access but do not eliminate dealer intermediation, and regulated markets are peripheral in volume but matter for retail access. Fragmentation is thus economic, not just a legal taxonomy.

Concentration Figure 8b uses venue groups rather than raw MIC codes, because Brexit led operators such as Bloomberg, Tradeweb, and MarketAxess to maintain parallel EU and U.K. MICs for the same trading infrastructure; Appendix D.11 details the grouping. On this grouped basis, every venue type is concentrated. The effective number of venue groups is only 2.4 for OTFs and 2.7 for MTFs, and the top three groups account for 97% and 91% of volume, respectively. SIs and regulated markets are less concentrated, with effective group counts of 9.4 and 6.5, but they are not diffuse.

Trading in Bunds is concentrated in a handful of bonds but fragmented across trading venues. Figure 9 combines the bond concentration and bond-venue sparsity facts. About 62 Bunds are outstanding on a typical day, but the liquid core is a handful of recently issued on-the-run Bunds with benchmark maturities and bonds eligible for Bund futures delivery. In Figure 9a, each on-the-run bond carries 6.5% of daily volume on average, against 2.9% for each other deliverable bond and 0.9% for each remaining bond, so the four on-the-run benchmarks and other deliverable issues together account for 56% of daily volume. Figure 9b summarizes, for each venue type's daily top venue group, how much of the outstanding bond universe actually trades there. OTC and MTF top venues usually cover almost the full outstanding universe (97% and 98% of bonds with at least one trade), but even there the set of bonds with at least five

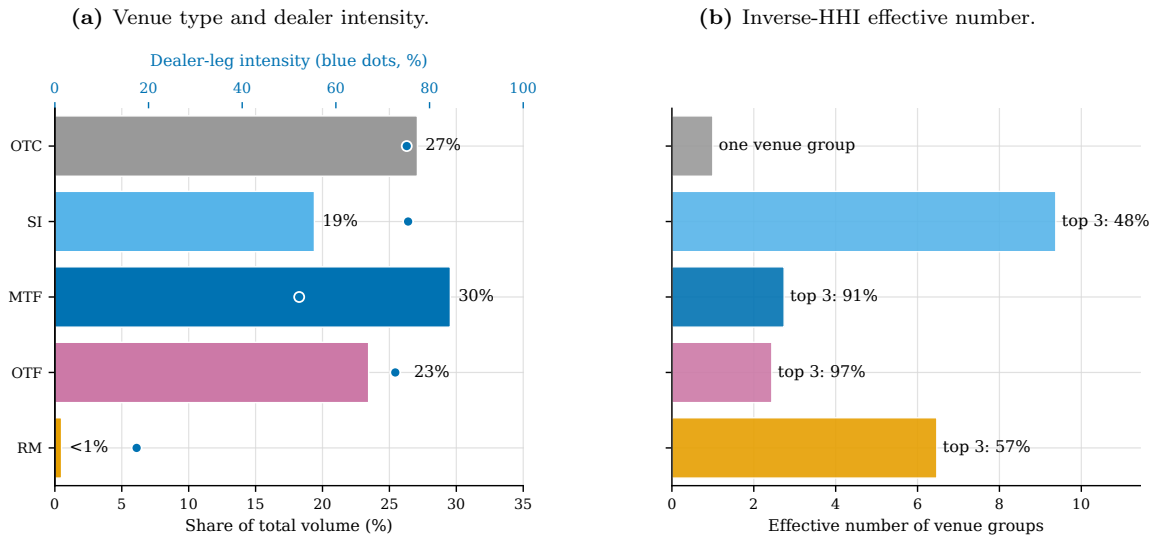


Figure 8: Venue architecture and dealer intermediation. The left panel reports each venue type's share of aggregate Bund trading volume (bars, lower axis) and dealer-leg intensity (blue dots, upper axis): the volume-weighted share of record legs held by investor groups in the primary dealer sector, in percent, so that a client-to-client record scores 0, a dealer-to-client record 50, and a dealer-to-dealer record 100. Volume bars use all positive-notional MiFID II executions with an identified venue type; dealer-intensity dots use the positive-notional trade records with both buyer and seller groups known, including records between two entities of one investor group. The right panel reports the inverse-HHI effective number of venue groups within each venue type, $1/\sum_g s_g^2$, where s_g is venue group g's volume share within the venue type. Venue groups merge MICs of the same type that belong to the same operator; text labels give the combined share of the top three venue groups. Bilateral OTC is treated as one venue group. MiFID II subsample.

same-day trades is smaller (63% and 69%). The other venue types are much sparser: the daily top SI, OTF, and regulated market venues cover only 48%, 34%, and 15% of outstanding bonds at the median. Fragmentation therefore does not mean the same liquid order flow is replicated everywhere; trading opportunities are specific to each venue for much of the outstanding Bund universe.

Figure 10 shows how venue choice depends on trade size. Bilateral OTC trading is widespread across ticket sizes; small trades (up to €100,000) are also executed on-exchange; medium trades (up to €50 million) are frequently executed on electronic platforms; and the largest trades concentrate on interdealer platforms and in bilateral execution with systematic internalizers and over the counter. The trade size patterns across venue types mirror evidence from the interdealer Bund market, where different protocols coexist and brokered OTC trading is particularly relevant for executing large, anonymous trades (de Roure, Moench, Pelizzon, and Schneider, 2026).

Segmentation The Bund market is *segmented*: investor sectors reach the market through systematically different venue types. Figure 11 shows that official sector institutions, investment funds, and ICPF trade mostly on MTFs (85%, 72%, and 70% of their client trades), and hedge funds trade 62% there. Banks split between MTFs and bilateral execution, over the counter or with a systematic internalizer. Retail households are the near mirror image, concentrated on the regulated retail exchanges and bilateral execution. This is about *who* trades, not *which* bond: within the same bond and day, knowing a client's sector removes 37.5% of uncertainty about

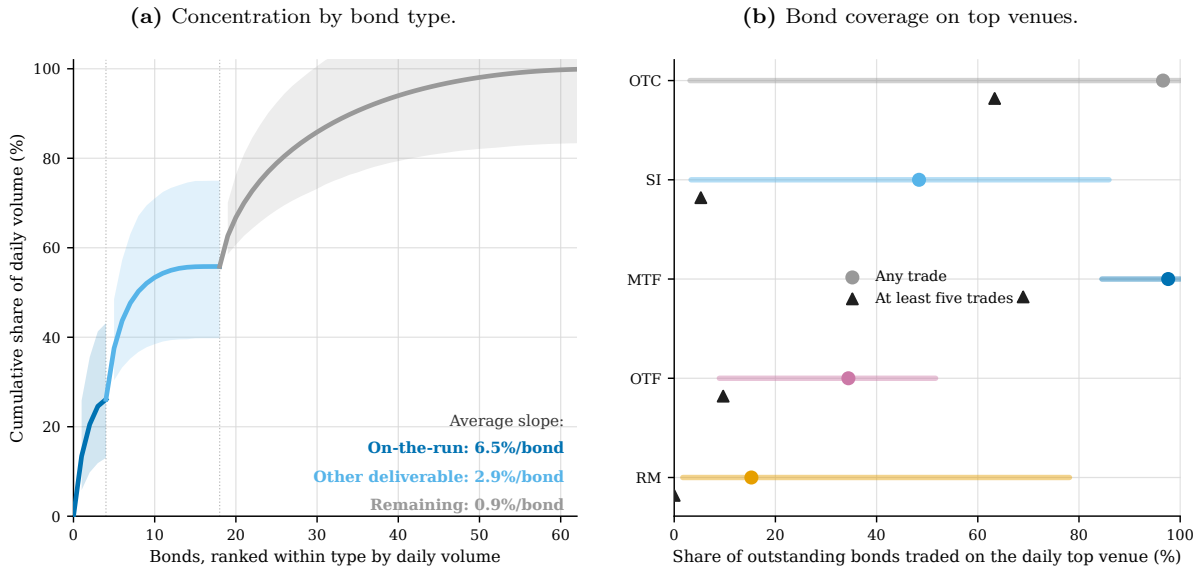


Figure 9: Bond concentration and bond-venue coverage. The left panel splits bonds into three types—on-the-run, other futures-deliverable, and remaining—ranks bonds within each type by daily volume, and concatenates the mean cumulative share of daily trading volume each type contributes; shaded bands are the 5th–95th percentile range across days. The right panel summarizes bond coverage on each venue type's daily top venue group. Circles report the median share of outstanding bonds with at least one trade on that top venue; horizontal bands are the 5th–95th percentile range. Triangles report the median share with at least five trades. Volumes and trade counts are of executions. MiFID II subsample.

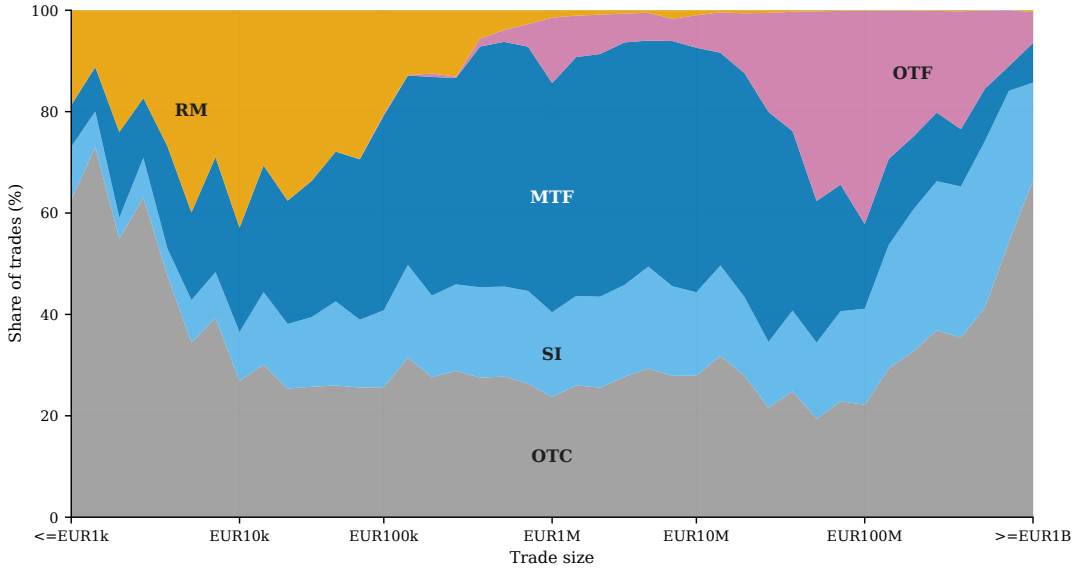


Figure 10: Fragmentation across venues by trade size. Venue composition of executions as a function of execution size.

venue type, against 16.8% under a no-segmentation null randomly reassigning sector labels within the bond–day. The 20.7-point gap lies far outside the null and survives conditioning on trade size, restricting to institutions, excluding the official sector, and the 2021 U.K.-venue coverage break.[12]

We read segmentation descriptively, not causally: we do not identify whether investors select venues for their information environment or are sorted onto them by access and intermediation. Its theoretical implication is that venue shares bundle client identity, dealer access, and the information content of flow, not just technology, consistent with evidence that investor identity interacts strongly with information, trading costs, and protocol choice (Czech et al., 2021; Pintér et al., 2024). A comparison of price impact or execution cost across venues therefore partly compares the sectors that trade on each venue (Section 5).

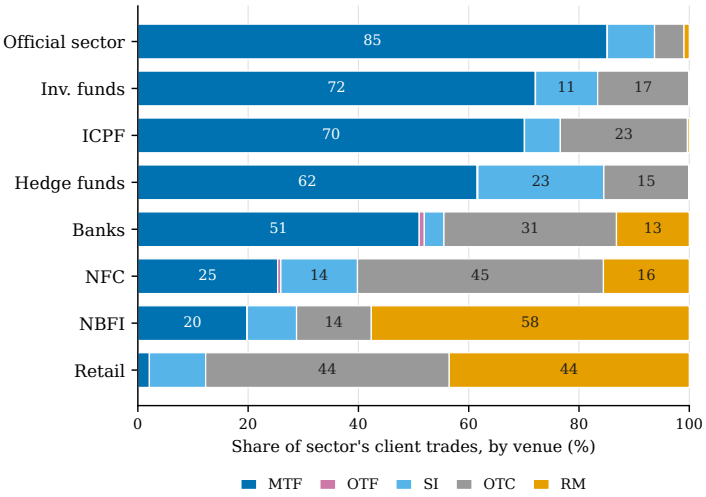


Figure 11: Venue composition by client sector. Venue composition of client legs in each investor sector's dealer-to-client trades (count shares), with sectors ordered by their MTF share. Each bar pools a sector's trades across all operators of a venue type. MiFID II subsample, investor-group sector classification, 3,408,392 client legs of dealer-to-client trades.

These facts identify the market structure: a small dealer core intermediates a segmented periphery through persistent relationships, across fragmented venues.

5 Information and Execution Costs

Market structure matters only if it shapes prices, flows, and trading costs. We therefore turn from who trades with whom and where to who moves prices, who supplies or absorbs stress flow, and who pays execution costs. The information, execution cost, and platform centralization analyses focus on dealer-to-client trades, unlike the market structure and clearing exercises. They require

[12]The statistic is the conditional mutual information $I(\text{sector}; \text{venue} \mid \text{bond–day})$ on client legs of dealer-to-client trades, normalized by the conditional venue entropy $H(\text{venue} \mid \text{bond–day})$, so it is the fraction of within-bond-day venue uncertainty resolved by knowing the sector. The permutation null preserves each bond–day's sector and venue marginals and breaks only their pairing; because finite-sample mutual information is biased upward in sparse cells, the null is strictly positive, and the observed-minus-null gap is the bias-corrected measure of segmentation. The unconditional sector–venue association is Cramér's $V = 0.34$.

client identity and transaction signs from the client's side; we do not separately model interdealer bargaining or dealer-to-dealer price formation.

5.1 Investor Flows and Intermediation

We first identify the buyers and sellers during sharp Bund price moves to assess whether Bunds were being bought as a hedge, sold for cash, or repriced because expected supply had changed. We therefore study three short event windows that isolate distinct shocks. March 2020 is a global cash shock: Bund prices fell as investors raised liquidity during the Covid dash for cash. March 2023 is a safety shock with the opposite price sign: Bund prices rose during banking-sector turmoil. March 2025 is a domestic fiscal news shock: Bund prices fell after news of substantially higher future German borrowing. The three episodes therefore separate the two safe asset roles emphasized by Duffie (2025) from a non-safe-haven supply repricing.

For each episode, we measure sector-level net Bund purchases over the acute window in which prices moved before the subsequent stabilization or reversal. The Covid window, 9–18 March 2020, ends at the ECB's pandemic emergency purchase program announcement. The banking-stress window, 9–17 March 2023, runs from the Silicon Valley Bank shock to the Friday before Credit Suisse's resolution weekend. For the fiscal news episode, the window is 5–11 March 2025, the first five trading days after the fiscal package was announced on the evening of 4 March. These short windows avoid mixing the dislocation with later policy responses, rebounds, or issuance dynamics. That separation is only feasible with high-frequency transaction flows; monthly or quarterly holdings would net the event against what happened after it.

Figure 12 reports results. Because every transaction has a buyer and a seller, the sector bars sum to zero once retail and unclassified reporting legs are included. We show the unclassified bar for accounting closure only: it captures reporting legs without a mapped investor sector, including missing account owner information, internal account reporting, and identifiers that do not map to an investor sector. In March 2020, the German 10-year yield rose 45 bps. The dominant net seller was the official sector, which sold €11.3 bn net; foreign official sector accounts sold €10.0 bn of it, consistent with reserve managers liquidating Bunds to raise dollars (dash for cash). Hedge funds were net buyers, and investment funds were small net sellers.[13] Primary dealers were also substantial net buyers in this cash shock.

In March 2023, the price move had the opposite sign: Bund yields fell 46 bps amid the banking turmoil (flight to safety). The flow signature nevertheless repeated. Foreign official sector accounts again sold, and hedge funds bought. The same foreign official seller therefore appears in a cash shock and a safety shock, but the buyers differ: primary dealers are small net sellers in 2023, and hedge funds are the main identified absorbing sector.

[13] Quarterly holdings of all debt securities show banks and investment funds trading procyclically and insurers and pension funds countercyclically (Timmer, 2018); in this acute window, insurers and pension funds were net sellers of Bunds, as were investment funds. Quarter-end holdings also include any rebound, while our daily Bund flows isolate the dislocation. A net seller took that side on balance, not necessarily in a forced sale, and primary dealer net purchases measure secondary market flow, not balance sheets.

The March 2025 fiscal repricing is different. Friedrich Merz, then Chancellor-designate, announced plans to relax Germany's constitutional debt brake for defense and infrastructure spending. Bund prices fell sharply, with the German 10-year yield rising 41 bps. This was not a global dash for cash, and it was not a flight to safety. It was news about future German debt supply. The flows line up with that interpretation. Primary dealers were the main net sellers. Mapped client sectors were net buyers, especially investment funds, banks, and ICPF. Hedge funds, which bought heavily in the March 2020 and March 2023 stress episodes, bought only modestly. Thus dealers were not absorbing customer sales, as in a liquidity shock. They were reducing Bund exposure and passing supply risk to end investors through secondary market trades. Recall that we exclude primary allocation legs, so these flows show the secondary market adjustment to expected supply. As Appendix A.1 shows, the same windows leave the split of client flow across the dealer hierarchy unchanged: the stress reallocation is between investor sectors, not across dealers.

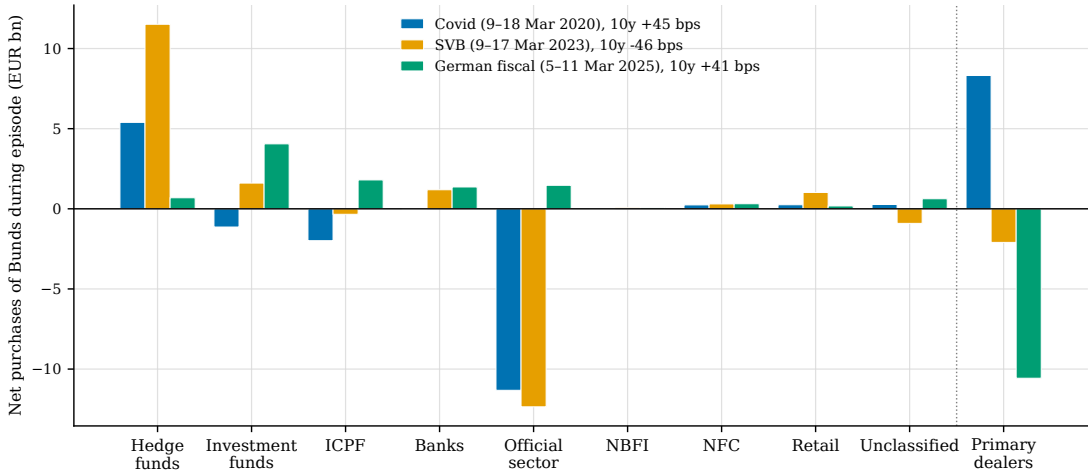


Figure 12: Net Bund purchases by investor sector and reporting category across three stress episodes. Bars report net purchases (trade volume bought minus sold, € bn) over the acute window before intervention in three stress episodes: the Covid dash for cash (9–18 March 2020, Bund yields rising); the March 2023 banking stress around Silicon Valley Bank (9–17 March 2023, Bund yields falling); and the German fiscal reform repricing (5–11 March 2025, Bund yields rising). Negative bars denote net sellers. Legend entries report the change in the German 10-year yield from the last close before each window to the close of its last day. Retail bars collect natural person legs; unclassified bars collect legs without a mapped investor sector. Primary market allocation trades to primary dealers are excluded by construction. The 2020 window predates the United Kingdom's departure from the MiFIR reporting perimeter on 1 January 2021: 60% of its volume is reported on U.K. venues, against 11% and 11% in the 2023 and 2025 windows, so cross-episode magnitudes are not directly comparable.

5.2 Who is the informed flow?

A central question in market microstructure is *whose* flow has permanent price content. Because the Bund data identify client sectors, we can rank sectors directly, whereas equity and Treasury data cannot (Brandt and Kavajecz, 2004; Hendershott, Livdan, and Schürhoff, 2015). For each dealer-to-client trade we sign flow from the client's side, aggregate to bond–day–sector net flow, and ask whether a sector's net buying predicts a subsequent decline in the bond's curve-relative

pricing error. ISIN fixed effects absorb the bond's average curve-relative pricing error, and maturity bucket×day fixed effects (remaining maturity under 1, 1–3, 3–7, 7–15, and over 15 years) absorb yield and fitted-curve moves common to bonds of similar maturity on a day, so the coefficient is a within-segment security selection effect. The regressions also control for the bond's cheapness before the flow day: its curve-relative pricing error on the previous trading day and the two preceding daily changes in that error. Investors tend to buy bonds that have just cheapened, and curve-relative cheapness mean-reverts, so these controls keep predictable reversion from being credited to the flow (Appendix A.3). A permanent response indicates price-relevant flow; a response that reverts indicates liquidity demand (Evans and Lyons, 2008).[14]

In Figure 13, hedge funds and the official sector stand out: their net purchases predict the most persistent richening, and their estimates are the most precise. Hedge fund net purchases predict permanent richening of 0.03 bps per standard deviation of net flow at the ten-day horizon ($t = 7.6$). Net purchases by the official sector predict the largest persistent richening, but we read this as policy and reserve management flow rather than private information of the kind associated with hedge funds. Official accounts tend to buy bonds that cheapened on the previous trading day: a one standard deviation net purchase follows a cheapening of 0.011 bps ($t = 2.9$). Part of the official impact estimated without the cheapness controls is therefore reversion of that cheapening (Appendix A.3), and such mechanical value buying fits the policy and reserve management reading. Investment fund flow carries an impact that peaks at three days and fades: 0.01 bps at ten days ($t = 1.9$) and 0.00 bps at twenty days ($t = 0.2$). The impact is transitory, not permanent, the signature of liquidity demand. Bank flow carries a short-lived impact: 0.03 bps at five days ($t = 2.0$), 0.00 bps at ten days ($t = -0.1$), and an estimate not distinguishable from zero at twenty days ($t = 0.0$). Unlike the hedge fund and official sector estimates, it does not carry over to future implied yield changes (Appendix A.3). ICPF, NBFI, NFC, and retail household flow carry no significant ten-day signal; the retail estimate is 0.01 bps ($t = 0.5$) and is not significant at any horizon from one to twenty days. The ranking is the contribution. The magnitudes are small, measured relative to the kernel ridge curve, and not a tradable strategy net of costs.

These results support the presence of price-relevant sector flow but do not identify whether the information is private research, inventory-revealing flow, policy demand, or dealer learning. The hedge fund result survives several checks: it holds when the March 2020 stress quarter is excluded, has the same sign in every year from 2018 to 2025 and is significant in most of them, survives replacing the curve-relative pricing error with future synchronized implied yield changes (Section A.3), and NFC flow, a placebo sector, carries no significant ten-day impact. The official sector estimate survives the same replacement. The official sector effect is concentrated among foreign reserve managers and is not a Bundesbank/ECB/Finanzagentur artifact.

This sector ranking sets up the pricing test. A simple adverse selection account would charge the highest effective half-spreads to the sectors whose flow has the most permanent price content. The Bund market does not.

[14] Appendix A.2 repeats this exercise by venue type. Because a venue's net flow pools the sectors that trade there, those estimates largely reflect which sectors trade where.

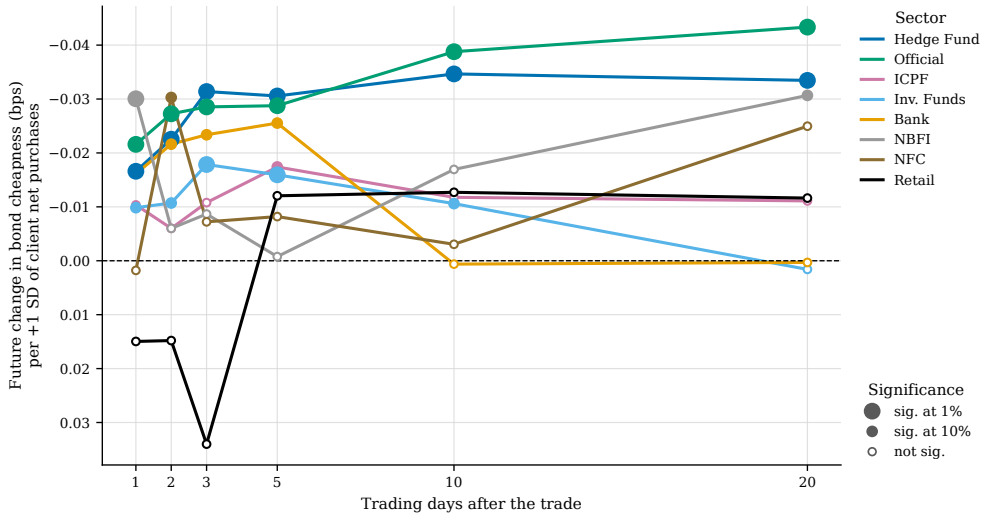


Figure 13: Permanent price impact by client sector. For each identified client sector, the cumulative future change in the curve-relative pricing error, defined as synchronized implied yield minus the kernel ridge fair yield, in bps, per net purchase of the bond equal to one standard deviation of the sector's absolute net purchases, by trading days after the trade. Sector-by-horizon regressions with ISIN and maturity bucket×day fixed effects and controls for the bond's lagged curve-relative pricing error and its two lagged daily changes; the y-axis is inverted so that upward denotes richening. Marker size encodes significance (ISIN-clustered standard errors): large filled = 1%, medium filled = 10%, small hollow = not significant. Dealer-to-client trades, MiFID II subsample (2018–2025).

5.3 Who pays higher effective half-spreads—and is it adverse selection?

The informed flow ranking has a natural counterpart: the price each sector pays for immediate execution.[15] For every dealer-to-client trade we measure the *effective half-spread*—the concession of the execution price relative to the pre-trade MTS top-of-book mid, signed from the client's side and reported in price basis points (Bessembinder et al., 2020; Pintér et al., 2024).[16] Classic adverse selection logic predicts this spread should be highest for the clients whose flow carries the most permanent price content.

Figure 14a shows otherwise: effective half-spreads vary sharply across client sectors. Retail households pay 2.48 bps, roughly 4 times the 0.65 bps paid by banks. The gap *widens* to +3.71 bps after absorbing ISIN×day, venue, and trade size fixed effects. Among institutional sectors the spread is not monotonically increasing in informedness, measured by the ten-day permanent price impact of Figure 13 (Figure 14b): hedge funds and official sector accounts—whose net flow has the most permanent price content—pay only middling spreads (0.70 and 0.65 bps), below ICPF and investment funds, whose permanent price impact is no larger. Retail households, who pay the most, carry no detectable permanent price impact (Figure 14b). Net of venue, size, and bond–day fixed effects, hedge fund flow trades at a spread below banks' (a

[15] We treat the client as the demander of immediate execution. The sign convention provides a check: a client sector that systematically supplied liquidity would show a negative average half-spread, whereas measured half-spreads are positive across sectors. The rules of the regulated retail exchanges make them quote-driven, with designated liquidity providers supplying continuous quotes against which retail orders execute.

[16] The reference mid is the last MTS top-of-book quote within 60 minutes before the trade; trades without a quote in that window are dropped, so the spread is never measured against a stale quote (median quote age is a few seconds; see Section 3.5).

−0.30 bps difference, $t = 5.2$). In Bunds, the effective half-spread is an execution cost that does not rise monotonically with permanent price content, not a price for adverse selection.

This is a statement about investor cost and the split of surplus, not about welfare: the effective half-spread nets no inventory, hedging, or matching cost, so it is neither a dealer rent nor, absent a model of matching, an efficiency loss. The two firm findings are the retail-versus-institutional cost gap and the absence of any extra charge on the high permanent impact flow; the raw rank correlation between cost and information across the eight client sectors is weak ($\rho = -0.07$) and far from significant. The comparison is across sectors, not a trade-by-trade test. The official sector is read as policy and reserve manager footprint. The sample is the MiFID II period (2018–2025, 3,313,054 dealer-to-client trades).

(a) Effective half-spread by sector. **(b)** Half-spread and permanent price impact.

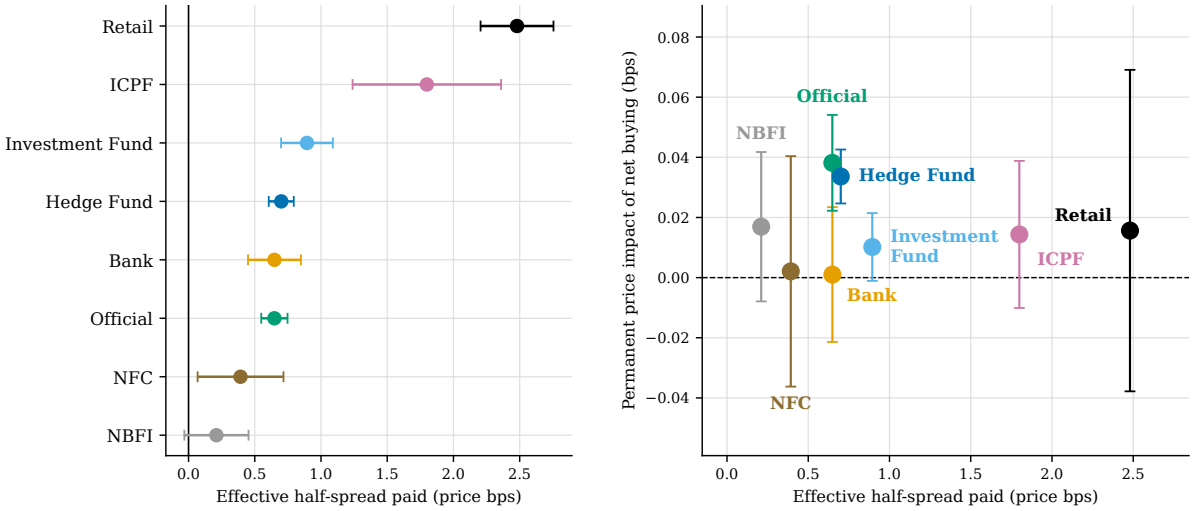


Figure 14: Effective half-spreads and permanent price impact by client sector. The left panel reports the volume-weighted effective half-spread paid by each client sector—the concession of the execution price relative to the pre-trade MTS top-of-book mid, signed from the client's side and reported in price bps; error bars are 95% ISIN-clustered confidence intervals. The right panel plots each sector's half-spread (x) against its ten-day price impact (y), the ten-day estimate of Figure 13 re-estimated on the trades with a pre-trade MTS quote; positive values denote richening. Vertical bars are 95% ISIN-clustered confidence intervals; half-spread intervals are in the left panel. Dealer-to-client trades, MiFID II subsample (2018–2025).

5.4 The price of size in Bunds

How does the cost of execution vary with trade size? Pintér et al. (2024) separate two offsetting forces in non-anonymous bond data: across clients, larger trades receive a pooled *size discount*, while within a client, larger trades pay a *size penalty*—and they document the pooled discount even in a sovereign market (Gilts). When we measure the client's cost the same way (the signed concession of the execution price to the pre-trade MTS mid) and use the same pooled-versus-within-client contrast, Bunds *confirm the size penalty but, unlike Gilts, exhibit no size discount.* Figure 15 shows effective half-spreads *rising* monotonically with trade size: net of ISIN×day, sector, and venue fixed effects, blocks above €25 million pay an extra 1.36 bps. The coefficient on log notional is 0.138 bps ($t = 13.7$), equivalent to 0.096 bps for a doubling of trade size.

The penalty is genuinely within-investor, and the absence of a discount is not a control artifact. The *raw* cross-client relation—the variation in which Pintér et al. (2024) recover a discount—is essentially flat (−0.002 bps for a doubling of trade size): the pooled-discount force is too weak to turn the net relation into a discount. Absorbing client-*group* (individual investor) fixed effects isolates one investor trading different sizes, the cleanest test of the heterogeneity concern of Pintér et al. (2024). The coefficient stays positive (0.143 bps on log notional, $t = 16.7$), equivalent to 0.099 bps for a doubling of trade size.

That Bunds preserve the within-client penalty but show *no* pooled discount is itself notable. A size discount is one of the more established regularities of over-the-counter trading, usually rationalized by bargaining models in which dealer market power falls with trade size. Its absence here is intuitive for a market that is fragmented yet extremely liquid: dealers are not competed into granting large Bund orders a relationship discount, so the inventory and price impact cost of warehousing a block is passed back to the client. As above, a size penalty is the execution cost of a block, not evidence of mispricing or dealer rents. The sample is the MiFID II period (2018–2025) and the 3,313,054 dealer-to-client trades of Figure 14.[17]

(a) Effective half-spread by trade size.

(b) Size slope by specification and sector.

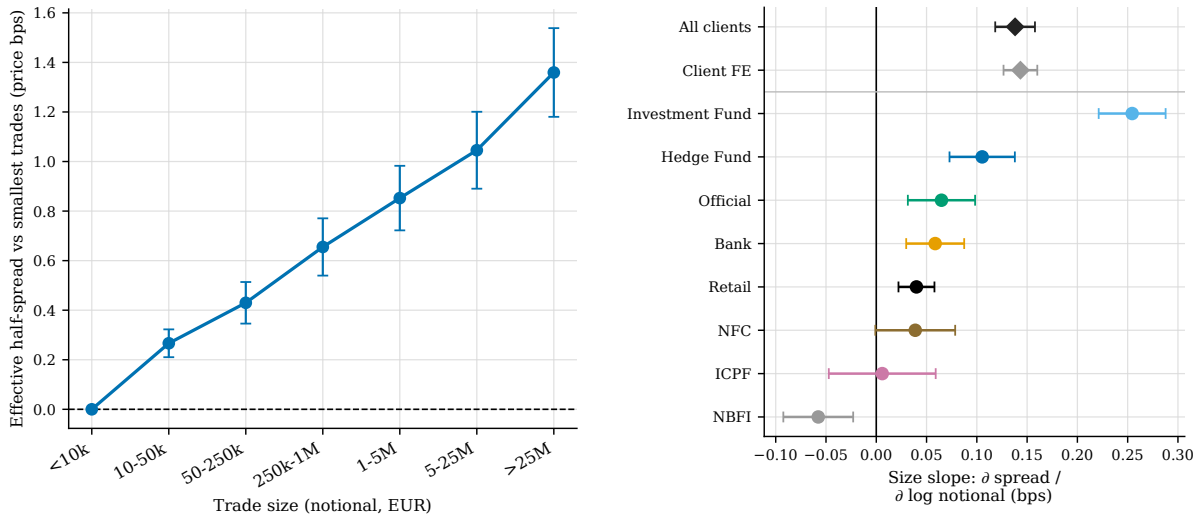


Figure 15: Effective half-spreads by trade size. The left panel reports the effective half-spread paid relative to the smallest trades, by trade size bucket, estimated from size-bucket dummies absorbing ISIN×day, client sector and venue fixed effects (so the profile is composition-free). The right panel reports the all-client size slope with sector and venue fixed effects, the all-client slope with client-group and venue fixed effects, and sector-specific slopes estimated with ISIN×day and venue fixed effects; positive means larger trades pay more. Regressions are unweighted. In both panels the error bars are 95% ISIN-clustered confidence intervals. Dealer-to-client trades, MiFID II subsample (2018–2025).

[17] These execution costs are, in turn, essentially all of the within-bond *price* dispersion: Appendix A.4 shows that a trade's venue, dealer–client segment, and client sector explain no more of the within-bond-day synchronized-yield dispersion than a same-cardinality random partition and do not predict it out of sample, so market structure disperses the cost of trading, not the level of prices.

6 Reform Counterfactuals

We use the market structure facts to evaluate two reforms targeting the dealer core at different stages of the trade. Mandatory central clearing is a post-trade settlement counterfactual: dealer concentration and fragmented settlement determine how much netting at a clearing house can reduce dealers' net settlement obligations. Platform centralization is a pre-trade matching counterfactual: platform access, persistent relationships, and dealer competition determine whether broader access raises gains from trade once relationship surplus may be lost. We therefore evaluate clearing as mechanical netting and centralization as equilibrium matching.

6.1 Mandatory Central Clearing

Two facts matter for settlement: a small dealer core stands on most trades, and those trades settle across fragmented venues and counterparties rather than a common netting set. Central clearing has been widely proposed to reduce "daisy-chain" settlement failures and counterparty risk and to enable more all-to-all trading (see, for example, Duffie, 2025; Chaboud et al., 2025); here we isolate its mechanical netting effect. It novates trades to a CCP and nets obligations multilaterally. Holding the observed trade set fixed, we ask how much dealers' net settlement obligations would fall if mandatorily cleared Bund trades entered a CCP. The exercise follows the per-dealer netting of Fleming and Keane (2021) for U.S. Treasuries and Baranova et al. (2023) for U.K. Gilts; the three markets differ in their status quo, which we account for below.

We estimate these effects at the settlement-date level. The sample includes all trade records with at least one leg in the primary dealer sector and excludes intragroup records. Each trade is converted into signed dealer settlement positions at $T+2$: dealer-to-client trades generate one dealer position and dealer-to-dealer trades generate one position for each dealer. Each dealer leg therefore counts as one settlement obligation. Gross settlement obligations are the sum of absolute nominal obligations across dealer legs settling on date s, $\text{GSO}_s = \sum_{\ell \in s} |\text{Notional}_\ell|$. Status quo net settlement obligations net only trades that are already centrally cleared—records that name a CCP and records on three venues we treat as centrally cleared—each dealer against the CCP; all other obligations settle gross. In the mandatory-clearing counterfactual, each dealer nets its own signed positions by ISIN and settlement date against the CCP, as in Fleming and Keane (2021, p. 13) and Baranova et al. (2023, equation 5). We report a pre-Covid baseline, the settlement dates of January 2 to February 25 and April 1 to 30, 2020, which follows the non-crisis window of Fleming and Keane (2021), and the settlement days of the 15 February–20 March 2020 stress window. The pre-Covid estimate is representative of the broader sample: the median share of gross obligations that own-book netting saves is about 34% on settlement days before 2021 and about 33% from 2021.

Figure 16 mirrors this accounting. Bilateral settlement leaves dealers funding gross obligations across many counterparties. Central clearing novates trades to the CCP, and each dealer's purchases then offset its own sales. A clearing house settles each member's net separately and keeps a matched book, so one dealer's net purchase does not cancel another dealer's net sale:

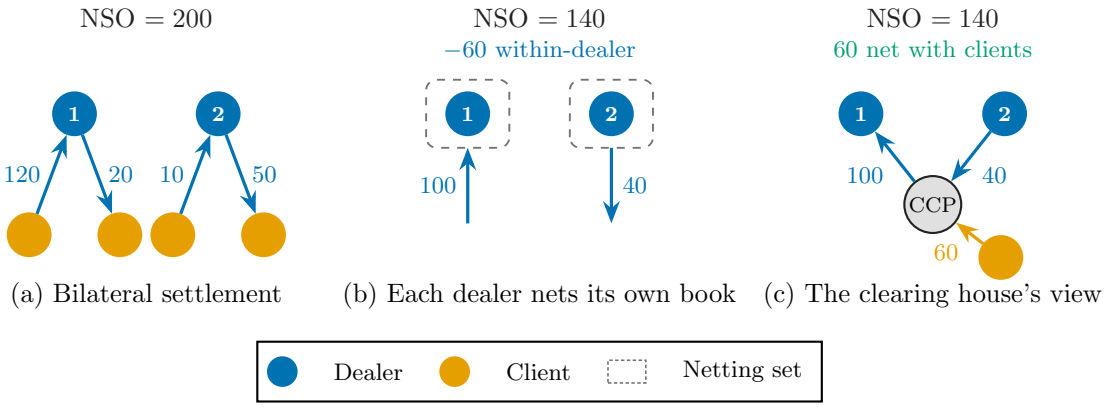


Figure 16: Central clearing and net settlement obligations (NSO) with two dealers. Arrows show deliveries of the security, and numbers are notional amounts. Dealer 1 buys 120 and sells 20; Dealer 2 buys 10 and sells 50. Panel (a) settles every dealer leg bilaterally. Panel (b) nets each dealer's purchases against its own sales. Panel (c) shows the clearing house's view: Dealer 2 delivers 40 to the CCP and Dealer 1 receives 100, so the dealers still owe 140. The 60 that remains after Dealer 2's net sale offsets part of Dealer 1's net purchase is the dealers' net trade with clients, which the CCP passes through on its matched book.

both dealers still settle with the CCP. Own-book netting at a CCP would lower dealers' pre-Covid net settlement obligations from €17.24 bn to €11.13 bn per settlement day. This cuts 35.4% of the status quo and 35.5% of gross obligations (Figure 17b). The saving of €6.10 bn per settlement day equals 0.57% of the €1.06 tn outstanding in the sample's bonds. Settlement instructions fall from 2,828 to 658 a day, a cut of 77%.

Fleming and Keane (2021) and Baranova et al. (2023) also net each dealer's own purchases against its own sales and add up what each dealer still owes the CCP. Their status quos differ from ours, so a reduction against the status quo is not comparable across markets: the Bund status quo nets 0.14% of gross obligations, against 32% in Treasuries, where interdealer trades are already cleared (Fleming and Keane, 2021, p. 11–12), and about 19% in Gilts, where dealers net bilaterally (Baranova et al., 2023, Table 5). Gross obligations are the common base. Against gross, clearing removes about 74% of dealer obligations in Treasuries from January to April 2020 (Fleming and Keane, 2021, p. 11 and Table 1), about 59% in Gilts before the March 2020 stress (Baranova et al., 2023, Table 5), and 35.5% in Bunds before Covid.[18] Bunds therefore benefit less from central clearing than either market. Against their own status quos, the printed reductions are 60% and 67% for Treasuries (Fleming and Keane, 2021, p. 13–14 and Table 1) and 48% and 53% for Gilts (Baranova et al., 2023, Table 5). Interdealer legs are dealer legs whose other leg is another dealer group or has no account owner, as on anonymous broker and venue trades; their share of dealer gross obligations is similar in Bunds (47%) and Treasuries (51%). But Bund interdealer legs net only 24% within dealer, against 63% (Fleming and Keane, 2021, p. 11 and p. 15), and 25% of Bund dealer gross obligations falls on dealer-bond-days with trades in one direction only, which cannot net. Across the settlement days of the 15 February–20 March 2020 stress window, the median own-book reduction is 35.8% of gross obligations.

[18] These figures count every reported leg, including pass-through and agency legs. Netting each dealer's positions within one execution before netting across its trades gives a reduction of 32.3% of gross obligations; netting per dealer legal entity instead of per banking group gives 32.4%; both corrections together give 29.2%.

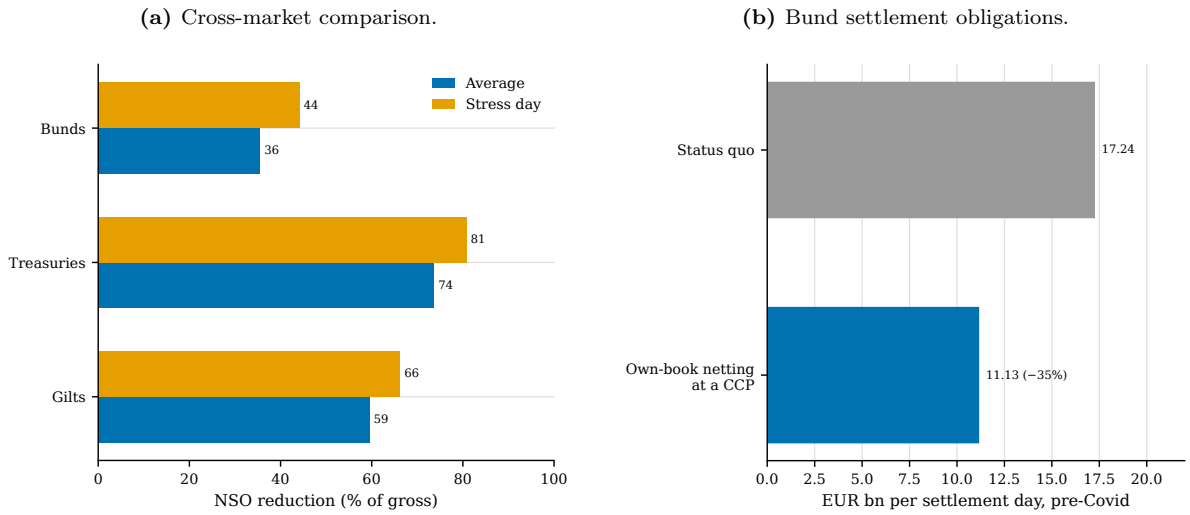


Figure 17: Mandatory central clearing and dealer settlement obligations. The left panel reports reductions in net settlement obligations under mandatory central clearing, with each dealer netting its own trades, as a percentage of gross settlement obligations. Treasury and Gilt bars are computed from the levels on p. 11 and in Table 1 of Fleming and Keane (2021) and in Table 5 of Baranova et al. (2023); Bund bars are based on our transaction data. Average bars: the Treasury window runs from January 2 to April 30, 2020, the only window with a gross figure; the Gilt window runs from January 2 to February 25, 2020; the Bund window runs from January 2 to February 25 and April 1 to 30, 2020. Stress-day bars: 28 February 2020 for Treasuries and 10 March 2020 for Gilts, the papers' peak trading days; 12 March 2020 for Bunds, the settlement date of the trades of 10 March, which is also the largest Bund reduction in the 15 February–20 March 2020 stress window. The right panel reports pre-Covid Bund obligations per settlement day: the status quo and own-book netting at a CCP.

Why would dealers care about net settlement obligations? For cash bonds, the balance sheet channel is weak. Settlement is delivery versus payment over a short window, so the counterparty exposure in an unsettled cash trade is small (Baranova et al., 2023, p. 16). Unsettled cash trades entered the leverage exposure measure only for dealers on trade-date accounting, and the measure now offsets cash receivables against cash payables whenever both trades settle delivery versus payment: under Basel III since January 2023 (Baranova et al., 2023, p. 10, 14, 33) and for EU banks since June 2021 (European Parliament and Council of the European Union, 2019, Article 429g(2)). Clearing such cash trades therefore no longer changes the measure. The material balance sheet gain that Baranova et al. (2023) find is in repo, which our data do not contain. What remains for cash Bunds is operational. Each settlement instruction is a delivery that can fail, and a failed delivery can propagate along a chain of dependent trades; in Treasuries, most fails in specific issues pair off and would disappear under full clearing (Fleming and Keane, 2021, p. 25–26). In our data, settlement dates are computed, not reported, and settlement fails are not observed, so we measure the instruction count, not fails.

Does clearing help most in stress? The share of gross obligations saved is about the same in stress as on other days: on the settlement dates of the trades of 9–18 March 2020, the dash for cash, the median own-book saving is 36.3% of gross obligations, against a median of 34.0% on other settlement days before 2021, and before 2021 the share is uncorrelated with the volatility of the 10-year Bund yield (correlation of −0.031). The euro saving scales with trading volume, so the resilience benefit of clearing is the size of the saving on high-volume days

and the simpler settlement, not a countercyclical release of capacity. On those dash-for-cash settlement dates, own-book netting would have removed €8.8 bn of obligations per day, against the cumulative €8.3 bn of Bunds that primary dealers absorbed from clients over the eight trading days (Figure 12). Clearing nets the churn around that absorption, never the absorption itself.

	One CCP	Venue buckets	MIC silos
Status-quo NSO, pre-Covid (EUR bn)	17.24	17.24	17.24
Own-book NSO after clearing, pre-Covid (EUR bn)	11.13	13.49	14.38
Pre-Covid NSO reduction (%)	35.4	21.7	16.6
Share of one-CCP reduction retained (%)	100.0	61.4	46.8
Largest reduction, 15 Feb–20 Mar 2020 (%)	44.1	28.5	22.2
Settlement date of the largest reduction	2020-03-12	2020-02-21	2020-02-24
Largest reduction as share of one-CCP largest (%)	100.0	64.7	50.4

Table 3: Mandatory central clearing with one CCP and with restricted netting sets. In every column, each dealer group nets its own positions by settlement date and ISIN against the CCP. The venue-bucket sensitivity nets only within three buckets: all MTFs, bilateral OTC and SI trades, and one bucket for all other venue types. The MIC-silo sensitivity nets only within observed MIC venue groups, falling back to MIC and venue type when the grouped venue field is missing. The lower panel reports the largest reduction across the settlement dates of 15 February to 20 March 2020.

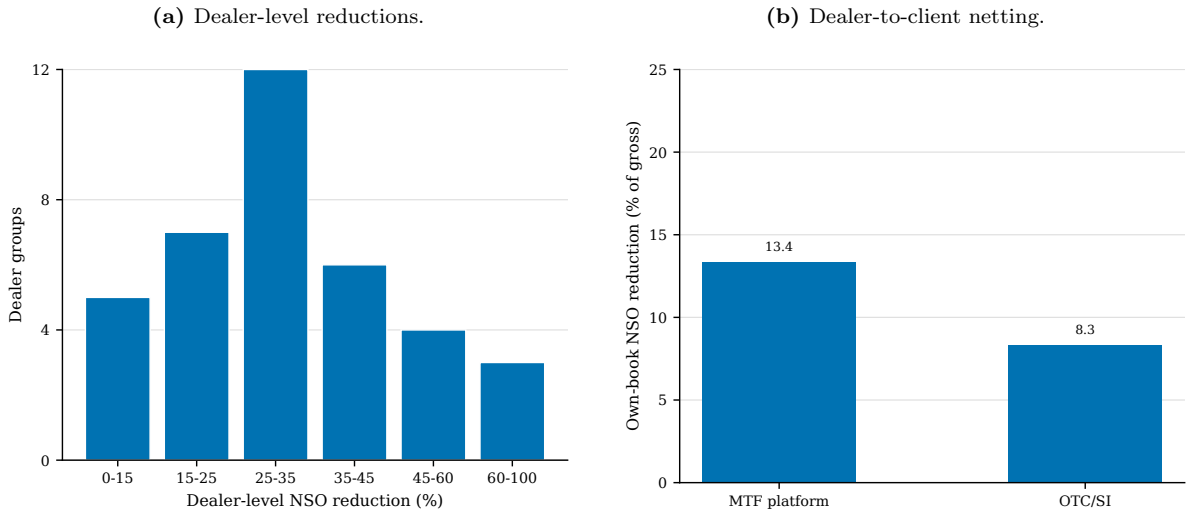


Figure 18: Dealer-level heterogeneity and dealer-to-client netting. The left panel reports the binned distribution of pre-Covid dealer-group NSO reductions under mandatory clearing. Bins are 0–15, 15–25, 25–35, 35–45, 45–60, and 60–100 percentage points. The right panel reports own-book reductions in net settlement obligations, as a percentage of gross obligations, for institutional dealer-to-client trades in the platform-centralization model sample, separately for MTF platform and bilateral OTC/SI trades.

One CCP for all Bund trades is an upper bound, because actual implementation could leave clearing fragmented across venue buckets or MIC-based venue groups. Table 3 therefore reports restricted netting sensitivities, not observed routing models or political transition paths: each dealer nets its own positions only within a venue bucket or a venue/MIC silo. The pre-Covid reduction in net settlement obligations declines from 35.4% with one CCP to 21.7% when netting is confined to three broad venue buckets (MTFs, bilateral OTC and SI, and all other

venue types), and to 16.6% under the conservative venue/MIC silo proxy. Netting gains are also uneven across dealers. Figure 18a shows five dealer groups with reductions below 15% and three with reductions above 60%; the dealer group median is 34%. Figure 18b gives a related but narrower diagnostic. Within institutional dealer-to-client trades used in the structural model of Section 6.2, MTF platform positions are more internally offset than bilateral OTC/SI positions: own-book netting removes 13.4% of gross obligations on MTF platforms, compared with 8.3% on bilateral OTC/SI trades. We read this as a sample diagnostic, not as causal evidence that rerouting a given bilateral trade to an MTF would lower settlement obligations. The point is institutional: clearing and platform centralization are distinct reforms, but the realized clearing gain depends on how broad the netting set is.

Interpretation These estimates are mechanical netting gains, not a full cost–benefit analysis of a clearing mandate. They exclude CCP fees, initial margin and liquidity costs, changes in trading flexibility, and private bilateral netting agreements that may already economize on settlement outside observed central clearing. Nor do they show that clearing would relieve a binding constraint on dealers. Implementation is also more complex in the euro area than in the United States because sovereign issuance, venues, clearing arrangements, and supervision are fragmented across jurisdictions. Consistent with this institutional constraint, European Systemic Risk Board (2024) recommends an incentives-based approach that harmonizes risk management between bilateral and CCP clearing, reduces prudential disincentives for NBFIs, broadens CCP access models, and strengthens CCP services. Our estimates therefore quantify the settlement efficiency gains available in the data before costs, while institutional design determines how much of those gains can be realized.

6.2 Venue Centralization

The centralization exercise uses the relationship and venue facts: platforms broaden dealer access and lower investors' effective half-spreads, but clients retain persistent dealer relationships that may carry non-price value. Platform centralization is therefore different from central clearing. Clearing changes settlement obligations after trades occur. Centralizing trading changes the matching problem before trades occur: investors may reach more dealers, dealers may compete more aggressively, and bilateral relationship value may be lost. Those are welfare questions, not mechanical netting questions.

We evaluate this reform by applying the dealer–investor framework of Allen and Wittwer (2023) to Bund dealer-to-client trades. The earlier facts set the mapping: bilateral OTC and SI trades are relationship-based; MTF trades broaden dealer access; clients have persistent home dealers; and platform use is already high among institutional investors. The counterfactual asks whether forcing more trading onto platforms improves matching enough to offset lost relationship surplus and other costs of restricting bilateral choice.

Sample and setup We use dealer-to-client transactions from 2023–2025 so that the estimation reflects the current venue structure. We collapse venues into two model categories: bilateral trading, which includes traditional OTC and SI activity, and platform trading, which includes MTF execution on venues such as Bloomberg and Tradeweb. This collapse follows the market structure evidence: bilateral trading is relationship-based, while multilateral platforms broaden dealer access and introduce competition among dealers (see Figure 4). It is a binary model mapping, not a claim that SI and traditional OTC execution are economically identical; both enter the bilateral outside option because they preserve dealer-specific access relative to multilateral MTF trading. Following Allen and Wittwer (2023), we exclude in-house trades between dealer groups and their own affiliates, rank group-level continuously active Bietergruppe dealers by dealer-to-client volume, and use the top 20 dealers with equal dealer weights as the preferred specification. Following the model, an investor is classified as institutional if it trades on a platform at least once during the estimation window and as retail otherwise. Table A2 documents this mapping: platforms account for about two thirds of all dealer-to-client trades and 70% among institutional clients, while SI trades are small by trade count but substantial by volume and are grouped with bilateral trading.

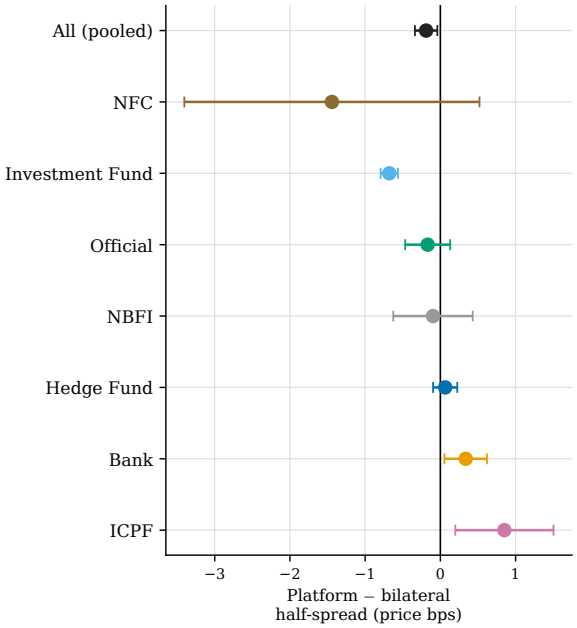
Despite high institutional platform adoption, bilateral and platform trading coexist. The model therefore has to allow platform access to improve dealer competition while preserving non-price relationship values such as account management, balance sheet access, tax and reporting support, or service during stress episodes. This non-price treatment is consistent with the matched home dealer check in Figure A6: for the model's canonical investors, home dealers do not offer a meaningful effective half-spread discount once we match on client, bond–day, size, and side.

The platform execution advantage Platform access delivers better execution for investors. Within the same client, bond, and day, matched on size and side, institutional clients pay a lower effective half-spread on MTFs than bilaterally. The differential is -0.19 price bps ($t = 2.5$) on the model diagnostic sample and -0.50 bps across the full 34-dealer cross-section, stable in leave-one-out checks (Figure 19). The advantage is concentrated among the genuine buy side—investment funds. It is not mainly an access-to-a-different-dealer effect: holding the dealer fixed leaves most of it intact (-0.18 bps), so the same dealer tends to quote more competitively on the platform.

This diagnostic constrains the model but does not by itself settle welfare. A lower effective half-spread is a transfer from dealer to investor unless it changes matching, participation, or the cost of intermediation; Allen and Wittwer (2023) document the same within-dealer venue wedge in the Canadian market and abstract from it by design (their Fact 2). The structural counterfactual measures welfare as gains from trade, so the platform wedge shows that platforms benefit investors privately, while the welfare result still depends on competition and relationship value.

Model outline and estimation The structural model features J dealers and N investors. Dealer j has investor-specific quality μ_{ij} and quotes a platform price equal to value plus a

(a) Matched platform differential, by client sector.



(b) Raw and matched platform differentials.

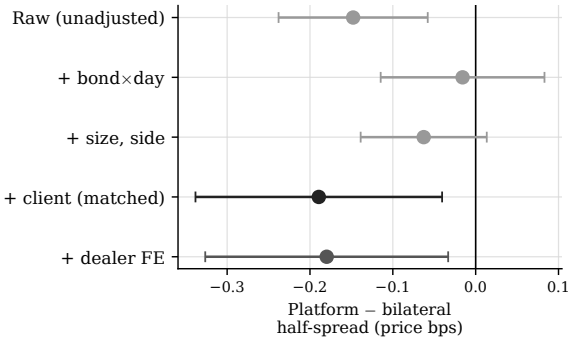


Figure 19: Platform and bilateral execution on the centralization analysis sample. The left panel plots the within-client effective half-spread differential between MTF (platform) and OTC/SI (bilateral) execution, matched on bond×day, trade size, and side, pooled and by client sector (95% ISIN-clustered confidence intervals; negative values mean lower effective half-spreads on the platform). The right panel reports estimates from the unadjusted comparison, the matched within-client specification, and the matched specification with dealer fixed effects. The effective half-spread is the client-signed deviation of the transaction price from the pre-trade MTS mid, in price basis points (negative = inside the mid). Diagnostic sample close to the Allen and Wittwer (2023) model sample (2023–2025: the ten largest dealers by trade count, all MTFs, platform-using clients in institutional sectors, 1,683 clients).

markup. Each institutional investor has a home dealer and chooses between bilateral execution with that dealer and platform execution with the dealer offering the best quote. A loyalty benefit r captures the non-price value of the bilateral relationship. The key parameters are demand sensitivity, dealer qualities, loyalty, liquidity shocks, and platform cost.

We follow Allen and Wittwer (2023) as closely as the Bund data allow: we assign home dealers, use platform prices and dealer shares to discipline competition and quality, and estimate the remaining parameters by the generalized method of moments. The main estimates are: demand sensitivity $\hat{\sigma} = 0.567$, loyalty benefit $\hat{r} = 1.01$ basis points, and platform cost $\hat{c} \approx -0.7$ basis points.[19] Dealer qualities are identified only relative to a reference dealer whose quality is normalized to zero; we use the largest dealer in the estimation sample.[20] For the demand elasticity we import the Canadian calibration of Allen and Wittwer (2023) ($1/\hat{\sigma} = 1.765$) and estimate the rest conditional on it. The Bund data do not identify this parameter: the auction-based instrument that identifies it in Canada has no power in Bunds, plausibly because dealers can lay off auction inventory in futures (Appendix A.6).

Welfare analysis Welfare is measured as model-implied gains from trade: centralization can improve matching by expanding access to efficient dealers, but can lower welfare when platform markups distort matching or investors lose valuable home dealer relationships. Table 4 reports the counterfactuals, which vary platform access (to institutional investors only, or to all investors), platform fees, full centralization, the interpretation of the loyalty benefit, and the intensity of dealer competition; the column and row definitions are given in the table note.

		Account I	Account IR	No Fee	Central
With loyalty	Imperfect	−6.01	−10.08	−11.38	−13.28
	Perfect	+9.07	+13.51	+14.52	+15.80
Switching cost	Imperfect	+19.21	+18.76	+22.29	+27.22
	Perfect	+28.21	+36.92	+40.55	+45.17
No loyalty	Imperfect	+26.19	+26.55	+29.54	+32.78
	Perfect	+50.44	+63.03	+66.99	+71.38

Table 4: Welfare counterfactuals relative to a fully decentralized market. Preferred-spec estimate following the Allen and Wittwer (2023) framework using Bund transaction data. Entries are percentage welfare changes relative to a fully bilateral market. Imperfect competition means dealers set quotes optimally; perfect competition means dealers quote at true values ($q = v$). Account I gives platform access to institutional investors only. Account IR gives platform access to all investors. No Fee sets platform fees to zero. Central forces all trades onto the platform. With loyalty means the home dealer benefit r enters welfare directly. Switching cost means r affects choices but is not valued in welfare. No loyalty sets $r = 0$.

[19] As in Allen and Wittwer (2023), the platform cost enters investors' venue choice but not the welfare measure; in their estimates it is close to the platform fee, a transfer. The loyalty benefit, by contrast, is the value of the relationship service itself. The estimated loyalty benefit is substantially smaller than the Canadian estimate of $\hat{r} = 2.37$ basis points, consistent with the higher platform adoption rate in Bunds. Dealer quality varies substantially across buy and sell sides and between institutional and retail investors.

[20] Allen and Wittwer (2023) report results for a single reference dealer. Because the estimated welfare changes depend on this choice, Table A3 also reports the centralization counterfactual with each of the second- to fourth-largest dealers as reference. The choice moves the welfare difference itself, not only the welfare of the fully bilateral market in the denominator.

The baseline sign is sharp. In the preferred specification, the home dealer benefit is direct surplus and dealers set platform quotes strategically. Relative to the current platform access regime, forcing all trades onto platforms reduces modeled welfare by 7.73%.

The positive cases in Table 4 are sensitivity exercises, not equally weighted outcomes. Welfare rises under perfect competition because platform access then improves matching without dealer quote distortions. It also rises when the home dealer premium is treated as a switching cost or set to zero, because centralization no longer destroys relationship surplus.[21]

The sign matches the Canadian estimates in Allen and Wittwer (2023); with the largest dealer as reference, the magnitudes are larger. Relative to the current access regime, full centralization lowers welfare by 7.73% in Bunds and by 1.75% in Canadian government bonds. Against the fully bilateral benchmark of Table 4, the current access regime is 6.01% below it in Bunds and full centralization 13.28%, compared with 1.29% and 3.02% in Canadian government bonds. The magnitude is not pinned down by the model alone. Relative to a fully bilateral market, centralization lowers welfare unless the demand elasticity exceeds $1/\hat{\sigma} = 2.94$ to 3.09, at or just above the upper end of the 95% confidence interval of the imported estimate ($1/\hat{\sigma} = 1.765$, standard error 0.599); Table A3 shows that the size of the loss also depends on the reference dealer.

Figure 20 plots every counterfactual in Table 4 against the corresponding estimate in Table 6 of Allen and Wittwer (2023). Every Bund cell has the same sign as its Canadian counterpart and, with the largest dealer as reference, a larger magnitude.

Discussion The model's policy implication is therefore conditional, not a general ranking of market designs. Taking estimated relationship value, imperfect dealer competition, and the benchmark imported demand elasticity at face value, mandatory platform centralization would lower modeled welfare. The positive rows identify the assumptions under which that conclusion reverses, and the sensitivity exercises show that the imported elasticity is especially influential. The model is suggestive about resilience, not a resilience test. In stress, centralized and competitive platforms may help high-value dealers absorb or redistribute inventory across a broader client base; this lies outside the partial equilibrium dealer-to-client model and should be read alongside the separate mechanical netting exercise. Three limitations matter, and their direction of bias is unclear. The institutional settings for Canadian bonds and Bunds differ. The dealer-to-client model captures only a partial equilibrium analysis for Bunds, which are part of a large ecosystem of futures and dealer-to-dealer trading. Lastly, the model assumes a single bond whereas the Bund market consists of many bonds with different maturities, benchmark status, and futures-deliverability status.

[21] The interpretation of the switching cost and no-loyalty rows warrants caution. As noted by Allen and Wittwer (2023) in footnote 22, these rows cannot be interpreted as a policy recommendation because removing loyalty is akin to removing costs from a profit calculation.

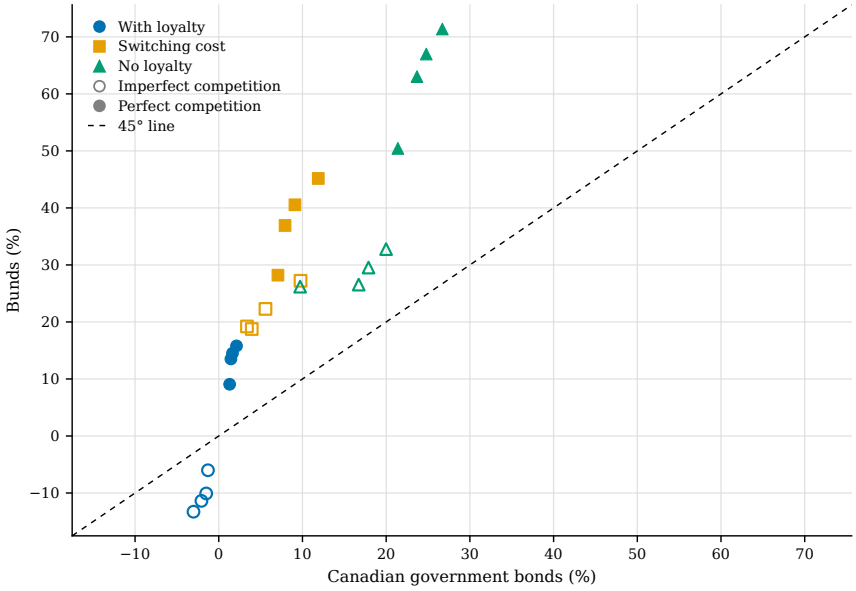


Figure 20: Welfare counterfactuals for Bunds and Canadian government bonds. Each point pairs one welfare change for Bunds from Table 4 with the corresponding Canadian government bond estimate in Table 6 of Allen and Wittwer (2023), in percent relative to a fully bilateral market. The horizontal axis reports Canadian estimates and the vertical axis reports Bund estimates. Hollow markers denote imperfect competition; filled markers denote perfect competition. The dashed line is the 45-degree line.

7 Conclusion

This paper studies the German Bund market using a new non-anonymous transaction-level dataset combining MiFID I and MiFID II transaction reports spanning 2008–2025. The data identify both sides of each trade and reveal the euro area's benchmark safe asset market structure. Trading is hierarchical around a small dealer core, relationship-based between dealers and clients, and segmented across investor sectors, with order flow fragmented across venues. We will maintain the cleaned dataset and its documentation as a reusable resource for authorized research within the Bundesbank.

That structure shapes stress flows, price discovery, and execution costs. Stress flows differ by shock: foreign official accounts sell and hedge funds buy in both the Covid dash for cash and the 2023 flight to quality. Primary dealers absorb material flow only in the cash shock and distribute Bunds after the 2025 fiscal news. In normal times, hedge funds and official sector accounts carry the most permanent price-relevant flow, but they do not pay the highest effective half-spreads. Retail households pay much more, and large trades pay a size penalty. The Bund market therefore rejects a simple adverse selection account in which the most price-relevant flow is charged the highest execution cost.

The reform implications follow from these facts. Mandatory central clearing is arithmetic: given observed trades, netting each dealer's book at a clearing house would cut dealers' net settlement obligations by 35%, less than comparable estimates for U.S. Treasuries and U.K. Gilts. Platform centralization changes dealer access, relationship surplus, and competition in equilibrium. In the benchmark structural counterfactual, forced platform centralization lowers

welfare: broader dealer access does not compensate for the loss of relationship surplus under imperfect dealer competition, in Bunds as in Canadian government bonds. Even in a benchmark safe asset market, market structure determines who moves prices, who pays execution costs, and which reforms create gains.

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Appendix
for
Anatomy of a Safe Asset: Bund Market Structure and Reform

by Mark Kerssenfischer and Willy Scherrieble

Section A provides additional results.
Section B details how we adapt the yield curve fitting method of Filipović et al. (2025) to Bunds.
Section C provides descriptive statistics and cross-checks of our dataset.
Section D details how we clean and process the MiFID I and MiFID II/MiFIR transaction reports, accounting for the complexity of the Bund market.

Appendix A Further Evidence

A.1 The Dealer System

This appendix supplements Section 4.2. Figure A1 shows the display behind the club fact: the Bietergruppe members' share of intermediated client volume by half-year, under the baseline behavioral screen and across screening configurations. The paragraphs below document two further properties of the dealer hierarchy: its geographic division of labor and its stability in acute stress windows.

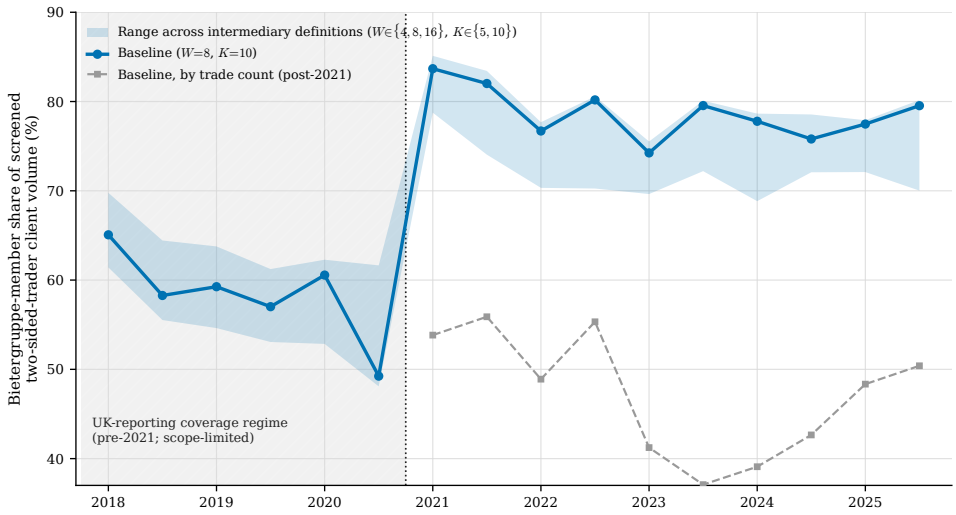


Figure A1: Bietergruppe members' share of intermediated client volume by half-year. The line is the baseline behavioral screen (round-trips in at least $W = 8$ ISIN-weeks and at least $K = 10$ counterparty groups per half-year); the shaded band spans the six screening configurations ($W \in \{4, 8, 16\}$, $K \in \{5, 10\}$) and reflects definitional robustness, not sampling uncertainty. The thin line is the members' share of trade count after 2020. The screened slice carries 25.8% of trading volume outside intragroup bookings (Table 2). Values before 2021 reflect the UK-reporting coverage regime (hatched) and are not comparable to later years.

The hierarchy sorts geography The division of labor across the hierarchy is geographic. German headquartered clients account for 19.4% of the top tercile dealers' institutional client volume but 55.6% at the bottom tercile—a top-minus-bottom gradient of -36.2 percentage points—negative in every half year after 2020 and within every client sector, and surviving comparison within venue, trade size, and maturity cells (-31.6 points; Figure A2). Part of the gradient reflects composition—the bottom tercile is disproportionately populated by small German headquartered dealers—but at most a third: holding dealer nationality fixed alongside venue, size, and maturity leaves a gradient of -25.4 points. The client sector mix differs across tiers as well: the dissimilarity index between top and bottom tercile sector shares is 14.0 percentage points, against a permutation benchmark of 3.1. The bottom tercile executes 1.8% of client volume, so the division of labor is a statement about whom the tail serves, not about how volume is split: the core of the hierarchy intermediates for foreign investors on electronic venues, while the tail is a domestic, voice-leaning niche. Geographic segmentation of bond portfolios is well documented at the investor level—home bias in sovereign holdings (Coeurdacier and Rey, 2013; Maggiori, Neiman, and Schreger, 2020) and gravity in cross-border asset trade (Portes and Rey, 2005)—and euro area capital markets are less integrated than headline holdings suggest (Beck, Coppola, Lewis, Maggiori, Schmitz, and Schreger, 2024). We locate the same segmentation one

layer deeper, in who intermediates for whom: even in the euro area's most homogeneous safe asset, the dealer system sorts clients by geography.

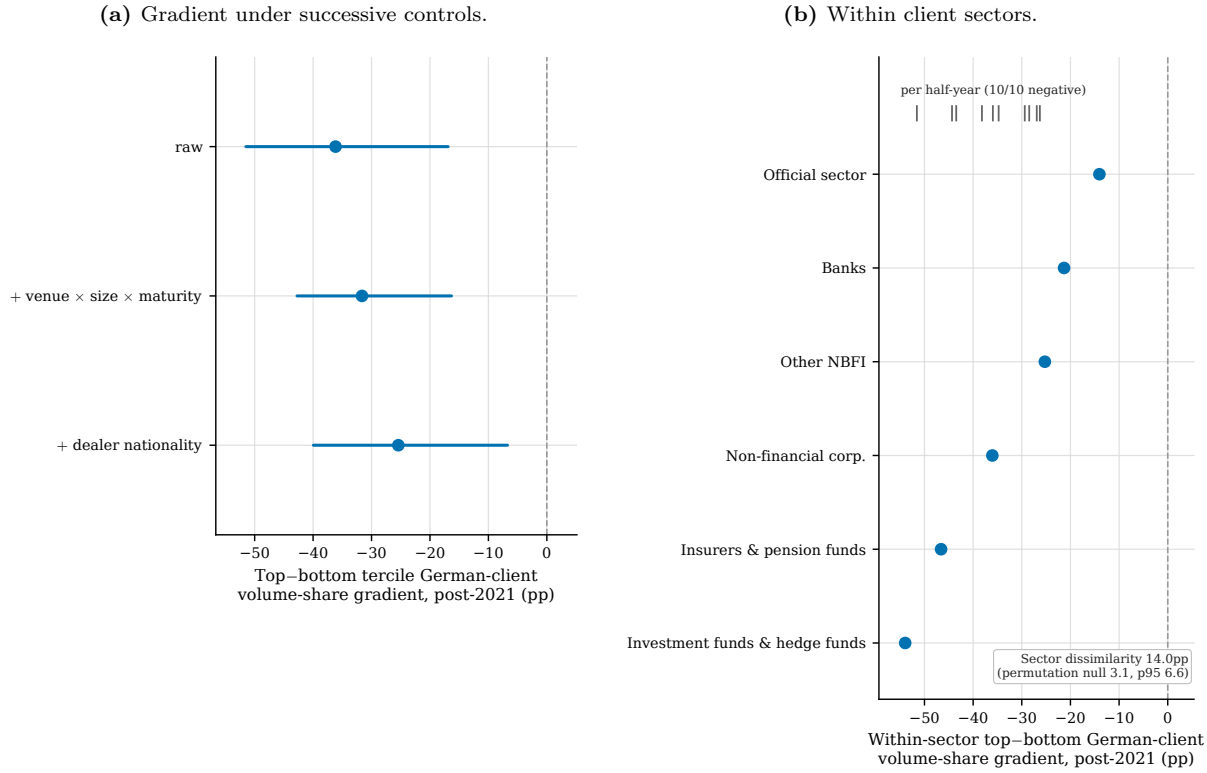


Figure A2: Dealer hierarchy and client geography after 2020. Panel (a): difference between top and bottom centrality terciles in the German headquartered share of institutional client volume—raw, within venue–size–maturity cells, and additionally holding dealer nationality fixed—with 95% dealer-clustered confidence intervals. The control ladder bounds composition rather than decomposing it: a shift-share split attributes between roughly none and a third of the raw gradient to dealer-nationality composition. Client and dealer nationality are the countries of the investor group's ultimate parent, so German arms of foreign parents count as foreign. Panel (b): the same gradient within each client sector, shown descriptively without confidence intervals; inference is in panel (a). The small "other" sector is omitted; its gradient is the smallest.

The core under stress In the three acute stress windows studied in Section 5.1, the dealer hierarchy does not systematically move. The top tercile's share of institutional client volume changes by 2.0 percentage points relative to calm weeks before each episode (95% confidence interval $[-3.2, +7.3]$; Figure A3). The calm weeks are the 40 business days that end one week before each episode, leaving out the days without account owner information (Section D.2); for 2025 they start on 18 November 2024. The level is high in both states: the top tercile intermediates 75.1–81.8% of institutional client volume in calm weeks and 77.3–84.5% in stress weeks. Placebo March windows in calm years move by similar amounts. The stress episodes therefore reshuffle who buys and sells Bunds, not which dealer tier intermediates the flow. This is not a claim that dealer balance sheets are unconstrained; it is a narrower null on reallocation within the dealer hierarchy.

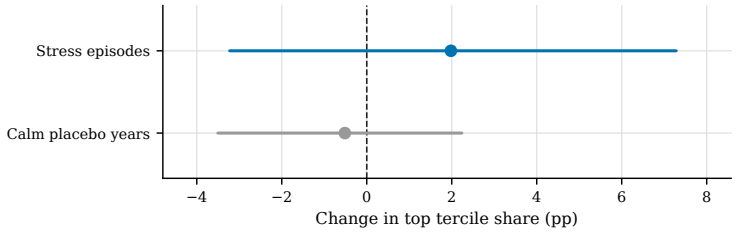


Figure A3: Top tercile dealer share in stress and placebo windows. Dots report pooled changes in the top tercile's share of institutional dealer–client volume relative to the 40 business days that end one week before each episode, with 95% dealer group bootstrap confidence intervals. The stress estimate pools the three acute stress windows; the placebo estimate pools same calendar March windows in calm years.

A.2 Permanent Price Impact by Venue Type and Venue Protocol Family

The informed flow test in Figure 13 can also be cut by where trades execute. We repeat the permanent impact exercise by venue type and, as a secondary descriptive cut, by venue protocol family: for each category, we sign dealer-to-client flow from the client's side, aggregate to bond–day–category net flow, and test whether net buying predicts a subsequent decline in the curve-relative pricing error over one to twenty trading days, absorbing ISIN and maturity bucket×day fixed effects and controlling for the bond's lagged curve-relative pricing error and its two lagged daily changes, as in Figure 13. These cuts are diagnostic, not operator-level claims: venue type and venue protocol family are MIC-level classifications pooled across venues and dealers, and no venue operator is named.

The venue protocol family does not restate the venue types (Table A1). A MIC enters the RFQ family or the order book family only when the venue's own rulebook or its operator documents that model; venue type alone never assigns either family. Systematic internalizers, venues that combine an order book with RFQ trading, and venues without a source are unknown; they account for 20.5% of dealer-to-client volume (Appendix D.11). Among the classified families, RFQ family net flow has a permanent impact of +0.037 bps at the ten-day horizon ($t = 6.8$), and net flow in the residual other family, mostly off-venue reports and OTF trading, has +0.005 bps ($t = 1.0$). Order book family flow, 1.3% of classified volume and almost entirely on records without an identified client, is close to zero at that horizon (+0.007 bps, $t = 0.4$). These estimates compare venue families; they do not show how individual trades executed.

Venue type	RFQ	Other	CLOB	Unknown
MTF	87	4	2	7
OTC	0	100	0	0
SI	0	0	0	100
RM	0	45	40	15

Table A1: Venue protocol family by venue type. Each row reports the composition of a venue type's dealer-to-client volume by venue protocol family (row-normalized percentages over all four labels). The family is a MIC-level label: RFQ family, order book family (CLOB), other (residual), or unknown. Systematic internalizers are unknown by construction. Pooled across all operators of each venue type. MiFID II subsample.

Venue Type The venue-type cut compares multilateral electronic platforms (MTFs), bilateral over-the-counter venues (OTC), single-dealer systematic internalizers (SIs), and regulated markets (RMs), mostly retail exchanges; OTF trades enter only the protocol family cut, in the residual other family. Figure A4 reports the permanent price impact of each venue type's net flow. Net flow on MTFs, including platforms such as Bloomberg and Tradeweb, predicts a permanent price impact that builds over the first ten days—0.022 bps after one day, 0.039 bps at the ten-day horizon ($t = 7.6$)—and remains at 0.030 bps after twenty. Bilateral OTC and single-dealer SI flow are also informative and permanent (0.020 and 0.033 bps at ten days; $t = 3.5$ and 5.0). Regulated market flow, by contrast, contributes little: the RM price impact is close to zero, −0.022 bps ($t = 1.1$), and is not distinguishable from zero at every horizon.

Each venue type's net flow pools the sectors that trade there, so these estimates largely reflect which sectors trade where. Within hedge fund flow alone, the MTF impact is 0.039 bps ($t = 6.6$) and the OTC impact 0.022 bps ($t = 2.8$). Each estimate scales net flow by the standard deviation of that venue type's flow, so the two are not comparable per euro; we do not test their difference, and the hedge fund trades routed to MTFs may differ from those executed over the counter. The regulated market null may partly reflect lower power—RM is only 0.5% of

volume—but the point estimate is near zero through the ten-day horizon, not merely imprecise. The results do not depend on the anonymous interdealer records that the dealer-to-client filter admits: restricted to records with an identified non-dealer client, MTF, OTC, and SI flow remain significant and permanent at every horizon and regulated market flow is still not distinguishable from zero, although SI flow then carries as much information as MTF flow.

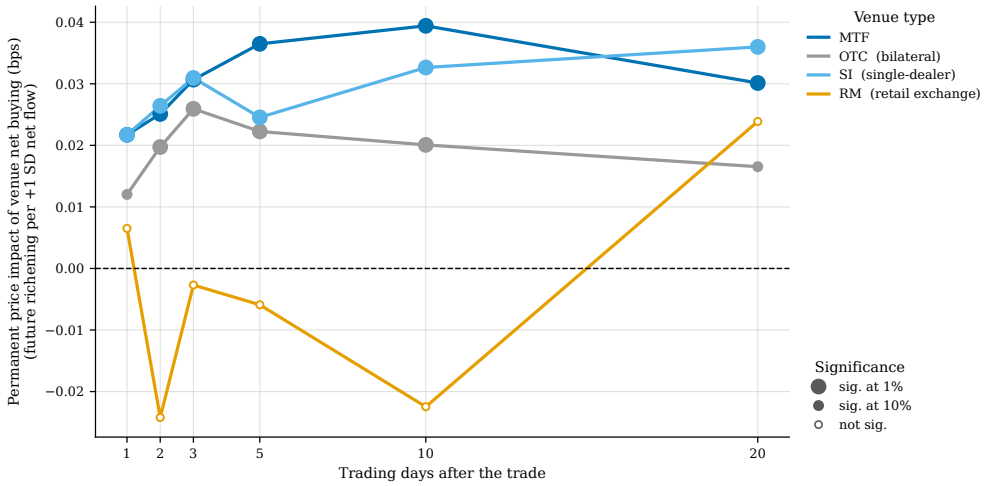


Figure A4: Permanent price impact by venue type. For each trading venue category, the figure reports the cumulative future change in the curve-relative pricing error, defined as synchronized implied yield minus the kernel ridge fair yield, in bps, per net purchase of the bond equal to one standard deviation of the venue type's absolute net purchases, by trading days after the trade, signed so that positive values denote richening. Venue-type-by-horizon regressions absorb ISIN and maturity bucket×day fixed effects and control for the bond's lagged curve-relative pricing error and its two lagged daily changes; marker size and fill encode significance using ISIN-clustered standard errors. Dealer-to-client trades, MiFID II subsample (2018–2025).

A.3 Informed Flow Measurement Robustness

The benchmark informed flow test in Figure 13 uses future changes in the curve-relative pricing error, because the economic question is whether a sector selects bonds whose curve-relative pricing error subsequently declines. For sector k and horizon h, we estimate $e_{i,t+h} - e_{i,t} = \beta_{k,h} f_{i,t,k} + \gamma' z_{i,t} + \alpha_{m(i,t),t} + \alpha_i + \varepsilon_{i,t}$ by unweighted least squares on the bond-days with net flow from sector k, where e is the curve-relative pricing error, $f_{i,t,k}$ is the sector's signed euro net purchases divided by the standard deviation of its absolute net purchases and capped at five, $z_{i,t}$ holds the lagged pricing error and its two lagged daily changes, $\alpha_{m(i,t),t}$ is a fixed effect for each day and bucket $m(i,t)$ of the bond's remaining maturity (under 1, 1–3, 3–7, 7–15, and over 15 years), and α_i is an ISIN fixed effect. The maturity bucket by day effects absorb yield and fitted curve moves common to bonds of similar maturity. Horizons count the bond's subsequent trading days in the sample, so they skip days without trades in the bond and the days without account owner information (Section D.2). Standard errors are clustered by ISIN. Throughout, a one standard deviation net purchase means $f_{i,t,k} = 1$.

Figure A5 repeats the same sector-level permanent impact exercise with the future change in the bond-day mean synchronized implied yield as the outcome, which does not use fitted curves. The outcome averages the trade-level yields unweighted, the same aggregation as the

benchmark pricing error; the fixed effects and the lagged pricing error controls are those of the benchmark.[22]

The hedge fund result carries over: hedge fund net purchases predict a future yield decline of 0.05 bps at the ten-day horizon ($t = 6.8$). So does the official sector result: official sector net purchases predict a decline of 0.06 bps ($t = 3.8$). Investment fund flow, transitory in the benchmark, is not distinguishable from zero here either (0.01 bps, $t = 0.6$). Bank and ICPF flow do not carry over: their ten-day estimates are not distinguishable from zero ($t = 0.7$ and 0.9). Retail household flow has no estimate different from zero at one to five days, 0.11 bps at ten days ($t = 2.1$), and an estimate not distinguishable from zero at twenty days ($t = 1.0$). We keep the curve-relative pricing error as the benchmark because raw yield changes are less directly tied to within-segment security selection.

The benchmark specification controls for the bond's cheapness before the flow day because some sectors buy bonds that have just cheapened, and cheapness mean-reverts. Hedge fund and official sector purchases follow a cheapening of the bond on the previous trading day (0.013 and 0.011 bps per standard deviation of net flow, $t = 4.7$ and 2.9); investment fund purchases do not ($t = 0.1$). On the same sample without these controls, the ten-day estimates are larger for the official sector (0.07 bps, $t = 6.7$) and investment funds (0.03 bps, $t = 3.9$) and essentially unchanged for hedge funds (0.04 bps). Controlling instead for the flow day's own pricing error and its changes on the flow day and the day before, which the flow itself moves, gives a lower bound: 0.02, 0.03, and 0.01 bps for hedge funds, the official sector, and investment funds ($t = 6.0$, 3.3, and 1.8). Under all three control sets, hedge fund flow has the most precisely estimated impact and official sector flow the largest.

With day and ISIN fixed effects in place of the maturity bucket×day fixed effects, and the same controls, the ten-day estimates are 0.04 bps for hedge funds ($t = 8.2$), 0.05 bps for the official sector ($t = 6.2$), 0.02 bps for banks ($t = 2.0$), and 0.08 bps for retail households ($t = 3.0$). Bank and retail purchases coincide with moves common to bonds of similar maturity; within the maturity segment their estimates are not distinguishable from zero. The retail estimate with day fixed effects is flat for three days and then rises into a hump that is not significant from forty days on.

A.4 Price dispersion

A natural question is whether the market structure taxonomy also organizes transaction prices: do trades in the same bond on the same day price systematically differently across venue types, dealer–client segments, and client sectors? Decomposing the within-bond-day dispersion of synchronized implied yields, we find that it does not. A naive variance decomposition attributes a sizable share of the dispersion to these groupings, but that share is almost entirely a counting artifact: a placebo that randomly reassigns each bond–day's trades to the *same number* of groups reproduces nearly all of it. The genuine, incremental contribution of the real groupings over this same-cardinality null is at most a few percentage points—and is negative for the dealer–client segment, whose observed share falls below the random benchmark. The groupings do not predict dispersion out of sample, and the largest systematic within-bond-day yield gap between any two venue, segment, or sector groups is around 0.1 bp.

[22] A volume-weighted bond-day yield is pulled toward the flow day's large trades and their price concession (Figure 15), so part of the regressor sits in the outcome: a sector's net purchases predict the flow day's gap between the volume-weighted and the unweighted yield for investment funds, hedge funds, and the official sector, and not the gap ten days later. On the volume-weighted outcome, the investment fund estimate is -0.06 bps ($t = -5.1$), a yield rise after their purchases, while hedge funds and the official sector keep 0.02 and 0.01 bps. Measured from the day before the flow, the volume-weighted estimates are 0.03, 0.04, and 0.06 bps for investment funds, hedge funds, and the official sector.

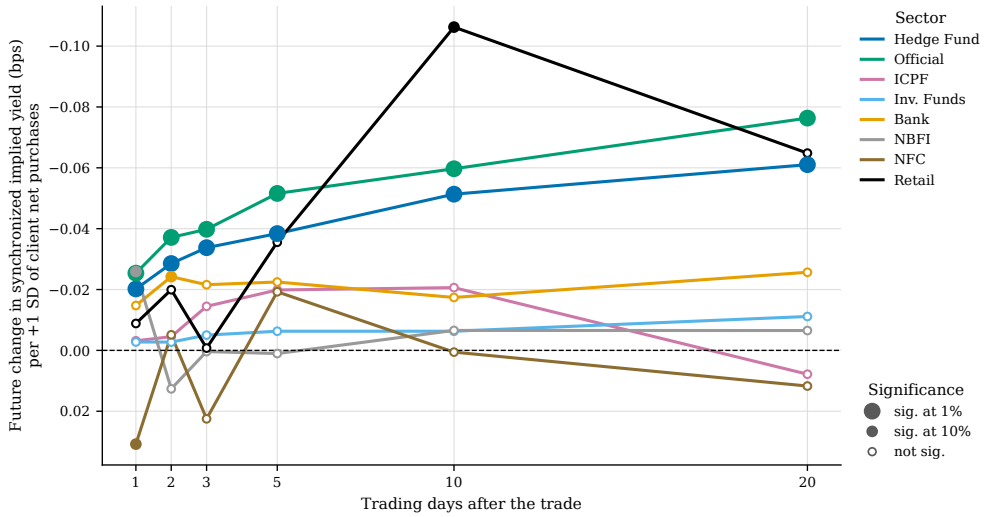


Figure A5: Permanent price impact using future synchronized implied yields. For each identified client sector, the figure reports the cumulative future change in a bond's synchronized implied yield, averaged without weights over the bond-day's trades, in bps, per net purchase of the bond equal to one standard deviation of the sector's absolute net purchases, by trading days after the trade. Sector-by-horizon regressions absorb ISIN and maturity bucket×day fixed effects and control for the bond's lagged curve-relative pricing error and its two lagged daily changes; the y-axis is inverted so that upward denotes richening through a lower future yield. Marker size encodes significance (ISIN-clustered standard errors): large filled = 1%, medium filled = 10%, small hollow = not significant. Dealer-to-client trades, MiFID II subsample (2018–2025).

For trades synchronized using the bond's own pre-trade and 11:45 MTS mids, the synchronized price equals the common fixing mid plus the transaction's deviation from its pre-trade mid. Within such a bond-day, price dispersion therefore reflects variation in transaction concessions. The yield transformation and observations synchronized using fallback quotes qualify this identity for the full sample. The dispersion exercise nevertheless concerns differences among trades within a bond-day; it does not identify whether the bond's common price level is close to fundamental value.

We stress what this is *not*. It is not a claim that trading is frictionless—the spread component is real and falls unevenly across investors (Figure 14). Nor is it a statement about futures anchoring. Bund futures plausibly anchor the *level* of cash prices across each maturity neighborhood through cash–futures basis arbitrage, but anchoring concerns where the reference mid sits relative to fundamental value, whereas within-bond dispersion concerns variation among transactions around their daily center—a different moment, silent on the level. Empirically, dispersion is essentially flat across the maturity curve: it is no tighter at the two-, five-, ten-, and thirty-year deliverable segments than in the gap between the ten- and thirty-year futures, which trades about half as often—so the futures footprint shows up in trading *activity*, not in execution cost uniformity. Testing the anchor itself requires the cash–futures basis, a separate question that this exercise does not take up; nor is it an independent test of the law of one price across venues.

A.5 Venue Centralization: Diagnostics and Sensitivity

This appendix collects diagnostics for the centralization counterfactual of Section 6.2: the venue mapping behind the model's two trading categories, two checks of its relationship-pricing premises, and the sensitivity of the welfare estimate to design choices.

Venue mapping Table A2 reports the composition of the structural estimation sample across the venue groups the model retains: bilateral trading pools OTC and single-dealer SI trades, and platform trading pools the Bloomberg and Tradeweb multilateral platforms, each of which spans several MIC codes of the same parent. Venue types the two-category model does not represent—most importantly OTF trading, which is close to an interdealer broker market, and the retail exchange—are excluded from the estimation sample. As in Allen and Wittwer (2023), quotes, values, markups, and the loyalty benefit are in yield basis points; a dealer's weekly platform quote is the average of its platform transaction yields across bonds after removing ISIN-by-week fixed effects.

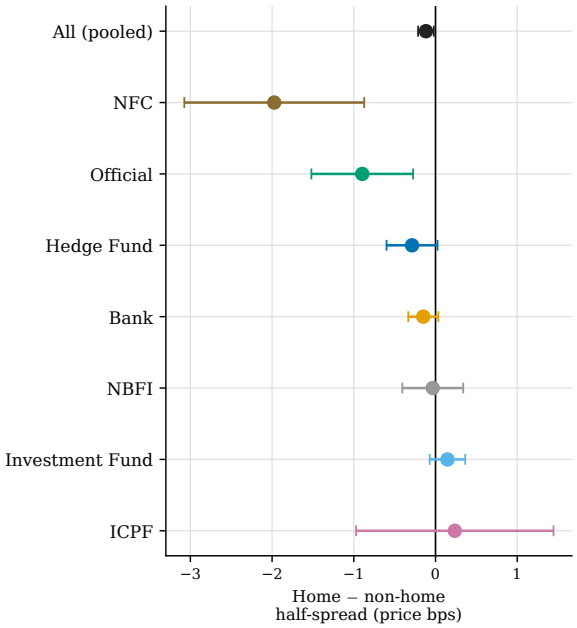
Venue group	Model	Trades	Share	Vol. (EUR bn)	Share	Customers
Panel A: all dealer-to-client trades						
OTC	Bilateral	271,058	26.4%	465.8	11.1%	6,416
SI	Bilateral	88,869	8.7%	928.1	22.2%	1,770
Bloomberg MTFs	Platform	288,186	28.1%	772.0	18.5%	1,941
Tradeweb MTFs	Platform	378,270	36.9%	2,014.6	48.2%	1,336
Bilateral total		359,927	35.1%	1,393.9	33.3%	7,587
Platform total		666,456	64.9%	2,786.6	66.7%	2,827
Panel B: institutional customers						
OTC	Bilateral	218,678	22.8%	387.2	10.2%	923
SI	Bilateral	71,932	7.5%	618.6	16.3%	671
Bloomberg MTFs	Platform	288,186	30.1%	772.0	20.4%	1,941
Tradeweb MTFs	Platform	378,270	39.5%	2,014.6	53.1%	1,336
Bilateral total		290,610	30.4%	1,005.8	26.5%	1,152
Platform total		666,456	69.6%	2,786.6	73.5%	2,827

Table A2: Venue mapping for the platform centralization model. The table reports the composition of the dealer-to-client structural estimation sample before collapsing venue groups into the model's two categories. Bilateral trading combines OTC and SI trades. Platform trading combines MTF trades on Bloomberg and Tradeweb. The Bloomberg and Tradeweb MTF groups each cover multiple MIC codes belonging to the same parent company. Institutional clients are those that trade on a platform at least once during the estimation window.

Matched home dealer half-spread differential The centralization model treats home dealer loyalty as a non-price relationship value rather than a separate bilateral price discount. To check that premise, we compare bilateral trades with a client's home dealer to the same client's other bilateral trades on the Allen and Wittwer (2023) model sample, using a leave-one-out home definition and matching on bond×day, trade size, and side. The model sample differential is small but statistically distinct from zero (-0.12 price bps; 95% confidence interval $[-0.21, -0.02]$), and the canonical investment fund investors show essentially no home discount (+0.20 bps). The full dealer cross-section gives a more precise but still small discount (-0.16 price bps), so the evidence supports treating relationship value as primarily non-price.

Affiliated-fund execution with the parent dealer The most captive clients provide a direct test of relationship pricing: dealers' asset management and insurance arms trading with

(a) Matched home dealer differential, by client sector.



(b) Raw and matched estimates.

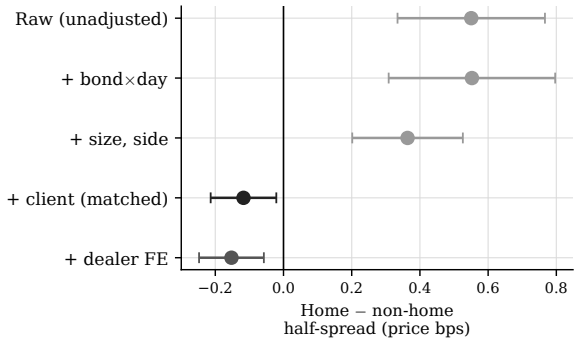


Figure A6: Matched home dealer half-spread differential. The left panel plots the within-client effective half-spread differential between a client's home (largest-volume) dealer and its other dealers, matched on bond×day, trade size, and side, pooled and by client sector (95% ISIN-clustered confidence intervals; negative values mean lower effective half-spreads with the home dealer). The right panel shows the same differential as controls are added, from the unadjusted comparison to the matched within-client specification. The effective half-spread is the client-signed deviation of the transaction price from the pre-trade MTS mid, in price basis points (negative = inside the mid). Bilateral dealer-to-client trades on the Allen and Wittwer (2023) model diagnostic sample (2023–2025, platform-using institutional clients, 545 multi-dealer clients).

their parent. These arms belong to their parent's investor group but not to its dealer desk, so an in-house trade is a dealer-to-client trade in which the client's investor group equals the dealer's. Within a given dealer, bond, and day, these affiliated funds pay their parent 0.52 price bps more on the effective half-spread than the dealer's other fund clients on OTC trades ($t = 1.4$; 95% confidence interval $[-0.23, +1.27]$), and no premium distinguishable from zero on MTFs or with systematic internalizers (Figure A7). The estimate draws on 21 parent groups; dropping any one of them gives estimates between -0.12 and 0.66 price bps. From the client side the gap is similar: within a given fund, bond, and day, the fund pays 0.86 price bps more when it routes to its parent than to an outside dealer ($t = 2.7$). The premium is imprecisely estimated, but the interval excludes any material discount: affiliated funds do not buy more cheaply from their own parent.[23] Bai et al. (2026) document an affiliation premium of 4–5 basis points inside bank holding companies in the Treasury repo market; in the Bund cash market the estimate is far smaller and confined to OTC trading. Bilateral relationships do not, on average, give affiliated clients a better price, and the most captive client, if anything, pays more, not less.

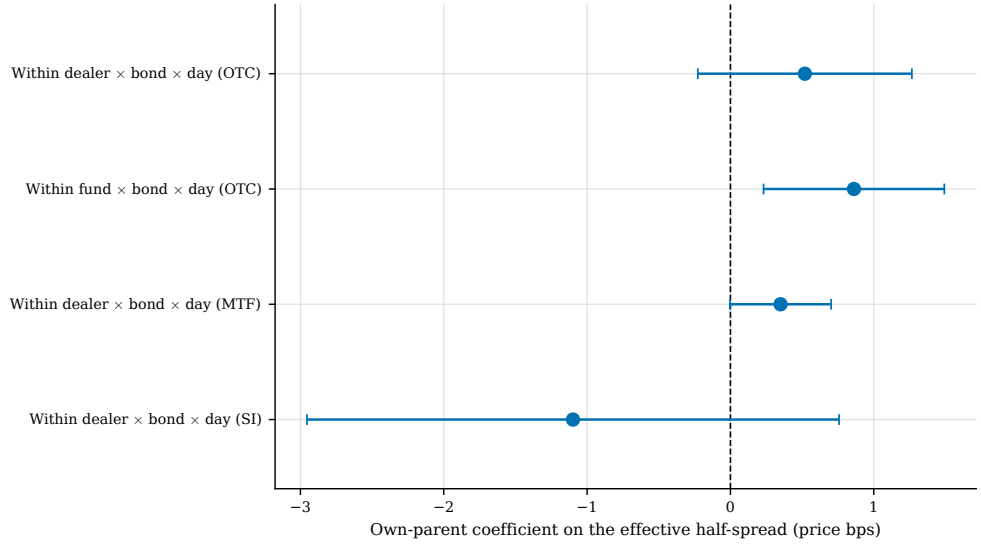


Figure A7: Own-parent coefficient on the effective half-spread for dealer-affiliated funds, by venue and specification. Each marker is the coefficient on an indicator for an affiliated fund trading with its parent dealer, from a regression of the client-signed effective half-spread, in price basis points, on that indicator and log trade size with the fixed effects listed; whiskers are 95% confidence intervals from standard errors clustered by ISIN. The within dealer specification compares an affiliated fund's trades with its parent to the parent's other fund clients on the same bond and day; the within fund specification compares the same fund's trades with its parent to its trades with other dealers on the same bond and day. Dealer-to-client trades of investment funds and insurers affiliated with a primary dealer, MiFID II (2018–2025).

Welfare sensitivity The welfare results in Table 4 hold the demand elasticity parameter at the imported value $1/\hat{\sigma} = 1.765$; Appendix A.6 documents why the Bund data do not identify this parameter. Table A3 reports the Central counterfactual under the "With loyalty" and imperfect competition specification, varying one design choice at a time: the dealer set, the dealer weights, the reference dealer, and the demand elasticity. The elasticity rows use the estimates of Allen and Wittwer (2023) for alternative specifications (0.68 and 1.44), the imported

[23] We read the estimate as descriptive. A positive wedge is consistent with the internal transfer price of Bai, Bostrom, Infante, and Ivashina (2026), but where an affiliated arm manages third-party assets the same wedge is instead a cost borne by outside investors; our data do not separate proprietary from third-party money, so we take no stand on which.

value, the upper end of its 95% confidence interval, and twice the imported value. Bisection places the reversal from loss to gain between $1/\hat{\sigma} = 2.94$ and 3.09, at or just above the upper end of the imported estimate's 95% confidence interval (standard error 0.599). Across the entire interval, centralization therefore lowers welfare relative to a fully bilateral market; its magnitude depends on the imported elasticity. These exercises do not evaluate centralization against the current access regime.

The loss is present in every dealer set, weighting and reference dealer we can estimate, but its size varies. With each of the four largest dealers as reference, the centralization effect at the imported elasticity lies between −13.28% and −1.61%. Trade size does not enter the model: welfare is a per-trade average, and every trade is priced at its dealer's weekly platform quote, so a trade size restriction changes only the estimation sample. Restricting the sample to trades of at most €50 m, the range over which the platform share of trades does not fall with size and every dealer in the sample trades on the platform, gives a Central effect of −14.15%; a cap of €100 m gives −13.08%. The loss therefore does not rest on the largest bilateral trades, for which the platform shows few comparable trades. Dealer sets of 30 or more group-level dealers do not identify the counterfactual: the added dealers appear in the transaction data, but for many weeks we do not observe the electronic platform price at which they compete, and full centralization would require imputing those prices.

Dimension	Specification	Central welfare (%)
Dealer set size	Top 10 dealers, equal weights	−20.80
	Top 15 dealers, equal weights	−19.38
	Top 20 dealers, equal weights (benchmark)	−13.28
	Top 25 dealers, equal weights	−32.01
	Top 30 or more dealers	Not identified
Dealer weighting	Top 20 dealers, volume weights	−12.09
Reference dealer	Largest dealer (benchmark)	−13.28
	Second-largest dealer	−13.16
	Third-largest dealer	−10.26
	Fourth-largest dealer	−1.61
Trade size	Trades up to €50 m	−14.15
	Trades up to €100 m	−13.08
Demand elasticity	$1/\hat{\sigma} = 0.68$ (no-loyalty estimate)	−20.44
	$1/\hat{\sigma} = 1.44$ (trade size estimate)	−16.90
	$1/\hat{\sigma} = 1.765$ (benchmark, imported)	−13.28
	$1/\hat{\sigma} = 2.94$ (upper 95% CI bound)	−0.10
	$1/\hat{\sigma} = 3.53$ (stress, 2× benchmark)	+6.09

Table A3: Sensitivity of platform centralization welfare to one design choice at a time. Entries are percentage welfare changes relative to a fully bilateral market for the Central counterfactual under the "With loyalty" and imperfect competition specification. The benchmark uses the top 20 group-level continuously active Bietergruppe dealers, excludes in-house trades, ranks dealers by dealer-to-client volume, applies equal dealer weights, normalizes the quality of the largest dealer to zero, and imports $1/\hat{\sigma} = 1.765$. Reference dealer rows instead normalize the quality of the second-, third- or fourth-largest dealer. Trade size rows drop the trades above the stated size from the estimation sample and re-estimate the model; the dealer numbering and each investor's platform access stay as in the benchmark.

A.6 Identification of the Demand Elasticity

Allen and Wittwer (2023) identify the demand elasticity from primary auctions: a dealer who wins an unexpectedly large share of the amount she demanded carries more inventory, and the induced variation in her platform quotes is plausibly unrelated to platform demand shocks. Their instrument is the share won of the amount demanded per dealer and week, with a first-stage F-statistic of 16; their appendix also validates a coarser variant that uses the amount won alone.

The Bund data contain the allotment side of this design but not the bids. MiFID II reports the primary allocation legs (Appendix D.6), which we aggregate to amounts won per dealer group and auction. The resulting allotments are well measured: ranked within half-years, they match the official Bietergruppe ranking with a Spearman rank correlation of 0.84, and the fit improves further when allotments are duration weighted, mirroring the weighted allotment criterion of the official ranking. Without the bids, the preferred share-won-of-demand instrument cannot be constructed; the amount-won variant and a surprise variant, which measures wins relative to the expectation implied by the previous half-year ranking, can.

These instruments have no power in Bunds. The win variation is real and survives dealer and week fixed effects, yet it is essentially uncorrelated with dealer platform quotes: first-stage F-statistics stay below 1.5 in the aggregate specification, with duration-weighted allotments, and in maturity-segment designs that match a win to quotes in the auctioned segment. The uninstrumented estimate is attenuated toward zero, as Allen and Wittwer (2023) also document in Canadian data. Measurement explains only part of the gap to the Canadian first stage: platform quotes are weekly averages of transaction yields whose cross-dealer dispersion is below one basis point, with a split-half reliability of roughly 0.25, so even a Canadian-strength pass-through would produce a first-stage F-statistic of only about five here. The observed pass-through is an order of magnitude below that benchmark.

The remaining gap is plausibly market structure. Bund dealers hedge cash inventory in the deep Bund futures market (Kerssenfischer and Helmus, 2025), so an auction allotment need not be worked off through cash quotes; the Canadian government bond market has no comparably deep futures segment. We do not test this mechanism here: the pass-through is absent in every dealer size tier, so no subgroup of dealers identifies σ in the Bund data.

We therefore import the Canadian point estimate and treat the demand elasticity as an identification limit of the Bund setting: Table A3 reports how the centralization welfare estimate depends on it. Estimating a Bund-specific elasticity would require dealer-level bid schedules from the auction system, which the transaction reports do not contain.

Appendix B Price Synchronization and Yield Curve Estimation

Price synchronization At the trade time and the 11:45 a.m. fixing, we select the most recent same-bond MTS quote within the preceding hour, preferring quotes with a bid–ask spread no greater than 10 yield basis points. We choose 11:45 a.m. to align with the Bundesbank yield curve fixing and a relatively quiet news period; see Kerssenfischer and Schmeling (2024). Where comparable bonds are available, we reject shifts that differ by more than 10 yield basis points from the median shift of up to four nearby maturities, requiring at least two. We also reject the bond-day when the median residual across at least three five-minute bins exceeds 10 basis points in absolute value. When a required quote is unavailable, we use a subsequent quote within an hour, subject to the observed trading session for quotes after the trade, or the median yield shift of nearby quoted bonds. These fallback observations are flagged separately and do not undergo the neighbor-shift check. Subsequent quotes and neighboring bonds are not substituted for the pre-trade same-bond benchmark in effective half-spreads.

Yield conversion and sample selection We convert synchronized prices into yields using the settlement cycle, TARGET holidays, irregular first coupons, and unadjusted coupon dates under the ICMA street convention. For older bonds without published first coupon and accrual start dates, we apply the federal convention for long first coupons: accrual starts on the settlement date, or on the preceding maturity anniversary when the bond settles within about two months after it, and the first coupon falls on the first maturity anniversary at least roughly 300 days after accrual starts. The yield sample excludes when-issued trades, bonds with less than three months of remaining maturity, and bonds with at most one year of original maturity. We screen individual prices against the bond's own daily yield center, using a tolerance of 25 yield basis points or the yield equivalent of 0.10 price points, whichever is larger. The center is the median synchronized yield across at least three distinct timestamp–price observations. With fewer observations, it is the yield of the bond's 11:45 MTS mid-quote. A tight 11:45 quote also replaces the median when the two disagree by more than the tolerance. The screen applies where a center is available; we retain a separate flag recording whether the center agrees with the quote. Observations without a synchronized implied yield or a Bundesbank curve yield cannot enter the yield sample. A separate gross-error guard excludes yields at least 100 basis points from the Bundesbank's Svensson (1994) curve. For spline and kernel ridge estimation, we use bond-days whose volume-weighted yield is within 35 basis points of that curve, while evaluating the fitted curves at all retained bond-days.

Yield curve estimation We implement the kernel ridge yield curve estimation of Filipović et al. (2025) using their publicly available Python code. We adapt it to the German Bund market. Several market-specific features require adjustments relative to the U.S. Treasury setting considered in their study. In particular, Bunds pay annual rather than semiannual coupons, and coupon schedules are often irregular, especially for earlier issues and bonds with long first or last coupon periods. We perform the estimation day by day using the volume-weighted average synchronized price of each bond, with pricing errors weighted by the inverse squared product of the bond's duration and dirty price, so that the fit approximately minimizes yield errors.

In choosing the three hyperparameters of the kernel ridge regression—namely the kernel decay parameter α, which governs the smoothness of the discount curve, the kernel shift parameter δ, which allows for short-end flexibility, and the regularization parameter λ (ridge), which controls overfitting—we follow the cross-validation approach of Filipović et al. (2025). We apply a 10-fold cross-validation scheme on each evaluation date, where bonds are stratified into maturity buckets (<0.25y, 0.25–1y, 1–2y, 2–3y, 3–4y, 4–5y, 5–7y, 7–10y, 10–20y, >20y) to

ensure an approximately equal representation of maturities across folds. All cross-validation is conducted on the last trading day of each month, yielding 216 evaluation dates from January 2008 to December 2025. For each evaluation date, we randomly partition the cross-section of bonds without replacement into ten non-overlapping folds, constructed such that each fold replicates the overall maturity distribution of the sample; nine folds are used for estimation and the remaining fold for evaluation, and this procedure is repeated ten times so that each bond is used exactly once for out-of-sample evaluation.

The hyperparameter grid consists of 100 (α, δ) combinations and 9 ridge values, for a total of 900 candidate specifications: $\alpha \in \{0.01, 0.02, \ldots, 0.10\}$, $\delta \in \{0, 10^{-4}, 5 \cdot 10^{-4}, 10^{-3}, 5 \cdot 10^{-3}, 0.01, 0.05, 0.1, 0.5, 1\}$, and $\lambda \in \{0.001, 0.01, 0.1, 0.5, 1, 5, 10, 50, 100\}$. Model performance is evaluated using the root mean squared pricing error (RMSE) in terms of prices (not yields), averaged across dates and folds. The optimal parameter set is $(\alpha, \delta, \lambda) = (0.02, 0, 1.0)$, with a mean RMSE of €0.205 per €100 of face value. The criterion is flat near this optimum: the ten best specifications lie within 0.4% of its RMSE. Relative to the baseline parametrization $(\alpha, \delta, \lambda) = (0.05, 0, 1.0)$ that Filipović et al. (2025) use for U.S. Treasuries, the mean RMSE is 1.4% lower, a modest improvement that holds on 59% of the evaluation dates.

Appendix C Descriptive Statistics and Cross-Checks

Since our transaction-level dataset is new to the literature, this section provides some descriptive statistics and cross-checks with other established datasets.

Figure A8 compares execution volumes, because a MiFID I record is one execution while a MiFID II execution can span several records (Section D.8). MiFID II covers about 6.2 times as much annual trading volume in Bunds as the earlier MiFID I dataset. MiFID II covers at least as much volume as the survey-based dataset of the Finanzagentur, Germany's debt management office, to which each member of the auction group reports its own trades (Branger et al., 2024): over the 80 months both datasets cover in full, MiFID II execution volume is 1.04 times the Finanzagentur total. Because each member reports its own side, a trade between two members may enter the Finanzagentur total once or twice; if twice, the ratio understates the coverage of MiFID II. Figure A8 compares the monthly series. A key difference is that the FA dataset covers only trades involving a primary dealer. Unlike our dataset, it misses trades between European non-dealers but may cover trades between two non-European investors, e.g. a trade between a U.S. branch of a German dealer bank and a U.S. investment fund.

Figure A9 shows an abrupt change in the relative shares of venue types in 2018. Some of this reflects transactions classified as OTC under MiFID I receiving more specific venue labels under MiFID II. The large increase in aggregate volumes, however, suggests that the primary driver is the expanded reporting of previously unobserved MTF, OTF, and SI transactions.

Figure A10 shows that Bund trading activity follows a distinct intraday pattern, with low volumes before 9 a.m., a midday lull between noon and 2 p.m., and a surge in trading around 5 p.m. followed by a sudden stop at 6 p.m. In the 2008–2017 sample, there is also a large spike shortly before noon.

Figure A11 shows the wide dispersion of transaction sizes. About 6% of MiFID II executions are smaller than €1,000, while 11% are €10 million or larger (MiFID I: 2% and 10%).

Figure A12 compares daily Bund yield curve fitting errors with the U.S. Treasury noise measure of Hu, Pan, and Wang (2013). The Bund series are similar in magnitude and time-series variation to U.S. Treasuries.

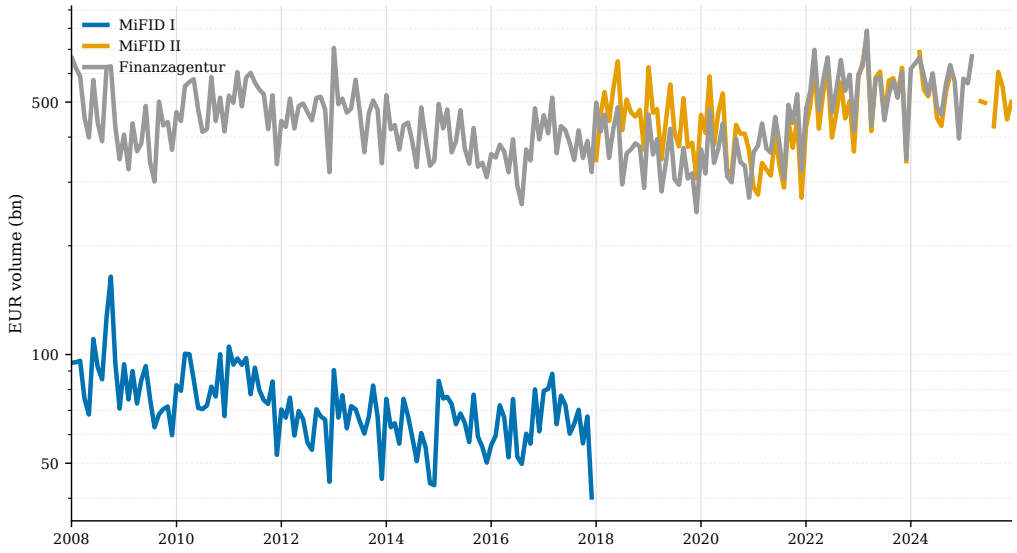


Figure A8: Monthly trading volume in MiFID and Finanzagentur data. MiFID volumes are execution volumes. MiFID II months that lose days without account owner information are left out (Section D.2).

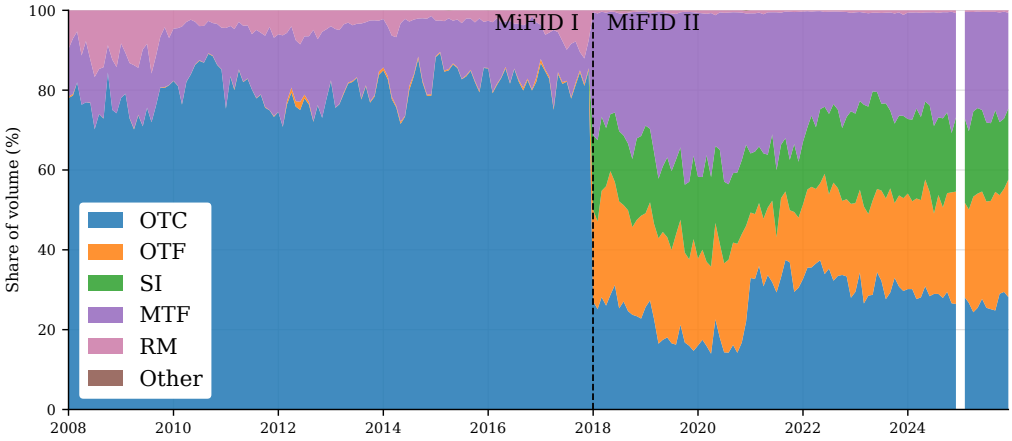


Figure A9: Volume shares across venue types. Shares of execution volume; an execution is assigned the venue type of its most specific venue record.

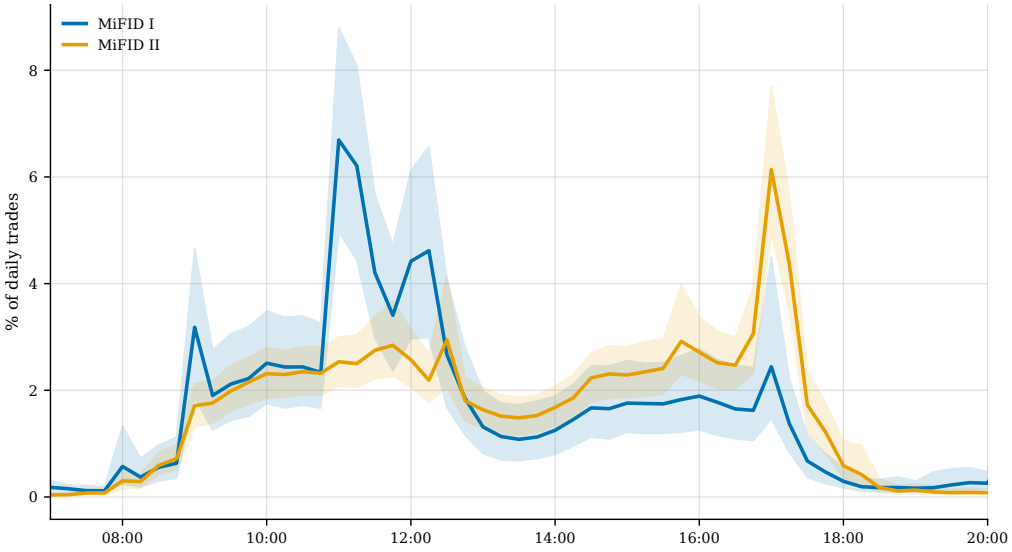


Figure A10: Number of trades across the day. Median and interquartile range across days of each 15-minute interval's share of the day's executions.

Additional market structure descriptives Securities holdings statistics already show who *holds* Bunds; what they cannot show is who *trades* them, with whom, and from where. A non-anonymous dataset linked to the GLEIF register reveals where the counterparties that trade Bunds are domiciled—a first-order question for a sovereign issuer.[24] We map each counterparty to its country of domicile and group countries into Germany, the rest of the euro area, the United Kingdom, the United States, and the rest of the world.[25] Figure A13 shows that Bund trading is predominantly cross-border: domestic German counterparties account for only 10% of trade-leg volume, and only 15% of client-side volume once primary dealers and other intermediaries are

[24]The sample covers reportable EU MiFIR transactions. U.K. and other non-EU venue activity is undercounted, especially after the reporting perimeter narrowed in 2021 (Figure A19). Domicile comes from the investor group's ultimate parent, but country shares still depend on this coverage.

[25]Country is assigned at the investor-group level from GLEIF, i.e. the domicile of the group's ultimate parent; a region is mapped for 83% of trade-leg volume. The unit is the trade leg (buyer and seller side), so a cross-border trade contributes to both regions.

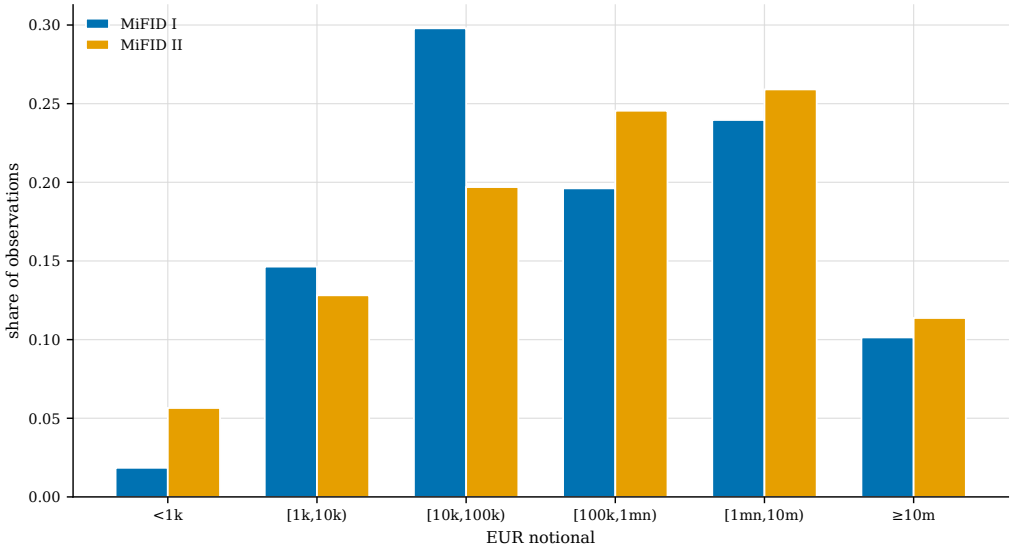


Figure A11: Distribution of transaction sizes. Share of executions by execution size in euros.

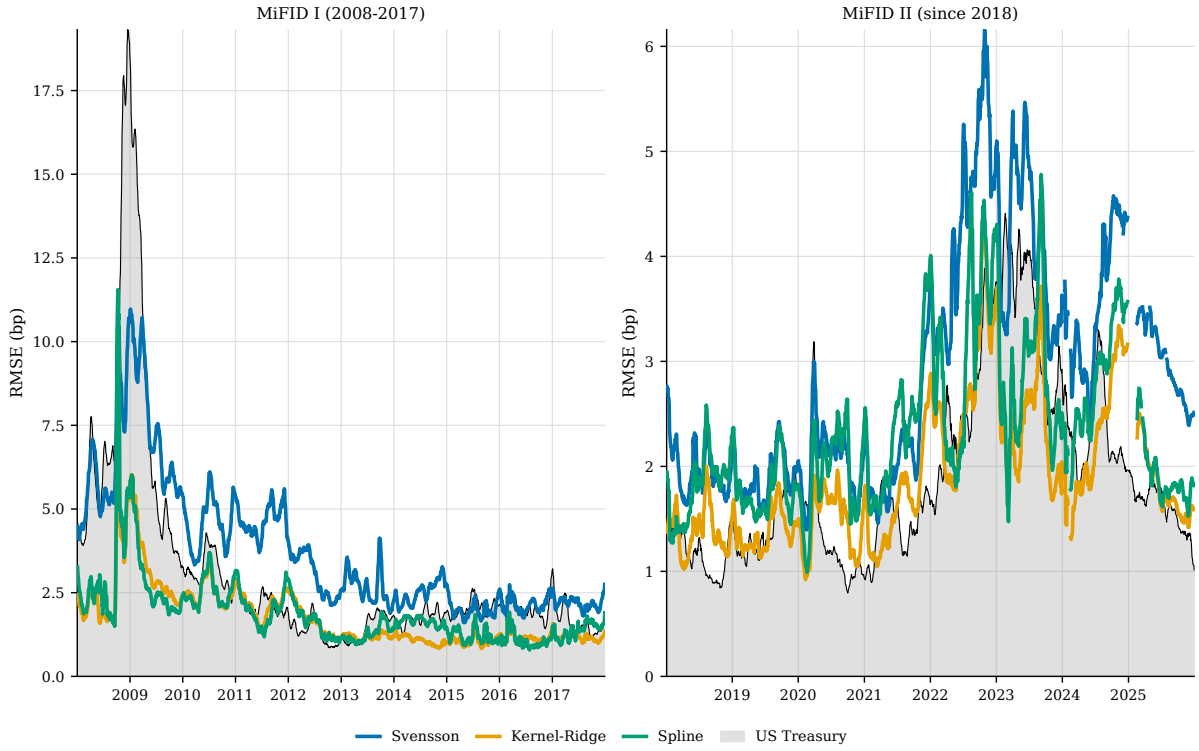


Figure A12: Bund yield curve fitting error over time. Root mean squared error (RMSE) between volume-weighted implied yields and three fitted curves: the kernel ridge curve of Filipović et al. (2025), the Svensson (1994) curve published by the Bundesbank, and a smoothed cubic spline. For each trading day, the RMSE is computed across all Bunds with at least three months to maturity and with observed cash market transactions. Section 3.5 provides details. For reference, the gray shaded area shows the U.S. Treasury noise measure of Hu et al. (2013) based on end-of-day quotes from CRSP, updated by the authors. They also fit yield curves via Svensson (1994), but in contrast to our Bund measures, they exclude bonds with less than one or more than ten years to maturity and bonds whose yield is four standard deviations away from the model-implied yield. All series are shown as 14-day moving averages.

excluded. The two views differ tellingly: excluding intermediaries, the United States share falls and the rest-of-world share rises sharply, so part of the headline United States footprint is dealer intermediation, not end-investor activity—a distinction between trading and ultimate ownership that transaction data can draw and holdings statistics cannot. Figure A13 and Figure A14 report the full regional split and the region-by-sector composition.

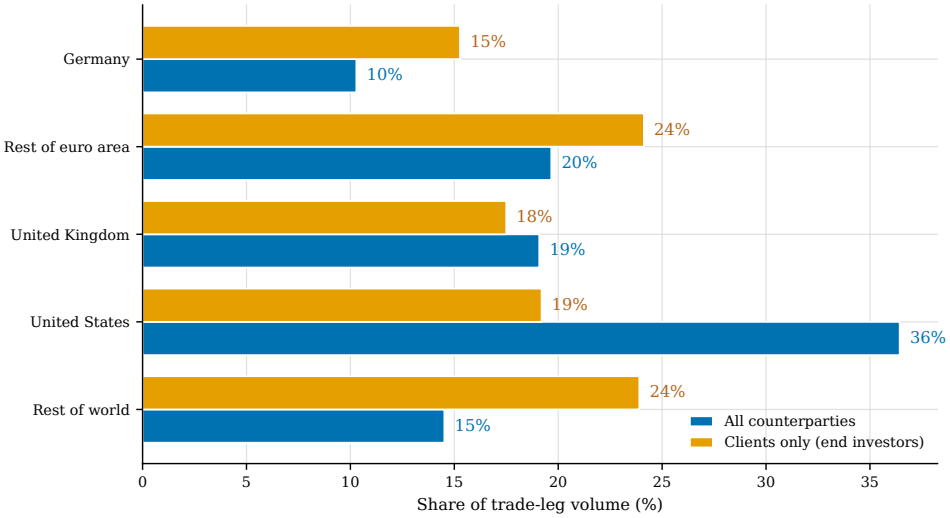


Figure A13: Cross-border participation. Share of trade-leg volume by counterparty region. Dark bars use all counterparties; light bars restrict to client legs, excluding primary dealers and other intermediaries to approximate end-investor geography. MiFID II subsample, investor-group (ultimate-parent) level.

In the all-leg view of Figure A13, the rest of the euro area accounts for 20% of trade-leg volume, the United Kingdom for 19%, and the United States for 36%; in total, 90% of activity involves a non-domestic counterparty. Restricting to client legs, 24% of client-side volume comes from the rest of the euro area, 18% from the United Kingdom, 19% from the United States, and 24% from the rest of the world—85% foreign in total.

Figure A14 reports how the end-investor sector mix varies across counterparty regions. Foreign official sector institutions and banks dominate some non-euro-area participation, while hedge fund activity is more concentrated in the United Kingdom and United States and euro area participation is more heavily weighted toward investment funds.

Dealer breadth and flow informativeness Is a client's dealer breadth a proxy for how informed its flow is? Figure A15 bins institutional clients by their number of distinct dealer counterparties and estimates, for each bin, the permanent price impact of the bin's net flow, exactly as in the sector-level test of Figure 13. In the cross-section the answer appears to be yes: single-dealer clients' flow carries no signal, while flow from the best-connected clients predicts permanent richening (Figure A15a). The gradient is mostly sector composition. The best-connected bins load increasingly on hedge funds and official sector accounts, and once these two sectors are excluded the gradient disappears except in the most connected bin, which keeps a smaller but significant impact (Figure A15b); within the remaining sectors, clients with fewer than ten dealers carry no detectable signal. In the Bund market, cross-sectional connectedness thus proxies for informedness mainly through sector composition: the informed sectors are the broadly connected ones. This complements Kondor and Pintér (2022), who propose a client's dealer connections as a proxy for private information but identify the effect from *time variation* within a given client, absorbing client fixed effects, not comparing connectedness levels across clients. Our cross-sectional result is consistent with that design choice, and it cautions against

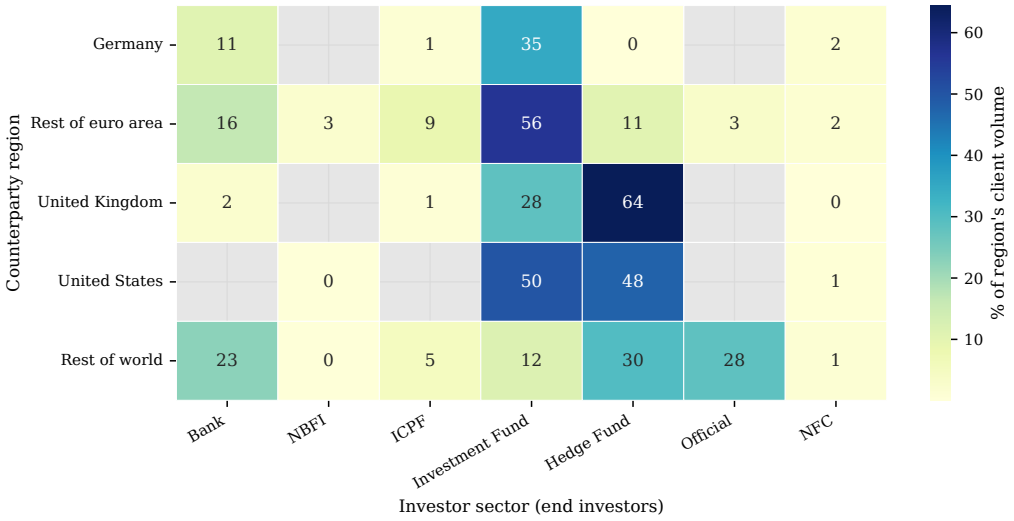


Figure A14: End-investor composition by region. Each cell shows a sector's share of the region's total client-side trade-leg volume (in percent). Intermediary sectors (primary dealers, interdealer brokers) and retail—structurally absent as account owners or without a legal entity identifier—are excluded. Gray cells are omitted; visible cells therefore need not sum to 100% within a row. MiFID II subsample, investor-group level.

reading the level of a client's dealer breadth as a sophistication index without conditioning on sector. Implementing their within-client design directly—regressing a client group's monthly trade performance on the number of dealers it faces that month, with client and month fixed effects, estimated separately before and after the 2021 reporting-perimeter break—we find no within-client effect either: the connections coefficient is small at every horizon and not distinguishable from zero at the five-day horizon (0.004 bps per dealer, $t = 0.9$), precisely enough estimated to rule out an effect of the magnitude documented for Gilts, and equally absent among the 159 largest client groups that account for 80% of institutional volume. In the Bund market, dealer breadth thus indicates who a client is, not how informed its trading currently is.

Whether post-crisis regulation has eroded dealers' willingness to warehouse risk is a central but hard-to-test question, because it requires observing dealers' trading capacity (He and Krishnamurthy, 2013; Duffie et al., 2023). The MiFIR trading capacity field lets us see it. Figure A16 decomposes dealers' volume with identified clients into principal (the dealer trades on own account and warehouses risk), matched principal (a riskless, simultaneous offset), and agency. The field describes the firm that files the record, so we use the records that dealers report themselves (85% of dealer-to-client record volume); on the others it describes the client. Principal trading dominates. Its share averages 97% in 2018–2020 and 93% from 2021 on, with a dip in 2022 and 2023, when the agency share peaks; it is 93% in 2025. The agency share rises from 2% to 5%, and matched principal never exceeds 3%. U.K. venues left the reporting perimeter in 2021, so we do not read the difference between the two periods as a change in dealer behavior. The shares describe how dealers book client trades, not the size of their inventories. They exclude records whose non-dealer side is empty, where the dealer faces an anonymous venue, a CCP, or an IDB (41% of dealer-to-client record volume): nearly all of the matched principal volume on these records belongs to executions that contain no dealer-to-client record with an identified client.

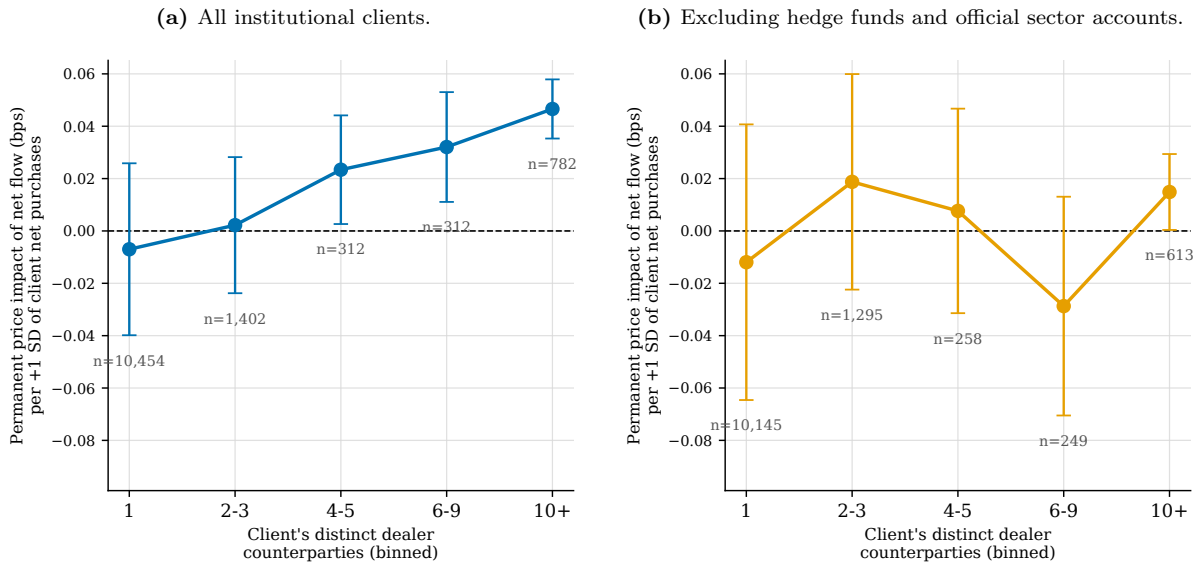


Figure A15: Permanent price impact of client net flow, by dealer breadth. Institutional client groups are binned by their number of distinct dealer counterparties. For each bin, the figure reports the ten-trading-day change in the curve-relative pricing error, defined as synchronized implied yield minus the kernel ridge fair yield, in bps, per net purchase of the bond equal to one standard deviation of the bin's absolute net purchases, signed so that upward denotes richening. Bin-level regressions absorb ISIN and maturity bucket×day fixed effects and control for the bond's lagged curve-relative pricing error and its two lagged daily changes; error bars are 95% ISIN-clustered confidence intervals. The left panel uses all institutional clients; the right panel excludes hedge funds and official sector accounts, on the same y-axis. Text labels report the number of client groups in each bin. Institutional client groups (valid legal entity identifier; retail clients and clients in the primary dealer sector excluded), dealer-to-client trades, MiFID II subsample.

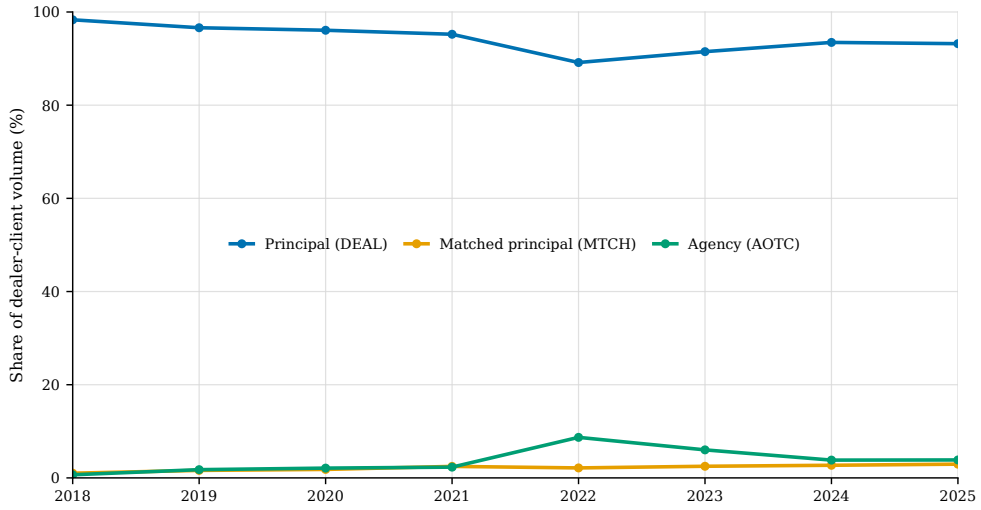


Figure A16: Dealer trading capacity over time. Annual share of dealer-to-client record volume traded in a principal (`DEAL`), matched principal (`MTCH`), or agency (`AOTC`) capacity, from the MiFIR trading capacity field. A record counts as principal only when it is reported as `DEAL` and the dealer is its buyer or seller. Dealer-to-client records reported by the dealer, with an identified client outside the primary dealer sector; MiFID II subsample.

C.1 Cash and Futures Trading Activity

To put the hypothetical scenario of centralizing the OTC market (Section 6.2) into perspective, Figure A17 compares cash market trading activity under the current fragmented structure with a counterfactual centralized market. The left panel shows trading volumes, the right panel trading frequency. For reference, both plots contain the equivalent activity measures for the four Bund futures.[26]

The figure highlights a stark scale difference: the futures market would dwarf the cash market even under full centralization. This suggests that the welfare and liquidity implications of venue centralization cannot be assessed in isolation from the cash–futures nexus. Bund futures embed institutional features that facilitate very high trading intensity—low initial margins, daily variation margining, and cross-product margining—so that a substantial share of price discovery and risk-sharing may be absorbed by the derivatives market.

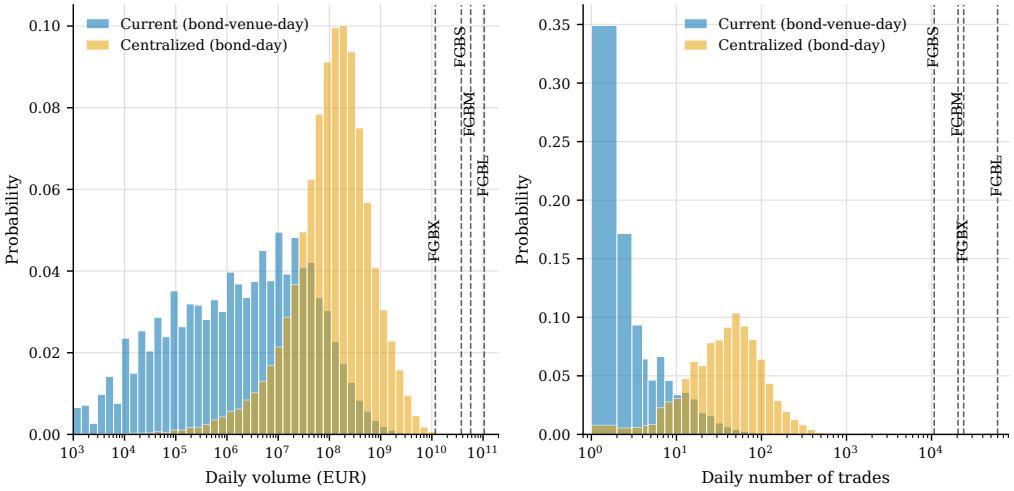


Figure A17: Trading activity in Bunds. The left histogram compares execution volumes in the current market setting (bond-venue-day cells) with a counterfactual fully centralized market (bond-day cells). The right histogram reports the number of executions instead of volume. In both panels, vertical dashed lines show the corresponding median values in the four major Bund futures.

C.2 Market Structure Dynamics

Because the dataset spans eighteen years and two reporting regimes, it can also document how the market's structure has *changed*. We highlight several dynamics, flagging where a shift reflects evolving data coverage, not economics.

Dealer concentration is cyclical, not secularly rising. Figure A18 plots the inverse-HHI effective number of dealers in dealer-intermediated volume. Within the full-coverage MiFID II window, concentration is far from monotone: the effective dealer count peaks at 16.8 in 2019, falls to 12.7 in 2023 during the inflation and rate-hike regime shift, and recovers to 16.3 by 2025. This is consistent with intermediation concentrating in the largest dealers when conditions tighten, not a secular decline in the number of active intermediaries.

[26] While there are three different contracts available for each Bund future, almost all trading is concentrated in the nearest-to-maturity contract, except for a few rollover days each quarter.

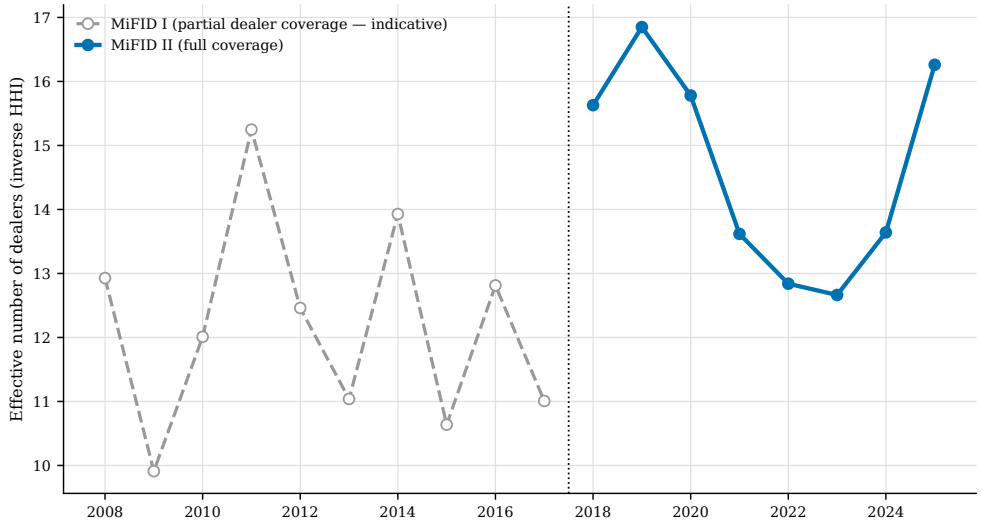


Figure A18: Effective number of dealers over time. Annual inverse-HHI effective number of primary dealers in intermediation volume, $1/\sum_j s_j^2$, where s_j is dealer j's share of primary dealer trading volume, counting each dealer's purchases and sales in all trades in which a primary dealer participates on either side. Lower values mean that intermediation is more concentrated among fewer dealers. The dotted vertical line marks the 2018 MiFID II data break. The MiFID I segment (2008–2017) is shown for completeness but is not directly comparable: that regime is dealer-side weighted and misses roughly eight active primary dealers—including five U.K.-reporting dealers absent throughout—which mechanically lowers the measured effective number.

The end of the Brexit transition reshaped venue coverage. Because the data separate *where* a trade executes from *who* trades, they reveal a sharp break at the end of the Brexit transition period (1 January 2021), when U.K. trading venues left the EU-MiFIR reporting perimeter. Figure A19 shows the share of Bund volume executed on U.K.-registered venues collapsing from about 64% before 2021 to 15% after. Much of this flow migrated to EU-registered venues, whose share rose from about 13% to 52% as operators such as Bloomberg, Tradeweb, and MarketAxess opened EU platforms. But the migration is incomplete: total on-venue volume fell from 79% to 69%, the remainder moving off-venue or out of view. The share of volume with a U.K.-*domiciled* counterparty fell far less (from about 24% to 16%), so the break was largely a relocation of trading *infrastructure* and a contraction of reporting scope, not an exit of U.K. participants. We therefore read the post-2021 venue composition as a change in coverage, not in economics, and treat Bund trading on U.K. and other non-EU venues as undercounted thereafter.

The return of the retail investor. The panel also captures a sharp swing in household participation. Figure A20 plots retail investors' share of Bund trades and of volume over the MiFID II period. Trades here are trade records, and a bank that aggregates household orders reports one record per client allocation. After ECB policy rates turned positive in mid-2022, retail's share of trade records jumped from about 11% to a peak of roughly 49% in 2024—more than a fourfold increase—as households returned to positively yielding Bunds after a decade of near-zero rates. Without the allocations of aggregated orders the share also rises more than fourfold, from 4% to 19% in 2023. Yet retail's share of volume stayed negligible (around 0.2%): the surge is in the number of trades, not in euros, reflecting very small ticket sizes. Retail trading also leans on regulated exchanges—about 36% of retail trades outside aggregated orders execute there (11% of all retail records), far more than for investment funds or insurers—so the episode lifted exchange trade counts without materially affecting market-wide liquidity or pricing.

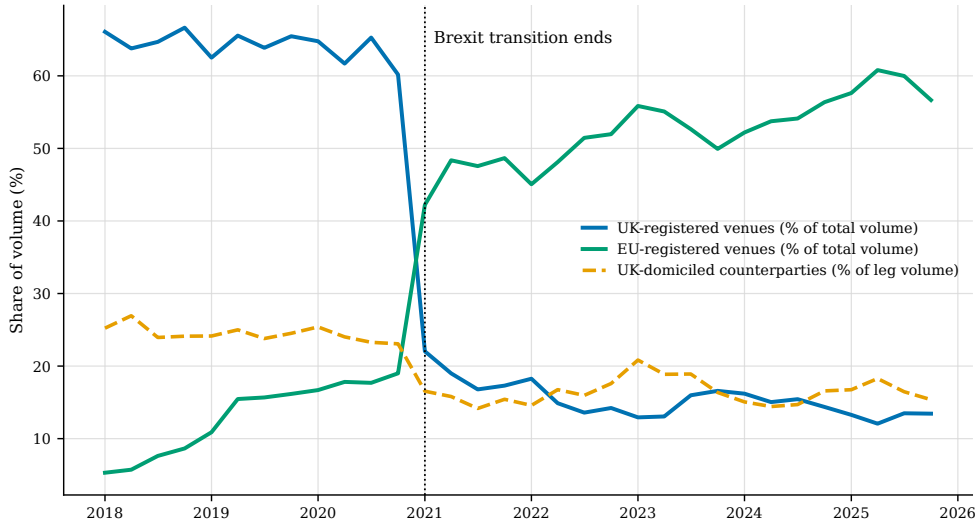


Figure A19: Venue and counterparty shares around the Brexit transition. Quarterly shares of execution volume on U.K.-registered and on EU-registered venues (MTF/OTF/SI/RM), and the share of trade-leg volume with a U.K.-domiciled counterparty (dashed). The dotted line marks the end of the Brexit transition period (1 January 2021). MiFID II subsample.

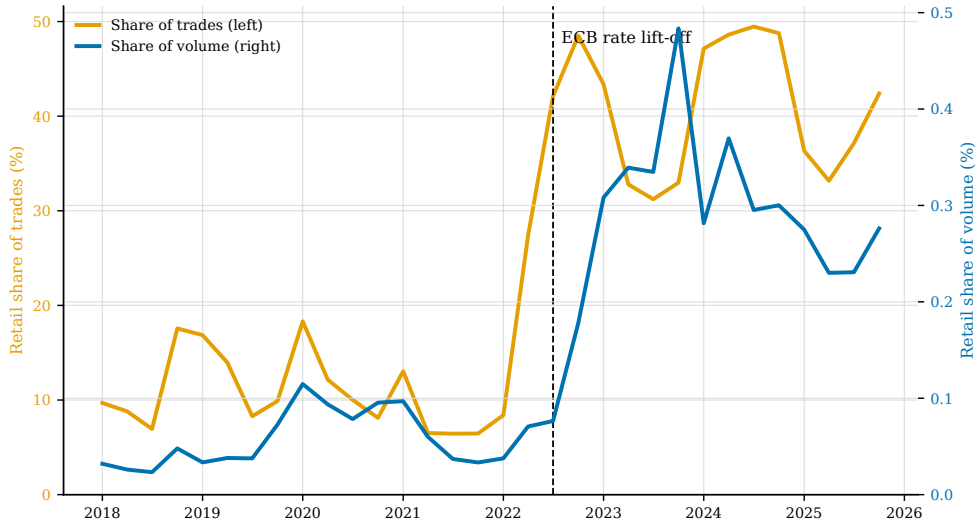


Figure A20: Retail participation over time. Quarterly share of Bund trade records, counting each client allocation (orange, left axis), and of trading volume (blue, right axis) with at least one retail (household) counterparty. The trade-count and volume series use separate axes. The dashed line marks the ECB's first policy-rate increase (July 2022). MiFID II subsample.

Algorithmic execution rose, but humans retook the wheel in stress. The MiFIR "execution within firm" field records whether each trade was executed by an algorithm or by a person, letting us track the machine's advance directly.[27] Figure A21 shows algorithmic execution climbing from roughly 25% of trades in 2018–19 to about 29% by 2025. Measured by volume the share is markedly lower—rising from about 10% to 19%—confirming that algorithms execute disproportionately small tickets while large blocks remain human-handled. The flag describes the firm that reported the record: among records filed by dealers the algorithmic share of trades moves from 42% to 46% over the sample, so the advance is concentrated in the records that clients file. The split by venue protocol family in Table A4 shows that algorithmic execution is concentrated in, but not confined to, order book venues: algorithms account for 55.1% of volume within the order book family, but also for 25.0% within the RFQ family and 10.7% within the residual other family. The advance is also not monotone. Around the March 2020 Covid dislocation the algo share fell from a 29% prior-six-month baseline to 24%, and around the September 2022 Gilt/LDI turmoil from 32% to 21%, as execution reverted to human traders when liquidity was thin; the March 2023 banking stress saw no retreat (27% against 23%). The pattern echoes the man-versus-machine evidence from Gilts (Guimaraes, Jurkatis, Pintér, and Wijnandts, 2025): algorithms have steadily gained share, but at the height of stress humans still step in.

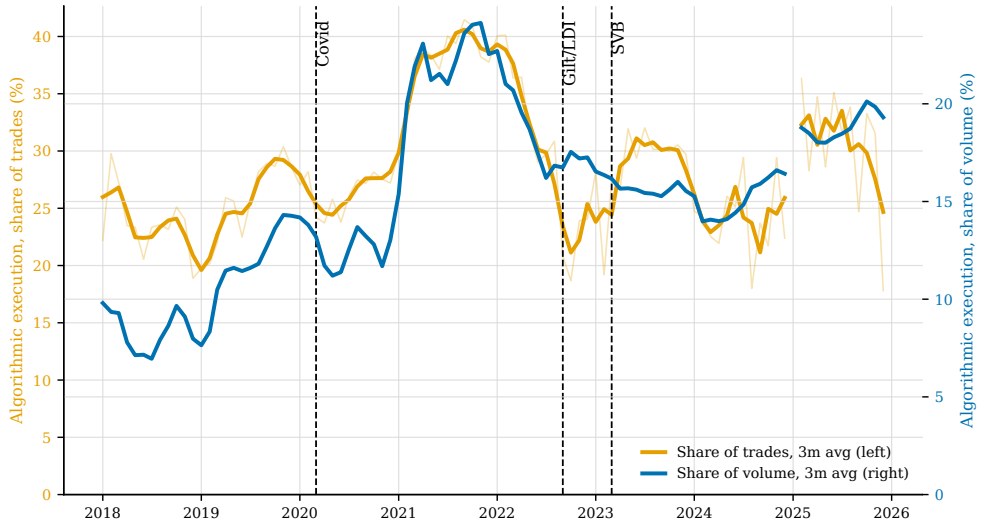


Figure A21: Algorithmic execution over time. Monthly share of Bund trades executed algorithmically (MiFIR "execution within firm" field), as a share of trade count (orange, left axis) and of volume (blue, right axis); both three-month centered averages, with the raw monthly count share in faint orange. Dashed lines mark three stress episodes. MiFID II subsample.

[27] The MiFIR "execution within firm" field (RTS 22) is populated on essentially every MiFID II trade record (100%) and distinguishes algorithmic (`ALGO`) from human (`PRSN`) or client-directed (`CLNT`) execution. Shares are computed on trade records as reported (Section D.8).

Venue protocol family	ALGO	PRSN	CLNT
RFQ family	25.0	72.2	2.7
Other (residual)	10.7	80.0	9.4
Order book (CLOB)	55.1	37.3	7.6
Unknown	11.0	83.7	5.3

Table A4: Execution type by venue protocol family. Each row reports execution-type shares of a venue protocol family's euro volume (row-normalized percentages) among trades with a populated MiFIR "execution within firm" field. MiFID II subsample.

Appendix D Data Details

Article 26 of Regulation (EU) No 600/2014 (MiFIR) requires investment firms to report complete and accurate details of each transaction to their national competent authority by the close of the following business day. For German sovereign bonds, the competent authority is BaFin irrespective of trading venue, so the data provide near-complete coverage of the Bund market. Because MiFIR was not designed for research use, the reports contain duplicated records, heterogeneous formats, and reporting errors. We follow the framework of Jurkatis (2024) for cleaning MiFID transaction data on corporate bonds, adapting it to the Bund market—in particular its far higher trading frequency: a single ISIN can generate up to 65,497 reports on a single day, making duplicate identification substantially more demanding.

Data availability Access to the non-anonymous MiFID I dataset can be requested via the Research Data and Service Centre (RDSC) at the Bundesbank and is subject to strict requirements. MiFID II/MiFIR transaction reports are not available to external researchers through the RDSC. We will maintain the final research dataset and its documentation as a reusable resource for Bundesbank researchers whose projects are authorized to use the underlying data. All other data are available directly from the data vendors.

Pipeline overview This Appendix describes the MiFID II cleaning pipeline in full. The MiFID I pipeline runs in parallel but differs in several respects. First, MiFID I records carry regime-specific administrative flags—regulatory exclusion codes, administrative correction markers, and name-to-follow placeholder legs—that are screened at the outset and have no MiFID II counterpart. Second, trading capacity is derived from proprietary reporting fields rather than an explicit DEAL/MTCH flag, so the report-splitting and aggregation steps described in Section D.3 do not apply. Third, deduplication applies the same cross-reporter record-linkage matching to proprietary (DEAL) pairs—though one-sided MiFID I reporting yields far fewer such duplicates—and adds intra-reporter matched principal pairing and venue-consistent cross-capacity matching. Fourth, investor identity is resolved via a BaFin-ID-to-LEI crosswalk, followed by the LEI enrichment. Both pipelines produce an identical output schema; all downstream processing is applied uniformly to the combined dataset. Figure 1 in the main text summarizes the construction: the raw reports from both regimes pass through the cleaning steps in this Appendix and the cleaned records are then enriched with bond-, investor-, venue-, and price-level reference data.

D.1 Reporting Requirements

MiFID I regime (2008–2017) Prior to MiFIR, transaction reporting in Germany was governed by MiFID I, implemented nationally via the Wertpapierhandelsgesetz (WpHG). Investment firms were required to report transactions in financial instruments to BaFin using BaFin-assigned institutional codes (*BaFin-IDs*) rather than standardized global identifiers such as LEIs. Reporting obligations were primarily one-sided: for a bilateral OTC trade, typically only the selling or executing firm submitted a report. This limits bilateral duplicate detection, which is a central step in the MiFID II cleaning pipeline; we adapt the deduplication logic accordingly. A second convention in MiFID I affects trade direction: the `g_art` buy/sell flag records the reporting party's own direction for proprietary trades but the client's direction for matched principal and agency client business. The pipeline therefore conditions on trading capacity and dealer status when constructing the buyer and seller account owner fields (`bao` and `sao`), swapping the two when the reporting party is the dealer side of a client trade. We resolve BaFin-IDs

to LEIs by combining a BaFin registry lookup, BIC-to-LEI resolution via RIAD, hand-curated assignments for major institutions, and exact and fuzzy name matching against the GLEIF register; no single source achieves full coverage. The resulting map covers more than 99% of MiFID I trade volume.

MiFID II/MiFIR regime (2018–2025) Under MiFIR, which took effect on 3 January 2018, each investment firm must independently report every transaction it executes, identified via LEI. Each report contains the instrument identifier, execution timestamp, price, quantity, venue, trading capacity, and counterparty identities. Because each firm reports its own transactions independently, a bilateral trade between two investment firms generates two reports of the same economic transaction, one from each side. A trade between an investment firm and a client that is not an investment firm, such as a fund, generates a single report. On an anonymous venue, each member reports its own side against the venue or its CCP, so one execution generates two one-sided reports. Trading venues report on behalf of unregulated participants. These asymmetries determine which trades appear once versus twice in the raw data and drive the duplicate-matching strategy in Section D.4.

D.2 Sample Selection

The first step extracts the relevant universe of Bunds. We collect the ISINs of all plain-vanilla German sovereign bonds from Finanzagentur and then filter the MiFIR data for these bonds. We exclude short-term Bubills, green bonds, inflation-linked bonds, and other non-standard bonds. We also exclude three groups of flagged reports. Securities financing transactions account for less than 0.2% of reported Bund volume in 2018 and 2019 and for 6–21% per year from 2020 on. From July 2020 on, they are mostly repos and securities loans between primary dealers and the official sector, which MiFIR covers when the counterparty is a central bank of the European System of Central Banks. Legs of complex trades, such as switches, curve trades and basis trades, account for 4–9% of reported Bund volume per year. We exclude them because their reported price can be the price of the whole package, not of the bond. Reports flagged as commodity derivatives or as changes in derivative notional account for less than 0.2% of volume. The MiFID II/MiFIR reports are as of 21–22 September 2026; the source keeps receiving new reports, corrections and cancellations, so a later extraction can differ.

When an instrument is bought or sold by a joint account, all account owners must be reported (see ESMA Guidelines, p. 42). Thus, one transaction can have multiple reported buyers or sellers. In 2019, 86.5% of the owners of joint accounts were reported as LEIs, 0.2% as natural persons, and the remaining 13.3% as internal accounts (INTC) or other identifier types.

Each account owner is assigned a "position", numbered from one to the number of owners of the account. To avoid double counting transactions involving joint accounts, we merge only the first owner (position = 1) into each transaction record. This preserves trade volumes but drops the other owners of a joint account.

In 2019, 0.06% of transactions involved a joint account, and merging all positions to the transaction data would increase the overall number of transaction records by 0.09%. Keeping only the first owner therefore loses almost no information.

The same rule applies to buyer and seller decision makers, although at a much smaller scale. In 2019, only 15% of transaction reports include a buyer or seller decision maker. As before, we include only the first decision maker for each transaction. We remove reports where the executing party is recorded as both the buyer and the seller.

On 64 trading days in 2024 and 2025, at least 20% of the MiFID II reports in our sample carry neither a buyer nor a seller account owner. On these days the account owner information

is missing at the source; the data owner has confirmed these data gaps, and their cause is still being investigated. Without account owners, the counterparties of these reports cannot be identified, and each report enters the data as two records without a counterparty (Section D.3). We therefore drop these days from all analyses. They hold 14% of the 2024–2025 records and include every trading day from 2 January to 5 February 2025, so monthly series have no observation for January 2025. Reports without account owners also occur on other days; they stay in the sample as records without a counterparty.

The raw extraction yields approximately 16.9 million MiFID II reports over 2018–2025 and 2.5 million MiFID I reports over 2008–2017. After the cleaning steps described in the remainder of this Appendix and without these days, these collapse to approximately 14.2 million trade records for 137 bonds and 2.2 million trade records for 146 bonds, respectively. The combined sample covers 16,350,195 trade records in 224 bonds spanning 2008–2025.

D.3 Normalization

The normalization step prepares the data for analysis by handling intermediary trades, aggregating reports, and identifying outliers in prices and quantities. These steps keep the dataset aligned with executed transactions rather than reporting artifacts. The statistics we report on these steps and on duplicate matching refer to the reports of 2019.

Splitting Reports

Some transactions are reported in a format where the executing party is neither the buyer nor the seller but acts as an intermediary. To standardize the dataset, we split these intermediary reports into two separate records—one where the executing entity is recorded as the buyer and another where it is recorded as the seller. This gives each transaction a uniform reporting structure.

Under MiFIR, firms are required to report the trading capacity of the transaction, that is, whether the firm was trading on its own account (`DEAL`), as matched principal (`MTCH`), or in any other capacity (`AOTC`). The latter two cases should list the executing party only as an intermediary while buyer and seller are distinct from it. Table A5 shows how the trading capacity flag aligns with the splitting criteria in this subsection.

cat_trading_capacity	split	%
DEAL	0	64.29
AOTC	1	28.5
AOTC	0	3.34
MTCH	1	3.07
DEAL	1	0.68
MTCH	0	0.12

Table A5: Trading capacity field accuracy.

Table A5 shows that, for 95.9% of transactions, the trading capacity flag aligns with whether the report requires splitting. Where it does not match, we treat the trading capacity flag as misreported and adjust it while retaining the original value. The dealer trading capacity shares in Figure A16 use the reported flag instead: a record reported as `MTCH` or `AOTC` counts as matched principal or agency even when the reporting firm is its buyer or seller, and a split `DEAL` report counts as matched principal. Overall, 32.2% of reports are split.

Aggregating Reports

As Jurkatis (2024) points out, firms sometimes report multiple trades with the same counterparty at the exact same time and price. This makes the duplicate report from the counterparty firm difficult to identify, because the counterparty may report them as a single large transaction. This problem partly arises from report splitting. Assume executing party A reports two matched principal transactions with quantities q_1 and q_2 that it intermediates between executing party B and clients C1 and C2. At the same time, executing party B reports only a single transaction with quantity $q_1 + q_2$ between itself and executing party A. Any duplicate-matching method will struggle to match these two reports because the quantities differ. Because report splitting turns the reports filed by executing party A into four reports, two for its trade with executing party B and two for transactions with A's clients, we can aggregate the former two without loss of information. The quantity of the aggregate report then matches that reported by executing party B, as should all other fields, and the two transactions can be identified as duplicates in the next step.

Around 21% of trades require aggregation. Of these, 84% are reports that were split in the previous step. Aggregation reduces the overall number of reports by around 16%. Table A6 summarizes the trading capacity, venue, buyer and seller account owner types, and trade volume of the aggregated trades. The affected trades are mainly off-venue trades with small volume, often involving a natural person or an internal account.

Variable	Split	% in agg trades	% in all data
Split	1	83.96	48.77
	0	16.04	51.23
Variable	Trading Capacity	% in agg trades	% in all data
Trading Capacity	AOTC	81.01	45.63
	DEAL	14.1	49.64
	MTCH	4.89	4.73
Variable	Trade Venue	% in agg trades	% in all data
Trade Venue	XOFF	59.75	28.84
	BMTF	13.54	27.65
	TREU	13.51	16.34
	Other	13.19	27.17
Variable	Counterparty Type	% in agg trades	% in all data
Counterparty Type	INTC	49.52	12.87
	LEI	44.02	77.25
	PRSN	4.99	7.98
	NaN	1.36	1.66
	MIC	0.11	0.24
Variable	Trade Size (mn EUR)	in agg trades	in all data
Trade Size (mn EUR)	Mean	4.9	5.44
	Median	0.11	0.5

Table A6: Characteristics of aggregated trades.

Not all fields of reports selected for aggregation match. We require the executing party, the buyer and seller account owners, the ISIN, the price, and the timestamp to be the same, and we

sum the transacted quantities. Some firms file one execution twice, under two report identifiers that agree on the timestamp to the nanosecond, the quantity, every reported field and the venue trade identifier. Summing such copies would double the execution, so we keep one of them; copies without a venue trade identifier stay summed and are flagged. For other fields, there is no clear aggregation rule. Table A7 reports the prevalence of disagreement between aggregated reports for each field. Overall disagreement is low and mainly concerns fields of minor interest for our analysis, so we keep the value from the first report in the aggregation group.

Column	Groups with differing Values (%)
eur_quantity	69.59
ind_investment_party	5.56
cat_trade_venue	5.37
id_tx_trade_venue	3.44
cat_short_selling	3.08
cat_exec_type	2.97
cat_trading_capacity	2.63
split	2.03
sdc_buyer_decision_maker	1.48
cat_price_type	1.03
cat_quantity_type	0.94
sdc_seller_decision_maker	0.87
sdc_transmitting_buyer	0.81
sdc_transmitting_seller	0.66
ind_transmission	0.14
cat_buyer_decision_maker_type	0.07
cat_inv_dec_type	0.06
cat_seller_decision_maker_type	0.0

Table A7: Disagreement among aggregated trades.

Price Outliers

We preliminarily flag price and quantity outliers so that they can be corrected using duplicate reports in the next step. If two reports are identified as duplicates and one has an unrealistic price or quantity, we replace the flagged value with the value from the paired report.

The data include a field for the unit in which the price is reported. Prices can be reported as percentages, yields, basis points, or monetary values. We confirm that, regardless of the reported unit, most prices are given as a percentage. Figure A22 reports the distribution of prices for each unit category. We disregard the reported unit and treat all prices as percentages.

We identify price outliers in two steps. First, an initial filter flags transactions with prices outside a reasonable range (below 20 or above 500 as a percentage of par). Next, a rolling median price is computed for each bond over a one-week window, and transactions with price deviations exceeding 50% from the rolling median are flagged as potential errors. If the price multiplied by 100 lies between 20 and 500 and within 50% of the rolling median, we treat it as a decimal misplacement, for example, a price recorded in units of 100 rather than percentage points, and correct it accordingly. This methodology adapts the corporate bond approach to sovereign bonds. In total, 0.2% of reported prices are flagged as outliers.

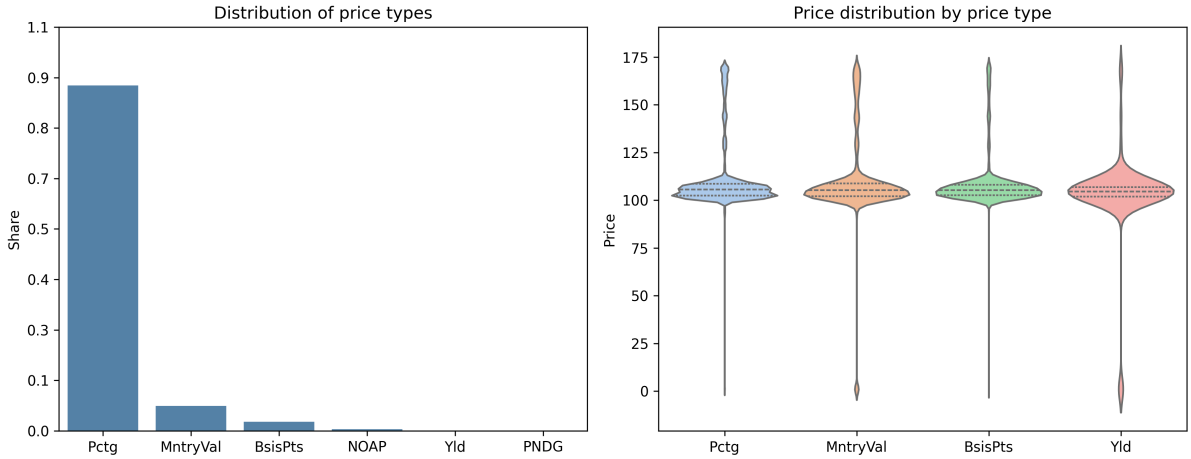


Figure A22: Price distribution by price type.

Quantity Outliers

As for prices, transaction quantities can be reported in different units. These are the nominal value and the monetary value. Again, whatever the reported unit, most quantities seem to be reported as nominal values, and we treat all as such (Figure A23).

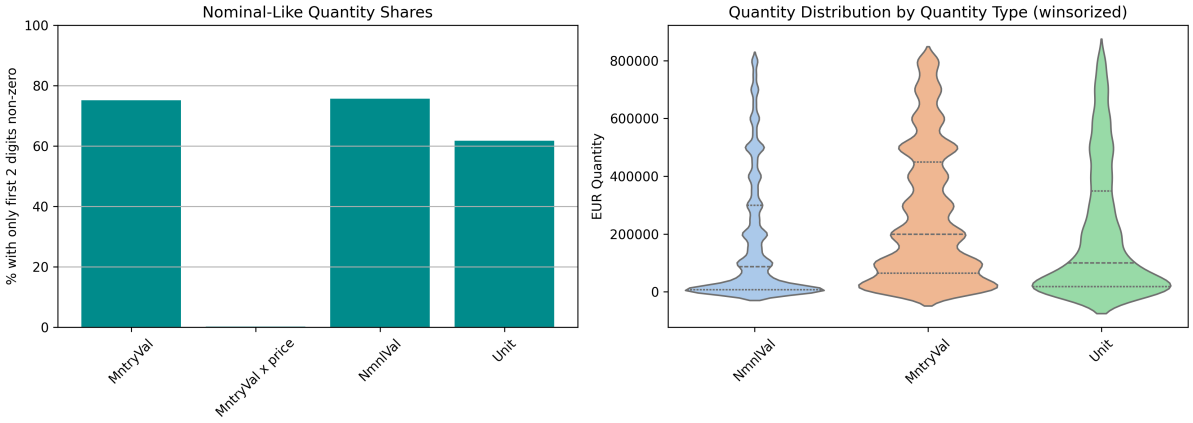


Figure A23: Quantities by quantity type. The left panel reports the share of observations in each category (including monetary value times price trades) that is divisible by a multiple of 10. The right panel reports the distribution of trade sizes for each category for trades below the median trade size.

We identify potential quantity outliers in the same spirit. The first pass compares trade sizes against reasonable thresholds. Extremely small ($\leq$€100) or excessively large ($\geq$€1 billion) trades are initially flagged as potential errors. A more refined pass then considers whether the transaction occurred before or after the bond's issuance date. Trades reported before issuance are compared against the total issuance volume, and those exceeding this limit are marked as outliers. For transactions near issuance, we divide the reported quantity by the amount issued and flag as an outlier any transaction above the 99.5th percentile. Post-issuance, we flag any trade whose size exceeds 5% of the gross amount outstanding or whose size-to-outstanding ratio exceeds the 99.5th percentile across that day's trades. Overall, 0.7% of quantities are flagged as outliers. The threshold choices are deliberately conservative and enter the pipeline as correction

flags, not as a broad deletion rule: flagged observations are first corrected from duplicate reports where possible, and the remaining global price-synchronization filter removes less than 1% of total observations (Section 3.5). The Appendix reports the affected shares at each stage so that readers can see the scale of the threshold-dependent steps.

Reasonable transaction sizes span a wide range, from small retail transactions to large interdealer trades, making clear outliers hard to identify. This preliminary screen is used only to improve duplicate detection; the analysis-specific outlier screen is applied later. These preparatory steps transform the raw data into a structured dataset suitable for further processing and reduce distortions caused by reporting anomalies. The duplicate-matching step then identifies and removes duplicate reports.

D.4 Duplicate Matching

Once the transaction data have been prepared, the next step is to identify duplicate records. Under MiFID II/MiFIR, both counterparties to a bilateral trade are required to report, so some transactions appear twice in the dataset; removing these duplicates is essential to avoid overestimating trading activity. Under MiFID I, reporting was predominantly one-sided, so fewer trades are reported twice. The duplicate-matching step adds two components specific to MiFID I—intra-reporter matched principal pairs reported by a single intermediary, and venue-consistent cross-capacity pairs—alongside the same cross-reporter record-linkage matching applied to proprietary trades, where it requires a total score of at least five and opposite reported directions. The following description focuses on the MiFID II logic; the MiFID I pipeline follows analogous but simpler steps.

We adapt the pairwise matching approach to sovereign bonds. For all trade reports with the same day and ISIN, we check agreement on price, quantity, timestamp, trade venue, buyer and seller, and whether the reports come from different executing parties. Because Bunds trade much more frequently than corporate bonds, computing all possible pairwise combinations of potential duplicates is infeasible. We therefore reduce the duplicate-search sample using the reporting requirements.

First, the reporting requirements apply only to investment firms, not to natural persons. A transaction side that involves only a natural person will therefore not have a duplicate report. This is the case when both the buyer/seller account owner and decision maker are natural persons, or when only the former is reported. We remove all transactions satisfying these conditions from the duplicate-search sample.

We apply the same logic when the buyer/seller account owner is reported as an internal account ("INTC"). As long as this side of the trade does not list an investment firm as account owner or decision maker, we exclude it from the duplicate search. This is possible because the report-splitting step leaves at most two parties to a trade, one of which is the reporting executing party. We therefore need only check whether the other party also reports.

Removing transactions involving a natural person reduces the sample by approximately 7.4%, while dropping internal account trades removes another 4.2%. These trades can occur in large same-time, same-ISIN batches: an executing party may report up to 7,724 trades at the same time for the same instrument. Removing trades that cannot have duplicates keeps the pairwise comparison problem computationally feasible.

Table A8 reports the characteristics of batches with more than 1,000 trades. They typically involve natural persons as counterparties and have small trade volumes. There are 30 such batches with 56,808 trades in total. Dropping trades that cannot have duplicates leaves 2,076 trades in batches of this size. About 31% of the dropped trades are part of a large batch. We use a Python record-linkage package to compute the pairwise comparisons. The package allows

Metric	Batched Trades	All Trades
Number of transactions	56808	1511701
Number of trade batches (>1000)	30	-
Average trade size (EUR)	15,458.55	6,444,532.68
Median trade size (EUR)	2,000.00	792,000.00
Minimum group size	1002	-
Maximum group size	7724	-
Mean group size	1893.60	-
Top counterparty types (%)		
PRSN	98.5%	8.8%
LEI	1.5%	84.8%
INTC	0.1%	4.4%

Table A8: Batched trades.

us to require compared trades to occur on the same date and in the same ISIN, which limits the number of pairs. A second necessary condition is that the executing parties reporting the two trades are distinct. The package does not directly support filtering pairs on the condition that a field differs, so we add this condition in a custom class to reduce both the number of pairs and the memory required for the comparison.

After these reductions, approximately 126 million pairs remain to be compared in 2019.

To identify duplicates among the selected pairs, we apply a set of comparison functions to assess similarity across key fields, including counterparties, trade venue, trade quantity, price, and timestamp. For each field, we assign a score to measure the similarity of the values reported by the two transaction records. This score can be either binary or continuous depending on the field.

Trade timestamps are matched within a 20-minute window, plus a shift of one or two hours. This accounts for possible reporting delays and time zone discrepancies. Although all investment firms are supposed to report in UTC, time zone discrepancies remain a potential source of error. The score assigned for this field is continuous. The score falls linearly from one at an exact match, or at a shift of exactly one or two hours, to zero at a discrepancy of 20 minutes from either.

Trade quantities and prices are checked for exact matches, as well as potential reporting errors involving common scaling factors, for example, misreporting by a factor of 2, 10, 100, or 1,000. Buyer and seller account owners are checked for exact matches. All four fields receive a binary score; a match after rescaling by one of these factors scores one.

Finally, the trade venue is compared. We usually require an exact match, unless one side reports an off-venue trade ("XOFF"). Because trades are split in the first step, the client side of a trade may still carry the trade venue of the market side when the two sides were initially reported as one trade. If the client also reports, the client will report only the client side, which occurs outside any trade venue.

The similarity scores from these comparisons are summed, and pairs with a positive timestamp score and a total above four out of six are candidate duplicate matches. This step differs from Jurkatis (2024), who use a probabilistic approach to filter for the most likely duplicates. Their approach accounts for the unconditional probability that a trade has a duplicate and allows for a more precise target rate of identified duplicates. Our threshold approach gives more direct control over match quality. We optimize the threshold by manually checking trades marginally above and below it. The final matched set is high-scoring: among accepted matches,

the average score is 5.64 out of six, and Table A9 reports field-level agreement rates, so the accepted threshold can be assessed directly rather than only described qualitatively.

The final step employs graph-based matching to ensure that each transaction is linked to at most one counterparty report, preventing duplicate pairings, while maximizing the overall score across all final matches. Figure A24 illustrates this approach.

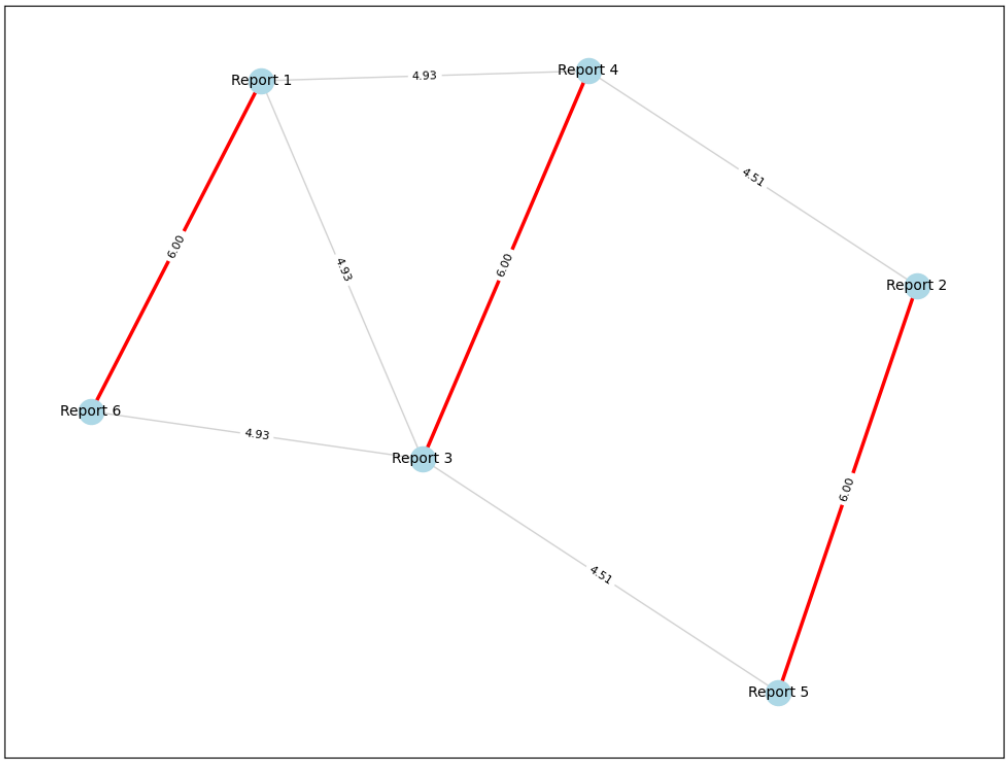


Figure A24: Maximum weight matching.

We find a duplicate for about 53% of all trades, or 60% of the trades included in the duplicate search. The average score of the final match set is 5.64 out of a maximum attainable score of six. Table A9 reports the average score across all matches for each field. Matched trades agree on all fields in 48% of cases, and the field-level agreement rate is at least 88% for every field. Disagreements most often concern buyer or seller information, for example because one side reports the parent company instead of the subsidiary with which it traded. Groups of fifty or more reports filed by the same executing party for the same instrument at an identical timestamp are retained and labeled with their group size. They are the shape MiFIR prescribes for an aggregated client order—one market execution followed by one allocation report per client at the same price—so we treat them as client allocations, not as spurious reports.

D.5 Correcting Reported Entries

After duplicate trades have been identified, the final step in the data cleaning process is to resolve inconsistencies within matched pairs. Errors in prices, quantities, counterparties, and trade venues are corrected using information from both records in a matched pair.

Buyer	Seller	Trade Venue	Quantity	Price	Timestamp	Total Score
0.88	0.88	0.92	1.0	0.99	0.97	5.64
0.0	0.0	0.0	0.0	0.0	0.0	4.0
1.0	1.0	1.0	1.0	1.0	1.0	5.0
1.0	1.0	1.0	1.0	1.0	1.0	6.0
1.0	1.0	1.0	1.0	1.0	1.0	6.0
1.0	1.0	1.0	1.0	1.0	1.0	6.0

Table A9: Match quality. Field scores and total score of accepted matches in 2019. Rows: mean, minimum, 25th percentile, median, 75th percentile, and maximum.

Price and Quantity Corrections

If only one transaction in a matched pair is flagged as an outlier, its price or quantity is replaced with the corresponding value from the non-outlier transaction. If both transactions in a pair are flagged but report the same price or quantity, the outlier flag is removed. Any remaining differences between prices or quantities after correction are recorded for later evaluation.

Counterparty Adjustments

Discrepancies in reported buyers and sellers are addressed by checking whether one transaction misidentifies the counterparty. If one report lists the executing party in the buyer or seller field while the other does not, the erroneous entry is corrected. If both reports list different counterparties inconsistently, they are flagged as unresolvable mismatches.

Trade Venue Standardization

If a matched transaction reports "XOFF" as the execution venue, this value is enforced across both records to maintain consistency. This adjustment ensures uniform classification of off-venue trades.

These corrections refine the dataset by leveraging duplicate reports to resolve inconsistencies, improving the accuracy of transaction-level data. The corrected transaction is therefore not an arbitrary kept side. Price and quantity use the non-outlier report when only one side is flagged; counterparty fields are corrected only when one side mechanically reports the executing party; and venue fields follow the off-venue standardization rule. Remaining disagreements are retained as diagnostics rather than resolved by discretion. Table A10 gives an overview of these remaining differences within a matched pair of transaction reports. Of each matched pair we keep the report with the earlier corrected execution time, then the earlier originally reported execution time. Ties favor a dealer report, then the lower report-identifier hash. Reporter-specific fields describe the retained reporting firm.

D.6 Primary Issuances

The MiFID II data include primary market allocations from Finanzagentur auctions and taps. These records appear as primary dealers buying bonds from the German federal government on auction day. Dealers report either Finanzagentur or the Bundesbank as seller; the latter acts as fiscal agent in the issuance process. Because the analysis targets secondary market trading, we drop observations in which Finanzagentur or the Bundesbank sells a bond on that bond's auction day to a primary dealer. We keep all other trades by Finanzagentur and the Bundesbank, which mainly capture market maintenance trades and the Eurosystem's asset purchase programs.

Field	% Disagreement
Price	1.13
Quantity	6.19
Trade Venue	21.4
Buyer	2.34
Seller	2.4
Timestamp	34.29

Table A10: Remaining disagreement between duplicates.

Figure A25 validates this filter in the raw transaction reports. In MiFID II, cumulative BBK/FA sales to primary dealers rise sharply after the auction close around 11:30 CET and reach most of the auction size by noon. In MiFID I, the same diagnostic is much smaller because some dealers do not report their primary market allocations and about eight primary dealers (including five U.K.-reporting dealers) do not report any trades.

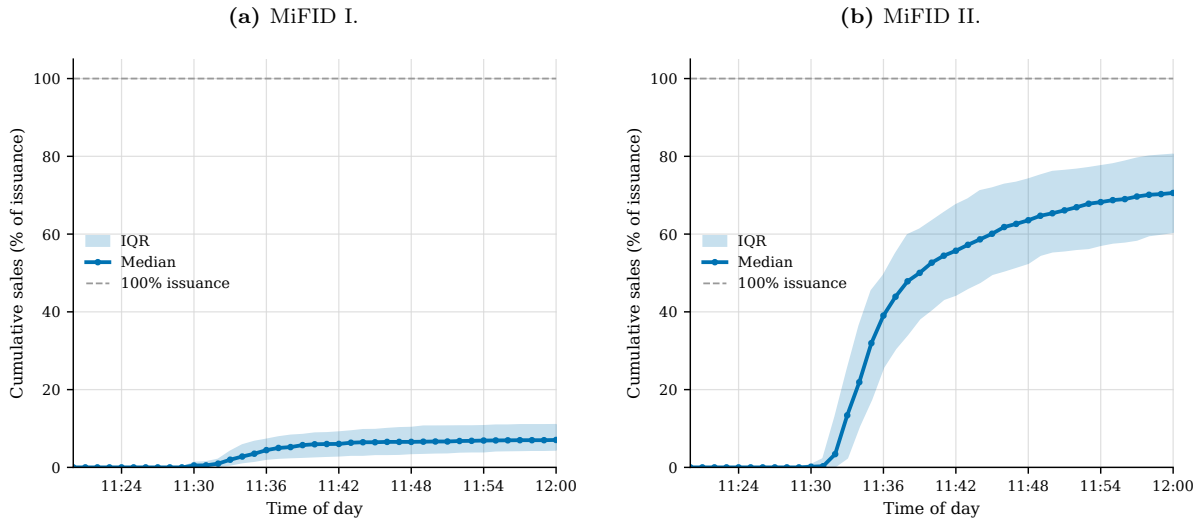


Figure A25: Primary issuance allocation legs in raw transaction reports. The panels plot, separately for MiFID I and MiFID II, the median and interquartile range of cumulative BBK/FA sales to primary dealers on auction days, normalized by the corresponding issuance volume. Each auction event is observed on a balanced one-minute grid from 11:20 to 12:00 CET. The horizontal dashed line marks 100% of issuance volume. The MiFID I panel is shown as a partial-coverage comparison under dealer-side reporting.

Figure A26 provides an independent, dealer-level check on these allocation legs across the full MiFID II sample. We aggregate each primary dealer's MiFID II auction allotments over a half-year, weighting each allocation by the bond's remaining maturity to match the Finanzagentur's own ranking basis, and compare the resulting ordering with the semiannual Bietergruppe ranking the Finanzagentur publishes. The maturity-weighted allotment share falls monotonically with the official rank (panel a), and the two orderings correlate strongly in every half-year (panel b). The residual gap reflects the ranking's broader scope: it covers all federal securities, including Bubills and inflation-linked and green bonds outside our nominal-Bund universe.

D.7 Futures Deliverability and Delivery Legs

Deliverability classification Each day, we flag whether each outstanding bond is deliverable into the front contract of the 2-year (FGBS), 5-year (FGBM), 10-year (FGBL), or 30-year

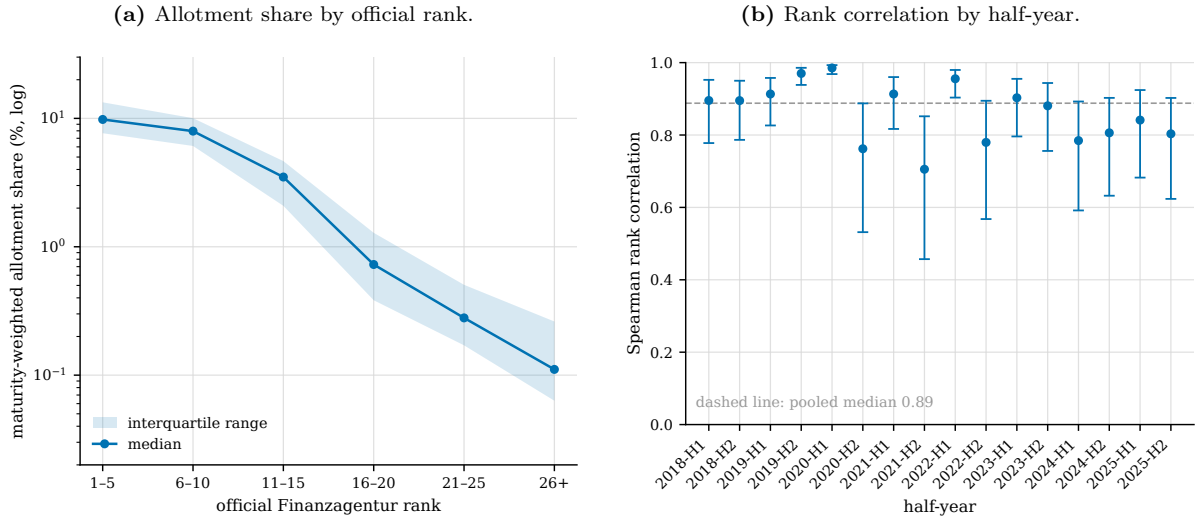


Figure A26: MiFID-derived primary-auction allotments against the official Finanzagentur Bietergruppe ranking over the MiFID II sample (2018–2025), both maturity-weighted (allotment volume times remaining years to maturity, matching the Finanzagentur's ranking basis). Panel (a): pooled median and interquartile range of each dealer's maturity-weighted allotment share of the half-year, by official-rank bin (bins of five). Panel (b): Spearman rank correlation between the MiFID-derived and official orderings per half-year, with 95% confidence intervals; the dashed line marks the pooled median.

(FGBX) Eurex Bund futures. Eligibility is based on remaining maturity at the futures delivery date: 1.75–2.25 years for FGBS, 4.5–5.5 years for FGBM, 8.5–10.5 years for FGBL, and 24–35 years for FGBX. Except for FGBX, deliverable bonds must have an original maturity of no more than 11 years. Since 1 December 2022, Eurex has also imposed a seasoning requirement: a bond must have been issued before the first calendar day of the previous delivery month. From September 2022 onward, we use the official list of deliverable bonds scraped from the Eurex website; before that date, we reconstruct deliverability by applying these eligibility rules to the universe of outstanding Bunds, counting a bond as deliverable only from its first auction onward.

Delivery legs The raw MiFID II reports also contain physical delivery legs of Eurex Bund futures. MiFIR excludes the delivery of an underlying after the exercise or expiry of a derivative from transaction reporting (European Securities and Markets Authority, 2016, Sections 5.6.2.6–5.6.2.7), so most firms correctly never report their Bund futures deliveries. A minority of clearing members report them nonetheless, typically as one large ticket per bond against Eurex Clearing, dated the futures contract's last trading day or the next business day. These rows are not secondary market cash trades. The pipeline therefore drops MiFID II rows dated either day when Eurex Clearing is the central counterparty, the bond is in the expiring contract's deliverable basket, and the quantity is at least €1 million. The size threshold separates delivery reports from the retail-sized exchange trades that Eurex Clearing also clears on the same days. We do not apply an analogous delivery-specific filter in MiFID I: in that regime, Eurex-related rows are handled through the general exchange and clearing-member normalization, and the raw data do not show a clean delivery-day signal that can be isolated with the same rule.

Figure A27 documents the filter. Figure A27a shows that raw rows involving Eurex Clearing have a sharp MiFID II spike on the last trading day, two business days before physical delivery, while the corresponding MiFID I series remains within its normal Eurex-related range. Figure A27b compares the volume selected by the MiFID II delivery filter with Eurex's public notified delivery volumes. The coverage distribution is a diagnostic, not an accounting reconcil-

iation: because most firms do not report deliveries, raw MiFIR usually captures only part of the notified face amount, and a small tail exceeds one-to-one coverage when both CCP-side reports or duplicate raw reports are visible.

(a) Delivery-window spike. **(b)** Coverage against Eurex notifications.

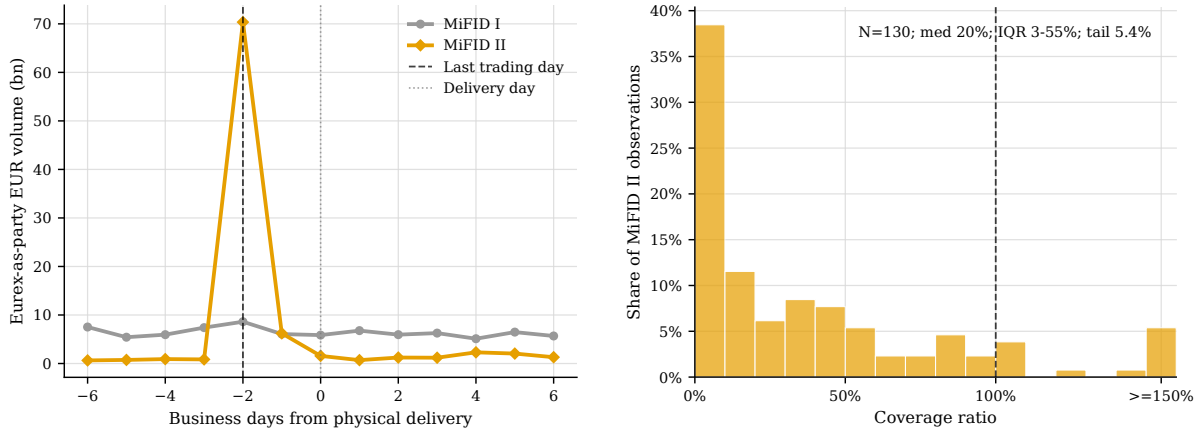


Figure A27: Futures-delivery filter diagnostic. The left panel plots aggregate raw € volume of rows involving Eurex Clearing around physical delivery dates, separately for MiFID I and MiFID II. Day 0 is physical delivery; the dashed vertical line marks the last trading day ($T = -2$). The right panel reports the MiFID II-only distribution of the coverage ratio, defined at the delivery-month–contract-root level as raw MiFIR € quantity satisfying the delivery filter divided by Eurex notified contracts times €100,000 face value. Bars are normalized to sum to 100%; bins are 10 percentage points and the final bin collects ratios of at least 150%. The histogram is a coverage diagnostic, not an exact reconciliation to delivered face amount.

D.8 Trades by CCPs, IDBs and venue operators

Because of reporting standards and our normalization of the data, "intermediaries" can appear as buyer or seller even if they do not take the bond into inventory. This is particularly relevant for CCPs and interdealer brokers (IDBs), and to a lesser extent venue operators. We pair the two sides of an intermediated trade where the reports identify the pairing, keep every other record, and record separately which records belong to one execution, so that reported-record volume can be distinguished from reconstructed execution volume. We proceed as follows.

Case 1: The intermediary is listed as buyer or seller and as the executing party of the transaction. This case concerns any transactions where the intermediary reports as an executing party in our data. We proceed in two steps:

First, we recombine reports split in the normalization step, provided they have not been aggregated in the meantime, joining the two records of one report on the report identifier and the trade date. A record that lists the executing party as both buyer and seller—a venue operator reporting against its own market identifier code—stands for the side its report lacks, and only for that side. We drop duplicates associated with either record of a recombined report.

Second, for the remaining reports, we group the intermediary's buy and sell records by bond, timestamp, and price. Within a group, a quantity with exactly one buy record and one sell record is paired into a single transaction between the two members, with the intermediary retained as executing party; Table A11 illustrates this case. Where one record faces several records on the other side whose quantities sum to no more than its own, each of the smaller records is paired with it at its own quantity, and any remainder stays as a residual record facing the intermediary. All other groups—several buy records against several sell records, or records

on one side only—are left as they are: each record keeps its member and faces the intermediary. Member reports that the duplicate matching linked to a paired record are deleted, since the pair already carries that trade; a member report linked to a record that remains unpaired stays until the final deduplication, which keeps one report per matched pair.

(a) Original reports.

Tx_id	Dup_id	Exe	Buyer	Seller	Time	Price	Qty
1	1	A	CCP	A	10:00:00	103	100,000
2	1	CCP	CCP	A	10:00:00	103	100,000
3	2	CCP	B	CCP	10:00:00	103	100,000
4	2	B	B	CCP	10:00:00	103	100,000

(b) Resulting report.

Exe	Buyer	Seller	Time	Price	Qty
CCP	B	A	10:00:00	103	100,000

Table A11: Case 1: original and resulting reports.

Case 2: The intermediary is listed as buyer or seller but not as the executing party of the transaction. The preceding steps concern reports where the executing party is an intermediary. Some reports instead list the intermediary as buyer or seller even though another party executed the trade, because the reporting executing party may not know the actual buyer or seller. These reports are kept as reported. In general, they should have a duplicate filed by the intermediary; if the duplicate-linking procedure identifies it, the final deduplication keeps one of the two reports. Otherwise the record stays as a one-sided trade: the intermediary is moved from the buyer or seller field into a separate counterparty field, and that side of the trade is left empty. The records that Case 1 leaves unpaired are treated the same way.

We keep the reported trading venue whenever it identifies a regulated market, MTF, or OTF. Trades intermediated by IDBs, such as BGC Brokers and GFI, usually carry the MIC code of the broker's OTF. When a report against an IDB or another venue operator instead carries an off-venue or systematic internalizer code, we assign the broker's most frequent trading venue. A central counterparty is not a trading venue, so reports against one keep their reported venue: LCH, for instance, clears trades executed both on MTS and on BrokerTec.

These steps do not eliminate pure intermediaries from the data as counterparties. Where the reports do not identify the pairing, the two members' reports of one intermediated trade survive as two one-sided records, each facing the intermediary. The execution layer described next records that they are one execution.

Trade records and executions The unit of the cleaned file is the trade record. One execution between two account owners can appear as two, three, or four records: the two records of an agency or matched principal report that the normalization step splits, an internal account booked in and out within the same second, or the two members' one-sided reports of an anonymous venue, CCP, or IDB trade. For example, when a bank buys a bond from a dealer as agent for a fund, its report becomes two records, dealer to bank and bank to fund; summing their quantities would count the bond twice. No record is deleted. Instead, each MiFID II record carries an execution identifier shared by linked records, a flag for whether the record's quantity counts toward execution volume, and the rule that linked it. The rules act on groups of records that

share a node—an identified account owner, an executing firm's internal account, or an empty side facing a venue, CCP, or IDB—in the same bond, second, and price. The two records of a split report are always linked, and the record on which the executing party sells stops counting. An identified party whose purchases and sales within the group balance, and that reported at least one of the records in an agency or matched principal capacity, is a pass-through: its selling records stop counting. A balanced pair that the party reports as principal on both sides is flagged but keeps counting on both records. For internal accounts and empty sides, the smaller side of the group by volume stops counting; without a venue trade identifier, this can merge two distinct opposite trades in the same bond, second, and price, so execution counts and execution volume are, if anything, understated. Where the reports carry a venue trade identifier, one-sided records of two reporters that share it in opposite directions are linked as one on-venue trade, and one-sided records that repeat the same identifier, party, direction, and quantity are counted once. When a split agency report and the other venue member's report form a three-record group, these rules can otherwise discount the same outgoing record twice and leave two records counting for one execution. We retain one counting record only when the raw-report lineage establishes the split and all three records agree exactly on price and quantity, with the venue-facing records sharing a known CCP or a specific MIC. This correction preserves the records and their economic fields; it does not collapse principal resale chains, partial fills, or larger ambiguous groups. Each final MiFID II record can be traced to its contributing raw reports through the cleaning transformations. Execution volume is the sum of quantities over counting records. MiFID I records are not linked and count once each.

Statistics that describe the size and venue composition of the market are computed on executions: trading volume and its comparison with MiFID I and the Finanzagentur, trade counts and sizes, venue-type shares, bond and venue concentration, and the venue shares around the Brexit transition. MiFID II has 7.7 million executions against 14.2 million records. The normalization step can record the client record of an agency chain, or one member's report of a venue trade, as off-venue, so an execution takes the venue type of its most specific venue record. Execution weighting lowers the OTC share of volume and raises the SI and OTF shares. Statistics defined by a side of the trade—sectors, counterparties, dealers, relationships, networks, net flows, settlement obligations, the dealer capacity mix with identified clients, and the regressions of execution costs and price impact—are computed on records: an execution has no single buyer or seller, and which record stops counting depends on the direction of the trade. There a "trade" is a record. Retail participation counts client allocations, one record per allocation. Execution type is read from each record as reported.

D.9 External transaction checks

We cross-check the regulatory reports against available MTS Bund transactions over 2008–2025 and records from Tradeweb, MTS BondVision, TP ICAP, and Fenics on six trading days in 2020 obtained for Kerssenfischer and Helmus (2025). After aligning price and quantity units and timestamps, we compare transaction prices, quantities, and execution counts across the sources. These comparisons check our reconstruction of executions, including the treatment of intermediated trades described in Appendix D.8. We use overlapping transactions and account for differences between commercial execution records and regulatory reports. Within the overlapping MiFID II samples, prices, quantities, and timestamps broadly agree with MTS, Tradeweb, and BondVision records.

D.10 Investor Sector Mapping

We construct an exhaustive mapping of LEIs to the following investor sectors: primary dealers, banks, interdealer brokers (IDB), investment funds, hedge funds, insurance companies and pension funds (ICPF), central counterparties (CCPs), official sector entities, non-financial corporations (NFC), and non-bank financial intermediaries (NBFI).

Table A12 lists our data sources. We prioritize high-quality regulatory data. The list of primary dealers participating in Bund auctions is available from Finanzagentur; we classify an LEI as a primary dealer if its auction-group membership was active during the MiFID II sample, that is, ending after 1 January 2018 or still open-ended, so that institutions that had left the auction group before then are not classified as dealers. We augment this with the European Central Bank's (ECB) Single Supervisory Mechanism (SSM) lists of Significant Institutions (SIs) and Less Significant Institutions (LSIs) and the ECB's list of Monetary Financial Institutions (MFIs), Investment Funds (IFs), Pension Funds (PFs), and Insurance Corporations (ICs). As the final regulatory data source, we use the Prudential Regulation Authority (PRA) register of U.K. regulated banks.

For non-European investors, reliable LEI-based investor classifications are not readily available. Hence, we proceed with two proprietary data sources. First, we use all available LEI-to-sector mappings from Moody's BvD BankFocus database. These data are especially useful for identifying large global banks. Second, we use Morningstar data to identify hedge funds. We used OpenAI's `gpt-6.1-sol` model in Codex with high reasoning effort to assist sector cross-checks for 362 observed identifiers, prioritizing large reported volumes and many reported records and consulting regulator and issuer sources where available; the authors adjudicated the final label decisions.

For any remaining LEIs, we first use the GLEIF Level 1 "Who is Who" register[28] and classify investors whose entity category is "FUND" as investment fund or as hedge fund if the fund is domiciled in the Cayman Islands. Lastly, we use the ECB's Register of Institutions and Affiliates Database (RIAD, see, for example, Perrella and Catz, 2020), which contains a mapping from LEIs to sectors (based on the European system of accounts framework).[29]

We use sector concentration within an ownership group as a diagnostic for classification discrepancies. Trading activity alone does not establish an entity's sector, so we retain sourced classifications and apply explicit corrections supported by regulatory or issuer evidence.

[28] https://www.gleif.org/en/lei-data/access-and-use-lei-data/level-1-data-who-is-who

[29] We avoid using RIAD where possible because it is notoriously unreliable. Apart from omitting many important investors, RIAD contains dubious sector classifications. For example, many Global Systemically Important Banks (G-SIBs) outside the euro area are classified as non-banks (ESA codes S.125-S.127), either because they are not depository institutions in the euro area or because they operate as holding companies.

Source Name	Description	Link
Finanzagentur	Official auction group list by Germany's debt management office	link
ECB SSM (SIs)	ECB list of systemically important credit institutions under direct supervision	link
ECB SSM (LSIs)	ECB list of smaller credit institutions under national supervision	link
ECB (MFIs)	ECB list of monetary financial institutions, including credit institutions and money market funds	link
ECB (IFs)	ECB quarterly register of investment funds by LEI and investment policy	link
ECB (PFs)	ECB register of pension funds by country of residence and LEI	link
ECB (ICs)	ECB register of insurance corporations by country of residence and LEI	link
U.K. PRA	U.K. Prudential Regulation Authority list of banks authorized in the U.K.	link
Moody's BvD BankFocus	Proprietary database incl. LEI-to-sector mapping	
Morningstar Hedge Funds	Proprietary database of hedge funds incl. LEI	
Manual Classification	Manually curated LEI-to-sector mapping	
GLEIF Funds	Global Legal Entity Identifier Foundation (GLEIF) Level 1 entities categorized as investment funds	link
ECB RIAD	ECB Register of Institutions and Affiliates Database (RIAD)	

Table A12: Investor reference data sources, hierarchical order.

Table A13 and Table A15 report descriptive diagnostics for this investor sector mapping. The unit is an account-owner side of a cleaned Bund transaction, so each transaction contributes once for the buyer and once for the seller. These diagnostics document the coverage and composition of the mapping used in the paper's downstream analyses.

Panel A: group-level sector coverage by sample

Sample	Sides	Vol. (EUR bn)	Class. sides (%)	Class. vol. (%)	Groups	Class. groups (%)
MiFID I	3,998,530	16,562.0	77.8	64.1	1,475	97.6
MiFID II	25,867,408	101,554.2	92.9	82.3	41,172	77.4
Full sample	29,865,938	118,116.2	90.8	79.8	41,482	77.5

Panel B: entity-source tier mix, MiFID II account-owner sides

Source tier	Sides	Side share (%)	Vol. (EUR bn)	Vol. share (%)	Groups
Regulatory lists	13,168,387	50.9	47,747.5	47.0	7,749
Hand/manual	5,692,808	22.0	32,186.1	31.7	1,337
Commercial data	598,315	2.3	2,461.8	2.4	3,466
GLEIF category	167,979	0.6	375.3	0.4	9,549
Name/MIC heuristic	284,334	1.1	344.5	0.3	769
RIAD fallback	231,566	0.9	563.2	0.6	18,874
Retail flag	3,966,190	15.3	71.5	0.1	0
Unclassified/no source	1,757,829	6.8	17,804.2	17.5	0

Table A13: Investor sector classification coverage and source provenance. Panel A reports group-level sector coverage by reporting regime for account-owner sides of cleaned Bund transactions in the analysis sample. Volume is side-volume, so each transaction contributes once for the buyer and once for the seller. Classified sides have a nonmissing group-level sector. Group coverage excludes retail sides without LEIs. Panel B reports the entity-level source tier recorded in the investor reference data for MiFID II account-owner sides. Source tiers are grouped for readability.

Counterparty coverage by regime The two regimes name different counterparties. A MiFID I report names the reporting firm and, by BaFin code, a counterparty that is itself subject to reporting; an end client usually stays unnamed. A MiFID II report names the buyer and the seller themselves. Table A14 shows the consequence for the client side of dealer-to-client records. In 2013–2017, 59% of its record volume carries no client identifier, 23% names only a CCP, interdealer broker, or venue, and 17% carries a sector. In 2018–2022, these shares are 15%, 22%, and 62%. The identified MiFID I clients are mostly banks and official sector entities: investment funds, hedge funds, insurers and pension funds, other non-bank financial intermediaries, and non-financial corporations together account for 1.6% of client-side volume in 2013–2017, against 39.2% in 2018–2022. Over the same two windows, the number of investment fund groups on the client side rises from 76 to 2,050, and the number of hedge fund groups from 6 to 389. Sector information therefore exists before 2018 (59–68% of buyer and seller volume carries a sector in each MiFID I year), but it describes regulated counterparties, not end clients.

Sector	Full sides (%)	Full vol. (%)	MiFID II sides (%)	MiFID II vol. (%)	Groups
Primary Dealer	34.5	54.9	35.5	56.7	38
IDB	0.0	0.0	0.0	0.0	0
CCP	0.0	0.0	0.0	0.0	0
Bank	21.9	3.9	23.0	2.8	2,118
Investment Fund	9.8	8.0	11.2	9.2	9,324
ICPF	1.3	0.8	1.4	0.9	1,546
Hedge Fund	2.0	6.8	2.3	7.9	732
Official	1.2	4.3	1.2	4.1	412
NBFI	3.8	0.4	2.1	0.2	2,597
NFC	0.7	0.3	0.7	0.3	15,578
Retail	15.6	0.4	15.3	0.1	0
Unclassified	9.2	20.2	7.1	17.7	9,345

Table A15: Investor sector composition of account-owner sides. The table reports the distribution of group-level investor sectors used in the paper's downstream analyses. The IDB and CCP rows are zero by construction: interdealer brokers and central counterparties are part of the entity-level mapping, but the normalization in Appendix D.8 pairs the two sides of the trades they intermediate or moves the intermediary into a separate counterparty field, so they do not appear as account owners in the cleaned file. The full sample spans 2008–2025; the MiFID II sample spans 2018–2025. Volume shares are side-volume shares.

Panel A: identification of account owners by year, share of record volume (%)

	All sides	Client side of dealer-to-client records			
		No client identifier			
Year	Sector	CCP, IDB, venue	None	Identifier, no sector	Sector
2008	62	16	64	1	19
2009	68	14	57	2	27
2010	66	13	61	1	24
2011	65	14	62	2	22
2012	67	14	57	2	27
2013	65	24	50	2	24
2014	63	29	53	1	17
2015	59	22	65	1	13
2016	60	22	65	0	13
2017	66	21	61	1	17
2018	80	26	12	0	61
2019	81	23	14	0	63
2020	83	16	20	0	63
2021	88	24	9	0	66
2022	86	22	18	0	60
2023	81	18	34	0	48
2024	80	20	30	0	50
2025	81	20	29	0	51

Panel B: client side of dealer-to-client records by sector

	MiFID I, 2013–2017		MiFID II, 2018–2022	
Client side	Volume (%)	Groups	Volume (%)	Groups
Banks	10.0	447	7.5	1,109
Investment funds	0.4	76	16.3	2,050
Hedge funds	0.2	6	20.1	389
ICPF	0.1	13	1.4	507
Official sector	5.0	32	15.6	160
NBFI	0.5	25	0.8	574
NFC	0.3	20	0.6	2,524
Households	0.0	–	0.0	–
Identifier without sector	1.1		0.3	
CCP, IDB or venue	23.3		22.3	
No client identifier	59.0		15.0	

Table A14: Counterparty identification in MiFID I and MiFID II. Buyer and seller sides of cleaned records in the analysis sample; shares of record volume, so each record counts once for the buyer and once for the seller. All sides: share of volume whose account owner has a group-level sector. A dealer-to-client record has exactly one side on a primary dealer's trading desk on the trade date; the other side is the client side. CCP, IDB, venue: the client side is a central counterparty, interdealer broker, or trading venue (Appendix D.8). None: no account owner, or an internal account, including reports filed without any account owner, which occur in every year and more often from 2023. Identifier, no sector: an LEI or BaFin code without a sector. Sector includes natural persons; banks include primary dealer sector banks outside their auction group window on the trade date. Panel B uses five-year windows; groups are distinct investor groups with at least one client-side record in the window.

Investor Grouping The group hierarchy uses GLEIF Level 2 direct and ultimate-parent relationships, augmented with ECB lists of Significant Institutions and Monetary Financial Institutions (the latter for banks). We prioritize ultimate-parent links, resolve remaining direct-parent chains recursively, terminate circular chains at the first repeated entity, and treat entities without valid ownership links as singleton groups. Only operating companies—entities GLEIF classifies as *general* or *branch*—are consolidated; fund vehicles (GLEIF category *fund*) stay out of the hierarchy. Consolidating funds would merge the many separately identified vehicles advised by a single manager into one investor: in our reference data BlackRock alone spans roughly three thousand distinct fund LEIs, including some 1,900 iShares ETFs, which consolidation would collapse into a single "BlackRock" investor. The asset-management companies themselves are still consolidated—a manager's discretionary trading arms are grouped—but the funds they advise remain distinct investors; interdealer brokers, by contrast, collapse to roughly one group per two dozen LEIs.

Within a primary dealer group, every entity except the subsidiaries whose own sector is asset management or insurance belongs to the dealer desk. It carries the dealer flag on every date on which an entity of the group is an auction member, whichever entity holds the membership. The excluded subsidiaries remain clients in their own sectors, so their trades with the parent's desk enter as dealer-to-client trades (the margin used in Figure A7). The same principle applies outside primary dealer groups: an asset manager keeps the investment fund sector at the group level even when an insurer, bank, pension fund, or non-financial corporation owns it, and execution subsidiaries that route a manager's fund orders to dealers are classified with the manager. A small number of manual corrections align group assignments with in-sample corporate structure: entities retired during the sample are attached to their in-sample successor (for example, Banca IMI to Intesa Sanpaolo), and Credit Suisse is kept separate from UBS, whose 2023 acquisition the current register snapshot would otherwise back-date onto the entire sample.

D.11 Venue Classification

We combine three external reference datasets on trading venues: the ESMA Financial Instruments Transparency System (FITS) register,[30] the U.K. Financial Conduct Authority (FCA) register,[31] and the ISO 10383 list of Market Identifier Codes (MICs).[32] We use the ESMA register as the primary source; for MIC codes not classified by ESMA, we rely on the FCA register; and for the remaining venues, we rely on ISO data. All venue types are harmonized to the ESMA taxonomy. A small number of venues only listed in the ISO data do not carry explicit type classifications. For these, we apply another hierarchical classification: (i) we match the LEI code of each MIC code to our sector classification (classifying venues run by interdealer brokers as OTF and those run by primary dealers as SI), (ii) we parse the "description" field for unambiguous keywords ("systematic internaliser", "approved publication arrangement", "electronic trading platform", and "organised trading facility"), and (iii) we classify based on well-known institution names (for example, Tradeweb and Bloomberg mapped to MTF, and major IDBs mapped to OTF). Approved publication arrangements and off-venue MICs are collapsed into OTC/off-venue, while residual unclassified MICs are retained as Other. This yields a harmonized classification of trading venues to venue types and a consistent mapping from MICs to LEIs.

We separately assign each MIC a venue protocol family. The family describes the trading model of a venue or segment; it does not record how an individual trade executed. A MIC is

[30] https://registers.esma.europa.eu/publication/searchRegister?core=esma_registers_upreg
[31] https://register.fca.org.uk/servlet/servlet.FileDownload?file=0150X000006gbb1
[32] https://www.iso20022.org/sites/default/files/ISO10383_MIC/ISO10383_MIC.xlsx

in the request-for-quote (RFQ) family or the order book family (CLOB) only when the venue's rulebook or its operator's description documents that model for the venue or segment itself; documents for a sister venue of the same operator in another jurisdiction are not transferred, and venue type alone never assigns either family. Venues whose documented model is neither, such as specialist and market maker segments of exchanges, and all OTC and OTF codes form the residual other family. The remaining MICs are unknown: every systematic internalizer, because SI is a regulatory status, not a trading model; venues documented as combining an order book with RFQ trading; and venues without a source. Each label holds for the whole sample period. The panel contains 276 distinct MICs. The 118 material MICs, which together account for 99.8% of execution volume, received a recorded disposition, and the curated RFQ, order book, and other labels rest on primary sources; the remaining MICs follow the residual rule. We used OpenAI's `gpt-6.1-sol` model in Codex with high reasoning effort to assist the source review of the material MICs; the authors adjudicated the final labels. The family labels are not taken from the ESMA, FCA, or ISO registers. Table A16 reports the share of execution volume and of executions in each family and evidence class, separately for MiFID I, MiFID II before 2021, and MiFID II from 2021; the 2021 split marks the exit of U.K. venues from the reporting perimeter.

		Venue protocol family				Evidence				
Period	Share of	RFQ	CLOB	Other	Unknown	Rulebook	Operator	Type	Mixed	None
MiFID I, 2008–2017	EUR volume	0.6	13.6	83.4	2.4	1.7	15.9	80.0	0.0	2.4
	Executions	0.4	3.2	90.5	5.9	15.7	27.7	50.7	0.1	5.9
MiFID II, 2018–2020	EUR volume	29.2	0.7	48.9	21.3	30.4	2.5	45.8	0.0	21.3
	Executions	47.3	2.7	33.1	17.0	45.5	6.4	31.1	0.7	16.3
MiFID II, 2021–2025	EUR volume	24.5	0.6	53.5	21.3	16.9	8.6	53.2	0.0	21.3
	Executions	28.0	5.2	45.4	21.5	22.3	17.8	38.3	0.1	21.3

Table A16: Coverage of the venue protocol family. Shares of execution volume (EUR volume rows) and of executions (Executions rows) by venue protocol family and by evidence class, in percent of the period total. Evidence classes: rulebook of the venue or segment; operator description; type (OTC or OTF residual); mixed (order book plus RFQ); none. Executions are reconstructed from trade records; MiFID I records are one execution each.

For venue group statistics, we merge MICs of the same venue type when they belong to the same operator. This grouping matters after Brexit because operators such as Bloomberg, Tradeweb, and MarketAxess maintain parallel EU and U.K. venues with distinct MICs but similar trading functionality. Statistics based on raw MIC codes would mechanically split those operator groups and understate concentration, especially for MTFs.